Fall 2025: Representations of Lie Groups

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*based on lectures by Roman Bezrukavnikov

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Introduction

The representations of simple Lie algebras over $\mathbb{C}$ is closed related to representations of compact Lie groups. The goal of this course is to generalize to noncompact Lie groups such as $\mathrm{SL}(n,\mathbb{R})$. Our focus will be real reductive Lie groups, in particular their infinite dimensional representations. Core technique will be algebraic (some algebraic geometry at the end), but we will also use some functional analysis.

The motivation for this subject comes from quantum physics. It also applies to theory of automorphic forms, which is related to local Langlands conjectures.

Reference will be Etingof's book "Representations of Real Groups" and Gaitsgory's Lecture Notes on "Representations of Real Reductive Groups".

1 Introduction

If G is a finite group, it has finitely many irreducible representations over $\mathbb{C}$. All of these are unitary: they all have invariant Hermitian forms. Note we have decomposition:

$$\mathbb{C}[G] = \bigoplus V_i^* \otimes V_i = \bigoplus \mathrm{End}_{\mathbb{C}}(V_i)$$

as (G, G)-bimodule. The identification is given by matrix coefficients:

$$v^* \otimes v \mapsto (m_{v^*,v} : g \mapsto \langle v^*, gv \rangle)$$

For each V_i, we have χ_{V_i} the characters. These form a basis for $\mathbb{C}[G]^G$. For $\varphi \in \mathbb{C}[G]$, we consider the action $\varphi \cdot v \coloneqq \sum_{g \in G} \varphi(g) gv$ for every $v \in V$ in a G-module. In particular, if we consider the regular representation, then:

$$\varphi \cdot f = \sum_{g \in G} \varphi(g) f(g^{-1} \cdot -)$$

is the convolution product. Note the unit for the convolution product is δ_1.

Now on $\mathbb{C}[G]$ we also have a G-invariant inner product:

$$\langle \varphi, \psi \rangle = \frac{1}{|G|} \sum_{g \in G} \varphi(g) \overline{\psi(g)}$$

Now note that:

$$\varphi(1) = \frac{1}{|G|} \mathrm{Tr}(\varphi, \mathbb{C}[G]) = \frac{1}{|G|} \mathrm{Tr}(\varphi, \bigoplus V_i^* \otimes V_i) = \frac{1}{|G|} \sum_i \dim(V_i) \mathrm{Tr}(\varphi, V_i)$$

As a consequence:

$$\langle \varphi, \psi \rangle = \frac{1}{|G|} \langle \varphi, \psi * \delta_1 \rangle = \frac{1}{|G|} \langle \psi^\dagger \varphi, \delta_1 \rangle = \psi^\dagger \varphi(1) = \frac{1}{|G|} \sum_i \dim(V_i) \mathrm{Tr}(\psi^\dagger \varphi, V_i)$$

where $\psi^\dagger$ is the Hermitian adjoint (note that the Hermitian adjoint of δ_g is $|G| \cdot \delta_g^{-1}$). In particular, if we take $\psi = \delta_g$, then:

$$\varphi(g) = |G| \langle \varphi, \delta_g \rangle = \sum_i \dim(V_i) \mathrm{Tr}(g^{-1} \circ \varphi, V_i)$$

More generally, let G be a compact connected Lie group, $\mathfrak{g} = \mathrm{Lie}(G)_{\mathbb{C}}$. If G is simply connected, then there are well-known bijections:

$$\mathrm{Irr}(G) \longleftrightarrow \mathrm{Irr}_{\mathrm{f.d}}(\mathfrak{g}) \longleftrightarrow X^*(T)^+ = \{\text{dominant integral weights}\}$$

Example. *Let $G = U(1) = S^1$. The irreducible representations are indexed by $\mathbb{Z}$. Note here the group is not simply connected.*

Example. *Let $G = SU(n)$, $\mathfrak{g} = \mathfrak{sl}(n, \mathbb{C})$. A weight is a character of the maximal torus $T = \{(z_1, \cdots, z_n) \mid \prod z_i = 1\}$. The weights are $\Lambda = \mathbb{Z}^n/\Delta\mathbb{Z}$. The dominant weights are indexed by nondecreasing sequence $\lambda \in \mathbb{Z}^n$. In particular, for $n = 2$, $\Lambda = \mathbb{Z}$, and $V_d = \mathrm{Sym}^d\mathbb{C}^2 = \mathbb{C}[x, y]_d$. Note that conceptually, $\mathbb{C}^2 \simeq V^*$ is the dual module (with right G)-action. This corresponds to dominant weight $\lambda = (d, 0)$: indeed, the highest weight vector is y^d with:*

$$\begin{pmatrix} z & \\ & z^{-1} \end{pmatrix} \cdot y^d = z^d \cdot y^d$$

For $SU(n)$, $\mathbb{C}[x_1, \cdots, x_n]_d = \mathrm{Sym}^d(V^*)$ is still an irreducible representation of $SU(n)$, corresponding to weight $(d, d, \cdots, 0)$.

Generalizing from finite groups, we hava map of $G \times G$ representations:

$$\bigoplus_\lambda V_\lambda \otimes V_\lambda^* \hookrightarrow C^\infty(G)$$

Theorem 1.1 (Peter Weyl)**.** *Let G be a compact Lie group. Then we have:*

$$L^2(G) \simeq \widehat{\bigoplus}_{\lambda \in X^*(T)^+} V_\lambda^* \otimes V_\lambda$$

Example. *For $G = U(1) \simeq S^1$. We recover the Fourier series expansion. Indeed let V_n be the representation $z \mapsto z^n$. Let $f : S^1 \to \mathbb{C}$ be any smooth function, then the action of f on V_n^* is given by scalar:*

$$\hat{f}_n = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta}) e^{-in\theta} d\theta$$

Hence this recovers

$$f(e^{i\theta}) = \sum_n \hat{f}_n e^{in\theta}$$

We say $C_{\mathrm{fin}}(G) = \bigoplus_\lambda V_\lambda \otimes V_\lambda^*$ is the space of G-finite functions, i.e. the subspace spanned by $g \cdot f$ is finite dimensional. Note $C^\infty(G)$ and $C_{\mathrm{fin}}(G)$ has two ring strucuters: pointwise or convolution. In the convolution case, this is a nonunital ring. We can work with a larger ring $\mathrm{Dist}(G)$, then δ_1 is the unit.

Moreover $C_{\mathrm{fin}}(G)$ has comultiplication coming from:

$$G \times G \xrightarrow{m} G$$

We thus have $m^* : C_{\mathrm{fin}}(G) \to C_{\mathrm{fin}}(G) \otimes C_{\mathrm{fin}}(G)$. This gives a complex affine algebraic group: $G_{\mathrm{alg}} = \mathrm{Spec}\ (C_{\mathrm{fin}}(G))$. Hence an irreducible reps of G gives an algebraic reps of G_{alg}. This is an equivalence of categories. Conversely, starting with a $\mathbb{C}$-algebraic group, we can recover G as a maximal compact subgroup in $G_{\mathrm{alg}}(\mathbb{C})$.

The formula for $f(1)$ and $\langle \varphi, \psi \rangle$ generalizes here.

The goal is to understand unitary (or "all") representations on noncompact groups.

Remark. *If G is simple, the unitary representations are infinite dimensional or trivial. Since if $\rho : G \to SU(N)$, the image must be closed, hence compact.*

We need to consider continuous action on a topological vector space over $\mathbb{C}$. For a unitary representation, we look for action on a Hilbert space. More generally, we need to consider Banach spaces or Fréchet spaces.

Example. *If G acts on a manifold X, then G acts on $C^\infty(X)$. If the action also preserves a volume form, we can consider $L^2(X, \omega_X)$.*

Definition 1.2. If $\rho : G \to \mathrm{Aut}(V)$ is a continuous action on a topological vector space. A subrepresentation is a closed invariant subspace. We can define the notion of irreducible representation.

We have Schur lemma. So if the representation has nontrivial automorphism, then it's not irreducible (for exampe in the case G acting on $L^2(G)$).

Example. *This is a basic example of an irreducible representation of noncompact groups.*

Let V be the space of functions on $\mathbb{R}^n$ acted upon by Heisenberg group H_n. Recall H_1 are upper triangular 3×3 matrices. In general, the Heisenberg group H_n is isomorphic to $\mathbb{R}^{2n+1}$ as a manifold. It has 2 copies of $(\mathbb{R}^n, +)$ and one copy of central $\mathbb{R} \cdot c$ inside of it, subject to relation $[v, w] = (v, w) \cdot c$. Hence $H_n/(\mathbb{R}, +) \simeq (\mathbb{R}^{2n}, +)$.

We now describe the action.

- *One copy acts by translation, namely $(v \cdot f)(x) = f(v + x)$*
- *The other copy acts by multiplication by a character: $(w \cdot f)(x) = e^{2\pi i(w,x)} f(x)$*
- *Hence we can determine the action of the central element. Suppose $(v, w) = t$, we know:*

$$\begin{aligned}(tc \cdot f)(x) = ([v, w]f)(x) = (wv^{-1}w^{-1}f)(x + v) = e^{2\pi i(w,x+v)}(v^{-1}w^{-1}f)(x + v)\\ = e^{2\pi i(w,x+v)}(w^{-1}f)(x) = e^{2\pi i(w,v)} f(x).\end{aligned}$$

Hence tc acts by multiplication by $e^{2\pi it}$.

In fact, the Heisenberg group H_n also acts on $L^2(\mathbb{R}^n), C^\infty(\mathbb{R}^n)$, and Schwarz space

$$\mathcal{S}(\mathbb{R}^n) = \{f \in C^\infty(\mathbb{R}^n) \mid |P(x)D(f)| \text{ bounded}\}$$

Example. *There is a finite analogue of the above example. Take $H_n(\mathbb{F}_p)$, a p-group of order p^{2n+1}. Let $V = \mathbb{F}_p^n$. Note $H_n = \mathbb{F}_p \times V \times V^*$ as a set, acts on $\mathbb{C}[V]$, where V acts by translations, $\xi \in V^*$ acts by multiplication by $\psi_\xi(v) = \exp(\frac{2\pi i\langle v,\xi\rangle}{p})$, $\mathbb{F}_p$ acts by scalar multiplication $a \mapsto \exp(\frac{2\pi ia}{p}) \cdot 1_V$.*

To check it is irreducible: it suffices to check $\mathrm{End}_{H_n}(\mathbb{C}[V]) = \mathbb{C}$.

Indeed, $\mathrm{End}_V(\mathbb{C}[V]) = \mathrm{End}_{\mathbb{C}[V]}(\mathbb{C}[V]) = \mathbb{C}[V]$. So if $\varphi : \mathbb{C}[V] \to \mathbb{C}[V]$, then $\varphi(g) = g * f$ for some function $f \in \mathbb{C}[V]$. But this map is also V^*-equivariant, namely:

$$(\xi g) * f = \xi(g * f)$$

Now we specialize to $g = \delta_0 \in \mathbb{C}[V]$ we have:

$$f(v) = f(v) \cdot \exp(\frac{2\pi i\langle v, \xi\rangle}{p})$$

So $f \in \mathbb{C} \cdot \delta_0$.

Remark. *Similar argument works for H_n acting on $L^2(\mathbb{R}^n)$.*

For a Lie group G acting on manifold X. If G preserves volume form, then $C_c^\infty(X)$ carries an invariant Hermitian pairing: $\int_X \varphi(x)\overline{\psi(x)}dx$.

More generally, can define this on *half-densities* $f(x)\sqrt{|dx|}$. Fixing a coordinate sysmtem on $U \subset X$, we identify half densities on U with functions; but for a different coordinate system $y = y(x)$, we must have

$$f(x)\sqrt{|dx|} = f'(y)\sqrt{|dy|}$$

so

$$f(x) = f'(y(x))\sqrt{\det(\frac{\partial y_i}{\partial x_j})}$$

Example. *Let $G = \mathrm{SL}(2,\mathbb{R})$, B be the standard Borel, U the unipotent radical of B. We have $G/U \simeq \mathbb{R}^2\backslash\{0\}$ and $G/B \simeq \mathbb{RP}^1$. The maximal torus $T \simeq B/U \simeq \mathbb{R}^\times$. So we have a T-fibration $p : G/U \to G/B$. Consider $\mathcal{O}(G/U)$ given by smooth functions, this is naturally G-equivariant.*

Now each character of T will give rise to a (G,T)-equivariant line bundle on G/U, which corresponds to a G-equivariant line bundle on G/B. Hence for character λ of T, we have:

$$L_\lambda = \Gamma(G/B, \mathcal{L}_\lambda) = \{f \in \mathcal{O}(G/U) \mid f(gt) = \lambda(t)^{-1}f(g)\}$$

Now return to the case of $\mathrm{SL}(2,\mathbb{R})$. The T-action on $G/U = \mathbb{R}^2\backslash\{0\}$ is just given by dilation. Each complex number s gives rise to a character $t \mapsto |t|^s$, this in turn gives a G-equivariant line bundle $\mathcal{L}_s$ on G/B, and:

$$L_s = \Gamma(G/B, \mathcal{L}_s) = \{f \in \mathcal{O}(\mathbb{R}^2\backslash\{0\}) \mid f(vt) = |t|^{-s}f(v)\}$$

Note for $s \in \mathbb{R}$, the line bundle $\mathcal{L}_s$ is precisely the bundle of $\frac{s}{2}$-densities. Indeed, as a space L_s is just functions on $\mathbb{RP}^1$, but the G-action (on the affine part) is given by: if $g^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then:

$$g \cdot f(x) = g \cdot f(x,1) = f(ax+b, cx+d) = f(\frac{ax+b}{cx+d})|cx+d|^{-s}$$

But this is precisley how $\frac{s}{2}$-densities transform:

$$g \cdot f(x)|dx|^{\frac{s}{2}} = f(\frac{ax+b}{cx+d})|d(x \circ g^{-1})|^{\frac{s}{2}} = f(\frac{ax+b}{cx+d})|d(\frac{ax+b}{cx+d})|^{\frac{s}{2}} = f(\frac{ax+b}{cx+d})|cx+d|^{-s}|dx|^{\frac{s}{2}}$$

Example. *We work in the above example for $s = 0$ but with smooth functions. It says $C^\infty(\mathbb{RP}^1)$ is naturally a $\mathrm{SL}(2,\mathbb{R})$ representation. The Fourier series expansion tells us that:*

$$C^\infty(\mathbb{RP}^1)/\mathbb{C} \times 1 \simeq L^+ \oplus L^-$$

where L^+ is the closure of the boundary of the functions $f \in z\mathbb{C}[z]$ on the unit circle; L^- is the closure of the boundary of the functions $f \in z^{-1}\mathbb{C}[z^{-1}]$ on the circle.

2 Recap on Functional Analysis

We now work with topological vector spaces over $\mathbb{C}$, Hausdorff and locally convex (i.e. there exists a a basis neighborhoods of 0 consisting of convex opens $\iff$ topology is given by a collection of semi-norms: $v(x) \geq 0$ for all x, homogeneous, and satisfy the triangle inequality)

If we assume sequential completeness and first countable, then the topology is given by sequence of seminorms $\nu_1, \nu_2, \cdots$. Moreover, in this case second countbale is equivalent to separable.

Definition 2.1. A *Fréchet space* is a Hausdorff, locally convex, sequentially complete, and second countable topological vector space. If the topology is given by one norm, it is called a *Banach space*.

Remark. *$L^p([0,1])$ for $p < 1$ is a nonexample. The only nonempty convex open is the whole space $\iff$ are no nonzero continuous linear functionals.*

A Fréchet space is metrizable: for ν_i seminorms, we define

$$d(x,y) = \sum \frac{1}{2^n} \frac{\nu_n(x-y)}{1+\nu_n(x-y)}$$

This is translation invariant.

Let $C(X) = C_0(X)$ be continuous function endowed with topology of convergence uniform on compact set. If X is compact, this is a Banach space (we have the sup norm). In general, if X is Hausdorff and second countable. then $C(X)$ is a locally compact Fréchet space: write $X = \cup K_i$ for K_i compact increasing, $\nu_i = \sup_{x \in K_i} |f(x)|$

For X a manifold, $C^k(X)$ uniform convergence of first k derivaties. It is Banach for X compact, $k \in \mathbb{N}$. This becomes Fréchet if X is noncompact or $k = \infty$.

Remark. *If X is noncompact manifold, $C_c^\infty(X)$ is not Fréchet (as it's not complete). But $\mathcal{S}(\mathbb{R}^n)$ is a Fréchet space.*

Lemma 2.2. *An action of a topological group on a Banach space V is continuous if and only if $G \to \mathrm{Aut}(V)$ is continuous in strong topology of $\mathrm{Aut}(V)$.*

Remark. *Recall $E_i \to E$ in strong topology if $E_i(v) \to E(v)$ for any $v \in V$.*

Proof. Use uniform boundedness. If $\sup_i |E_i(v)|$ is bounded for every v, then $\sup_i \|E_i\|$ is bounded.

$\implies$ Suppose the action is continuous, $g_i \to g \implies g_i v \to gv$, so continous in strong topology.

$\impliedby$ We need to show if $(g_i, v_i) \to (g, v)$, then $g_i v_i \to gv$. Indeed, we already know $g_i v - gv \to 0$ as $i \to \infty$. Hence it suffices to show $g_i v_i - g_i v \to 0$. Suppose not, then $\frac{|g_i(v_i - v)|}{|v_i - v|} \to \infty$, contradicting uniform boundedness. $\square$

Remark. *We can endow $\mathrm{End}(V)$ with the operator norm. But then $G \to \mathrm{End}(V)$ is not continuous for norm topology.*

3 Algebra of Measures

Let X be a Hausdorff locally compact second countable topological space. We define

$$C(X) = \varprojlim_{K_i \subset X \text{ compact}} C(K_i)$$

It has topological dual

$$C(X)^* = \varinjlim_{K_i \subset X \text{ compact}} C(K_i)^*$$

We equip $C(X)^*$ with weak topology. Then $\mathrm{Meas}_c(X) \coloneqq C(X)^*$ is defined to be the space of (compactly-supported) *measures*.

Given μ a measure, we define $\mathrm{supp}(\mu)$ as following: $x \notin \mathrm{supp}(\mu)$ if there exists $x \in U$ such that $\mu(f) = \int f d\mu = 0$ for all f with $f|_{X\setminus U} = 0$. Note this implies the support is always closed. In fact $\mathrm{supp}(\mu)$ is compact for all $\mu \in C(X)^*$.

Remark. *If X is compact, $C(X)^*$ has the operator norm $\sup_{|f|=1} \frac{|\mu(f)|}{|f|}$. But the topology on $C(X)^*$ is not the norm topology. Indeed if $x \to x_i$, then $\delta_x \to \delta_{x_i}$ in the weak topology, but $|\delta_x - \delta_{x_i}| = 2$.*

Example. *1. δ_x*

2. X manifold with volume form $|dx|$. We have $C_c(X) \subset C(X)^*$ via

$$f : \varphi \mapsto \int_X f(x)\varphi(x)|dx|$$

Definition 3.1. A topological vector space V is complete if for any embedding $V \subset W$ such that the topology of V is the topology induced from W, then V is closed in W

Lemma 3.2. *1. If (μ_i) is a Cauchy sequence in $C(X)^*$, then there exists a compact K such that $\mathrm{supp}(\mu_i) \subset K$.*

2. $C(X)^*$ is sequentially complete.

3. $C(X)^*$ is separable.

Remark. *$C(X)^*$ is not necessarily complete or first countable.*

Example. *Let $X = \mathbb{N}$, $C(X)$ are sequence in $\mathbb{C}$, $C(X)^*$ are finite sequences in $\mathbb{C}$. $\sigma^i \to \sigma$ if there exists a finite subset $S \subset \mathbb{N}$ with $\mathrm{supp}(\sigma^i), \mathrm{supp}(\sigma) \subset S$ and $\sigma^i_j \to \sigma_j$ for all $j \in S$.*

Note This space is sequentially complete, but is not complete (it's not closed in $C(X)^*_{\mathrm{alg}}$).

This is a also countable dimensional vector space but it has no countable basis in neighborhood 0. Indeed, if $U_1 \supset U_2 \supset \cdots$ is a base of neighborhoods of 0, set $\delta_m = e_m = (0, 0, \cdots, 1, 0, \cdots)$. Then there exists $c_m \in \mathbb{R}_{>0}$ such that $\frac{e_m}{c_m} \in U_m$ with $\frac{e_m}{c_m} \to 0$. Now choose $f = (c_1, c_2, \cdots)$, then $(\frac{e_m}{c_m}, f) = 1$, contradiction. This means this space is not first countable.

Let $\mathrm{Meas}^0_c(X) = \{\sum_{i=1}^n c_i \delta_{x_i}\} \subset \mathrm{Meas}_c(X)$.

Lemma 3.3. $\mathrm{Meas}_c^0(X) \subset \mathrm{Meas}_c(X)$ *is sequentially dense.*

Proof. Use partition of unity. Let $\mu \in \mathrm{Meas}_c(X)$, WLOG X is compact. For any open cover $X = \bigcup_{i=1}^n U_i$, we can find $f_i : X \to \mathbb{R}_{\geq 0}$ with $\mathrm{supp}(f_i) \subset U_i$ and $\sum f_i = 1$.

Now pick fine enough partition, $x_i \in U_i$ and set $\mu' = \sum \mu(f_i)\delta_{x_i}$. Now use the fact that X is metrizable and compact, so continuous function is uniformly continuous. Fix $\varepsilon = \frac{1}{n}$, U_i is contained in a ball $B(x_i, \varepsilon)$. One can check that $\mu'_n \to \mu$ as required. □

This enables us to define maps on $C(X)^*$ by extending from Meas_c^0 by continuity.

Definition 3.4. Let X and Y be nice spaces. We can define $\mu \boxtimes v \in \mathrm{Meas}_c(X \times Y)$ for $\mu \in \mathrm{Meas}_c(X), v \in \mathrm{Meas}_c(Y)$ by extending $\delta_x \boxtimes \delta_y = \delta_{x,y}$ (this map sends Cauchy to Cauchy).

For a locally compact topological group G, the space $\mathrm{Meas}_c(G)$ is an algebra. We have a continuous map $m : G \times G \to G$, the associative algebra structure is given by $\mu * v = m_*(\mu \boxtimes v)$. We have inclusions:

$$\mathrm{Meas}_c^0(G) = \mathbb{C}[G] \subset \mathrm{Meas}_c(G) \supset C_c(G)$$

Lemma 3.5. *Let V be a Fréchet topological representation of G, then $\mathrm{Meas}_c(G)$ acts continuously on V.*

Proof. First of all we need to check the $\mathbb{C}[G]$-action is continuous. For this, it's enough to check $(\mu_i, v_i) \to 0 \implies \mu_i(v_i) \to 0$. For a seminorm ν, we need to check $\nu(\mu_i(v_i)) \to 0$.

For $\mu = \sum c^j \delta_{x_j}$, we define $|\mu| = \sum |c^j| \delta_{x_j}$. Since $\mu_i \to 0$, we see μ_i is bounded in norm. Therefore we have:

$$\nu(\mu_i(v_i)) \leq \nu(|\mu_i|(v_i)) \to 0$$

□

Now let K be a compact Lie group, V a Fréchet representation. Recall $v \in V$ is K-finite if $K \cdot v$ span a finite dimensional subspace. We let V_{fin} be the space of K-finite vectors.

Proposition 3.6. V_{fin} *is dense in V.*

Proof. By Peter-Weyl, we have a dense inclusion

$$C(K) \supset C(K)_{\mathrm{fin}} = \bigoplus V_\lambda \otimes V_\lambda^*$$

But $C(K)$ is dense in $\mathrm{Meas}_c(K)$, by fixing a Haar measure on K. So can find a sequence f_i in $C(K)_{\mathrm{fin}}$ converges to δ_1. Now let $v \in V$ be arbitrary. We see $f_i \cdot v \to v$ as $i \to \infty$. One can check $f_i \cdot v$ is always K-finite. □

Corollary 3.7. *An irreducible Fréchet representation of a compact group is finite dimensional.*

Definition 3.8. Let G be a Lie group, we call $v \in V$ a *smooth vector* if $g \mapsto g(v)$ is smooth. We denote V_{sm} be the subspace of smooth vectors.

Example. *If $V = C(G)$, then $V_{\mathrm{sm}} = C^\infty(G)$. Note this holds for both left multiplication and right multiplication.*

Proposition 3.9. *Let G be a Lie group, V_{sm} is dense for any Fréchet representation.*

Proof. Let's find $f_i \in C_c^\infty(G)$ with $f_i dg \to \delta_1$. Again for $\mu = fdg$, $f \in C_c^\infty(G)$, we always have $\mu(z)$ a smooth point. □

Let G be a Lie group, $K \subset G$ maximal compact subgroup.

Proposition 3.10. *$V_{\text{fin}} \cap V_{\text{sm}}$ is also dense in V*

Proof. If μ_i is K-finite, μ_i' is smooth, with both converges to δ_1. Then $\mu_i * \mu_i' \to \delta_1$ is both K-finite and smooth. □

4 Admissible Representation

Let G be a connected Lie group, $K \subset G$ compact Lie subgroup (Soon we will assume G real reductive, K maximal compact). Let V be a topological representation of G.

Definition 4.1. We call V is *admissible* if for any irreducible representation ρ of K, $\text{Hom}_K(\rho, V)$ is finite dimensional.

Example. *Consider L_s to be the representation of $\text{SL}(2,\mathbb{R})$ induced from the character $\chi_s : t \mapsto |t|^s$ from T. Now the maximal compact subgroup is $K = \text{SO}(2)$, and K acts on $\mathbb{RP}^1$ transitively, so $K/T \simeq G/B$. Hence we know*

$$\text{Hom}_K(\rho, L_s) \simeq \text{Hom}_T(\rho, \chi_s)$$

has dimension ≤ 1, since each K-reps is determined by its character.

Last time we prove that $V_{\text{fin}} \cap V_{\text{sm}}$ is dense in V (in particular $V_{\text{fin}} \subset \overline{V_{\text{fin}} \cap V_{\text{sm}}}$). Hence

$$V_{\text{fin}} \cap V_{\text{sm}} = \bigoplus_{\rho \in \text{Irr}(K)} (V_{\text{sm}} \cap V_\rho \otimes \rho) \implies V_{\text{sm}} \cap V_\rho \otimes \rho \text{ is dense in } V_\rho \otimes \rho$$

Indeed, $V_\rho \otimes \rho$ is an isotopical K-type of V_{fin}, hence it is the image of an idempotent e_ρ in $C^\infty(K) \subset \text{Meas}_c(K) \subset \text{Meac}_c(G)$. If μ_i is smooth approximation to δ_1 in $\text{Meas}_c(G)$, then $e_\rho * \mu_i \to e_\rho$. So indeed smooth vectors are dense in $V_\rho \otimes \rho$. But if V is admissible, $V_\rho \otimes \rho$ is finite dimensional, so every vector is smooth.

Corollary 4.2. *If V is an admissible representation of a Lie group G with a fixed compact subgroup K. Then $V_{\text{fin}} \subset V_{\text{sm}}$.*

On V_{sm} we have an action of the Lie algebra $\mathfrak{g}$. The subspace V_{fin} is invariant under this action. Moreover the action map $a : \mathfrak{g} \otimes V_{\text{sm}} \to V_{\text{sm}}$ is G-equivariant:

$$a(g \cdot (x \otimes v)) = a(gxg^{-1} \otimes gv) = g(xv) = g \cdot a(x \otimes v)$$

Definition 4.3. If $\mathfrak{g}$ is a Lie algebra, K a Lie group, and a K-equivariant Lie algebra morphism $\text{Lie}(K) \to \mathfrak{g}$ (we equip $\mathfrak{g}$ with a K-action). A Harish-Chandra module is a space M with a locally finite action of K, an action of $\mathfrak{g}$ such that

$$\mathfrak{g} \times M \to M$$

is K-equivariant. Moreover, the two actions of $\text{Lie}(K)$ coincide.

Remark. *A $(\mathfrak{g}, K)$-module is nothing but a representation of $\mathfrak{g}$, and it is locally integrable to K when restricting to $\mathfrak{k}$. Note this is a purely algebraic object: this can be viewed as a $\mathfrak{g}_{\mathbb{C}}$-module, which becomes locally finite when viewing as a $\mathfrak{k}_{\mathbb{C}}$-module, and the $\mathfrak{k}_{\mathbb{C}}$-action can be upgraded to a algebraic action of $K_{\mathbb{C}}$.*

Example. *Let K be a compact connected group and $G = K_{\mathbb{C}}$, so that $K \subset G$ is a maximal compact (note we view G as a real group). Suppose I want to study admissible G-representations. This is the same as studying the $(\mathfrak{g}, K)$-module for $\mathfrak{g} = \mathfrak{k} \otimes \mathbb{C}$. Note in this case $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \oplus \mathfrak{g}$, and the natural map $\mathfrak{g} = \mathfrak{k}_{\mathbb{C}} \to \mathfrak{g} \oplus \mathfrak{g}$ is precisely the diagnonal map. So it is equivalent to study a $\mathfrak{g} \oplus \mathfrak{g}$ module, for which the diagnoal copy acts locally finitely, and can be upgraded to an algebraic representation of G.*

Example. *Let $M = U(\mathfrak{g})$. The first copy acts by $x : y \mapsto xy$, the second copy acts by $x : y \mapsto -yx$. G acts by adjoint action (since the diagonal acts by adjoint). This is not admissible, but*

$$Z(U(\mathfrak{g})) \simeq S(\mathfrak{h}^*)^W$$

we see $U(\mathfrak{g}) \otimes_Z \mathbb{C}_\chi$ for a character χ is admissible.

Definition 4.4. For a representation V, we have the matrix coefficient map:

$$V^* \otimes V \to C(G), \quad \xi \otimes v \mapsto (f_{v,\xi} : g \mapsto \xi(gv))$$

This is $G \times G$-equivariant.

Definition 4.5. We say $v \in V$ is weakly analytic if for any $\xi \in V^*$, $f_{v,\xi}$ is an analytic function on G.

Theorem 4.6 (Harish-Chandra)**.** *If V is admissible representation of G, then every $v \in V_{\text{fin}}$ is weakly analytic.*

To prove this, we need to use the theorem of elliptic regularity. Let X be a manifold, and $D(X)$ is the algebra of differential operators on X. There is a natural filtration on $D(X)$ by degree. We have:

$$\text{gr}D(X) \simeq \bigoplus_{n \geq 0} \Gamma(X, S^n TX)$$

For every differential operator D on X of degree n, we have the symbol $\sigma(D) \in \Gamma(X, S^n TX)$. For ever $x \in X$, $\sigma(D)_x$ is a homogeneous polynomial on $T_x^* X$

Definition 4.7. We say D is elliptic at x if $\sigma(D)_x(p) \neq 0$ for every $0 \neq p \in T_x^* X$. We say D is elliptic if it is elliptic at all points.

Theorem 4.8 (Elliptic Regularity)**.** *Suppose D is elliptic operator with real analytic coefficients on an open $U \subset \mathbb{R}^n$, and $f(x)$ is a smooth solution to the PDE*

$$Df = 0$$

on U. Then f is analytic.

Example. *Consider $n = 1$, $f^{(k)} + a_{k-1}f^{(k-1)} + \cdots + a_0 = 0$ has analytic solutions, if each a_i is analytic.*

Corollary 4.9. *The same holds on a real analytic manifold.*

Proof. (of Harish-Chandra analyticity) Fix a positive symmetric K-invariant bilinear pairing on $\mathfrak{g}$: $B \in \mathrm{Sym}^2(\mathfrak{g}^*)$ and denote $(x, y) = B(x, y)$. It exists by averaging trick. B^{-1} is a K-invarint element in $\mathrm{Sym}^2(\mathfrak{g}) \subset U(\mathfrak{g})$. So we obtain a K-invariant quadratic element C in $U(\mathfrak{g})$. Moreover, C defines a differential operator d_C on G. This operator is elliptic: we can identify $T^*G \simeq G \times \mathfrak{g}^*$ via right translation.

1. The symbol of d_C is $\overline{d_C}(g, \xi) = (\xi, \xi) > 0$ if $\xi \neq 0$.
2. $d_C(f_{v,\xi}) = f_{C(v),\xi}$
3. The action of C on V_{fin} commutes with the K-action, so it preserves every isotypic component $V_\rho \otimes \rho$. Fix ρ, let P be the minimal polynomial for C acting on V_ρ, then $P(d_C)$ kills $f_{v,\xi}$ for any $\xi, v \in V_\rho \otimes \rho$. Since d_C is elliptic, so is $P(d_C)$.
4. By elliptic regularity, we see $f_{v,\xi}$ is indeed analytic.

□

Let V be an admissible G-module.

Corollary 4.10. *1. The action of G on V is uniquely determined by the action of $\mathfrak{g}$ on V_{fin}.*

2. (closed) subrepresentations of V are in bijection with $(\mathfrak{g}, K)$-submodules of V_{fin}.

3. The functor $V \to V_{\mathrm{fin}}$ from G-reps to $(\mathfrak{g}, K)$-module is faithful (but not fully faithful). Moreover, it is exact and maps irreducibles to irreducibles.

Proof. 1. Since V_{fin} is dense, it suffices to specify $g(v)$ for any $v \in V_{\mathrm{fin}}$. By Hahn-Banach, $V \subset (V^*)^*$, it is enough to specify (gv, ξ). But this is real analytic on manifold G, so is determined uniquely determined by Taylor series at $g = 1$, which is the Lie algebra action.

2. $W \subset V$ is a proper subrepresentation. W_{fin} is dense in W. Obviously W_{fin} give es a proper submodule. Now check that for $M \subset V_{\mathrm{fin}}$ is a $(\mathfrak{g}, K)$-submodule, then $\overline{M}$ is a subrepresentation. It suffcies to know if $\xi \in V^*$, $\xi|_M = 0 \implies \xi(gv) = 0, v \in M, g \in G \iff f_{v,\xi} = 0$. So we only need to check its derivaties at $g = 1$ vanish. But these derivatives are of form $\xi(bw)$ for $b \in U(\mathfrak{g})$, hence it vanishes. Finally we see $\overline{M}_{\mathrm{fin}} = M$, since projection to $V_\rho \otimes \rho$ is continuous, and finite dimensional subspace is clsoed.

3. It follows from density.

□

Proposition 4.11 (Schur Lemma)**.** *If V is an admissible irreducible representation, then*

$$\mathrm{End}_G(V) = \mathbb{C}$$

Proof. By part two of the above corollary, we know V_{fin} is an irreducible $\mathfrak{g}$-module. Hence by part three of the above corollary, it suffices to show $\text{End}_{\mathfrak{g}}(V_{\text{fin}}) = \mathbb{C}$. But this just follows from the following general algeba result:

Lemma. *If M is an irreducible module over an associative $\mathbb{C}$-algebra with at most countable dimension, then Schur lemma holds.*

In fact, we know $D = \text{End}_A(M)$ is a disvision algebra. But if $\mathbb{C} \subsetneq \mathbb{C}$, then $\mathbb{C}(t) \subset D$. So D is uncountable dimensional, contradicting the countablity of dimension of M (we have an injection $\text{End}_A(M) \to M$ by choosing a generator of M). □

Remark. *For finite dimensional M, we have a simpler proof. For φ a nonscalar endomorphism, it has an eigenspace for some λ. But M_λ is a submodule.*

5 Structure Theory of Real Groups

5.1 Definition and First Examples

Definition 5.1. A *real semisimple group* G is a closed Lie subgroup in $\text{GL}(N, \mathbb{R})$, with semisimple Lie algebra $\mathfrak{g}$ (i.e. $\mathfrak{g}_{\mathbb{C}}$ is semisimple).

Remark. *For a real semisimple group G, we can define its complexification $G_{\mathbb{C}} \subset \text{GL}(N, \mathbb{C})$, and $G = (G_{\mathbb{C}} \cap \text{GL}(N, \mathbb{R}))_0$*

Definition 5.2. We call a subgroup $T \subset G$ is a *maximal torus* if $\mathfrak{t}_{\mathbb{C}}$ is a maximal torus inside $\mathfrak{g}_{\mathbb{C}}$. We call G *split* if it contains a maximal torus isomorphic to $(\mathbb{R}^\times)^n$ for some n.

Example. $\text{SL}_(n, \mathbb{R})$ *is split.* $\text{SO}(p, q)$ *is split when $p = q$ or $p = q \pm 1$.*

Definition 5.3. A *real reductive group* is one of the form $G(\mathbb{R})$, where G is a connected algebraic group defined over $\mathbb{R}$ for which $G_{\mathbb{C}}$ is reductive.

Remark. *Let K be any compact Lie group, and fix θ a $\mathbb{C}$-antilinear involution on $K_{\mathbb{C}}$. Then $G = K_{\mathbb{C}}^{\theta}$ is a real reductive group. For example, we can take $K = U(n)$, $K_{\mathbb{C}} = GL(n, \mathbb{C})$, $\theta(g) = \overline{g}$, then $G = \text{GL}(n, \mathbb{R})$ is a real reductive group.*

Over $\mathbb{C}$ a semisimple group is nothing but a reductive group with a finite center. Indeed, for any reductive G over $\mathbb{C}$, we have:

$$G' \times Z(G)^\circ \to G \to G/Z(G)^\circ \times G/G'$$

where G' is the derived subgroup and $Z(G)$ is the center. Both maps are onto with finite kernels.

Example. *Take $G = GL(n, \mathbb{C})$. Then $Z(G) = Z(G)^\circ = \mathbb{G}_m$, $G' = SL(n, \mathbb{C})$, $G/Z(G)^\circ \simeq \text{PGL}(n, \mathbb{C})$, and $G/G' = \mathbb{G}_m$. So the above maps read:*

$$\text{SL}(n, \mathbb{C}) \times \mathbb{G}_m \to \text{GL}(n, \mathbb{C}) \to \text{PGL}(n, \mathbb{C}) \times \mathbb{G}_m$$

Example. *Technically a real semisimple group need not be reductive. For example, we may consider* $\mathrm{PGL}(n,\mathbb{R})^+$ *which has semisimple Lie algebra. But it is not an real reductive group, since it is not the real point of a real algebraic group. In general, the problem is that for a real semisimple group* G, $G^\theta_{\mathbb{C}}$ *is not necessarily connected.*

Remark. *In general, given a complex algebraic group* $G_{\mathbb{C}}$ *with an anti-involution* θ*, the number of connected components of* $G^\theta_{\mathbb{C}}$ *is finite and generally small.*

Example. *Here is a more serious problem Consider* $\widetilde{\mathrm{SL}(n,\mathbb{R})}$ *with* $n \geq 2$*, this is semisimple but not reductive. Indeed, We have an extension:*

$$\mathbb{Z} \to \widetilde{\mathrm{SL}(n,\mathbb{R})} \to \mathrm{SL}(n,\mathbb{R})$$

Note it has simple Lie algebra, but it has no finite dimensional faithful representation. Indeed, suppose we're given $\widetilde{\mathrm{SL}(n,\mathbb{R})} \to \mathrm{GL}(N,\mathbb{R})$*, it induces an injective homomorphism on the Lie algebra* $\mathfrak{sl}(n,\mathbb{R}) \to \mathfrak{gl}(N,\mathbb{R})$*. We complexify and lift back, we see the central piece* $\mathbb{Z}$ *acts trivially.*

5.2 Cartan Involutions and Classification

For reference, see Etingof 18.755 lecture 15.

Definition 5.4. Let G be a real Lie group. A *Cartan involution* is an involution θ of G such that the form $B(x,y) = -(x,\theta(y))$ is positive definite, for $(x,y) = \kappa(x,y)$ the Killing form.

Remark. *Cartan involution preserves Killing form:*

$$(\theta(x),\theta(y)) = -B(\theta(x),y) = -B(y,\theta(x)) = (y,\theta^2(x)) = (y,x) = (x,y)$$

Note if $K = G^\theta$, then K acts on $\mathfrak{g}$ with finite kernal and preserve B. Hence we can $G^\theta \to \mathrm{SO}(\mathfrak{g})$, and the image is a closed subgroup of $\mathrm{SO}(\mathfrak{g})$, hence is compact. We will see this is in fact a maximal compact subgroup.

Example. *Let* $G = \mathrm{SL}(n,\mathbb{R})$*,* $\theta(g) = (g^t)^{-1}, K = \mathrm{SO}(n,\mathbb{R})$*.*

Example. *Let* $G = K_{\mathbb{C}}$*,* $G^\theta = K$*. The involution* θ *acts on* $\mathfrak{g} = \mathfrak{k}\otimes\mathbb{C}$ *by complex conjugation.*

Theorem 5.5. *Let* G *be semisimple Lie group, then it has a real algebraic Cartan involution* θ*, i.e. the form*

$$B(x,y) = -(x,\theta(y))$$

on $\mathfrak{g}$ *is positive definite. The Cartan involution is unique up to conjugacy.*

Definition 5.6. Let $G_{\mathbb{C}}$ be a complex Lie group. A *real form* of $G_{\mathbb{C}}$ is a real algebraic group G whose complexification is $G_{\mathbb{C}}$.

Remark. *For a complex Lie group* $G_{\mathbb{C}}$*, its real forms are classified by the anti-holomorphic involution on it.*

Remark. *We have*

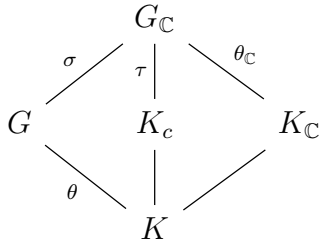
$$0 \to G_{\text{ad}} \to \text{Aut}(G) \to \text{Out}(G) = \text{Aut}(D) \to 0$$

Two forms with the same image in $\text{Aut}(D)$ *are called the* inner twists. *It is a fact that every form is an inner twist of a unique quasi-split form.*

A form is quasi-split if there exists a Borel subgroup defined over $\mathbb{R}$. *A split form is always quasi-split. A complex group viewed as real group* ($G_{\mathbb{C}} = G \times G$) *is quasi-split, not split.* $SU(p,q)$ *is quasi-split if* $p = q$ *or* $|p-q| = 1$.

Theorem 5.7. *Let* $G_{\mathbb{C}}$ *be a complex semisimple group. The real form of* $G_{\mathbb{C}}$ *are in bijection with holomorphic involutions.*

The idea is the following:



We're given a real group G. We pick a complexification $(G_{\mathbb{C}}, \sigma)$. The claim is that we can find K_c maximal compact in $G_{\mathbb{C}}$ determined by τ, and σ commutes with τ. Then $\theta_{\mathbb{C}} = \sigma\tau$ determines our intended involution: it descends to a Cartan involution when restricting to G. In other words, in the theorem we will send the real form σ to $\theta_{\mathbb{C}}$.

Example. *Let* $G_{\mathbb{C}} = SL(n,\mathbb{C})$. *Note every order 2 element defines an involution* $\theta_{\mathbb{C}}$. *In particular, if we choose* $\theta_{\mathbb{C}} = \text{Ad}_{\text{diag}(1,1,\cdots,-1,-1,\cdots)}$, *it determines* $K_{\mathbb{C}} = \text{SL}(p) \times \text{SL}(q)$. *This tells us* $K = \text{S}(U(p) \times U(q))$. *We can pick* $K_c = \text{SU}(n)$, *and one check* $\theta_{\mathbb{C}}$ *commutes with* τ. *We then know* $\sigma = \theta_{\mathbb{C}}\tau$ *determines* $G = \text{SU}(p,q)$. *One can then check the restriciton of* $\theta_{\mathbb{C}}$ *to* G *determines* K.

Similarly, if $\theta_{\mathbb{C}}$ *we choose is the outer automorphism* $g \mapsto (g^{-1})^t$, *then it determines the split form* $\text{SL}(n,\mathbb{R})$.

5.3 General Principle of Descent

The general principle of attacking these type of problems is the following. To classify objects X over a field F having a fixed shape after base change (i.e. to find all X such that $X_E \simeq X_0$ for some Galois extension E/F), we need to look at first Galois cohomology

$$H^1(\text{Gal}(E/F), \text{Aut}(X_0))$$

Example. *We start with a real group* $G = G_{\mathbb{C}}^{\theta}$, *and look for* G' *such that* $G_{\mathbb{C}} \simeq G'_{\mathbb{C}}$. *Suppose* $G' = G_{\mathbb{C}}^{\theta'}$. *We can write* $\theta' = \alpha \circ \theta$ *for* α *algebraic/holomorphic involution. So*

$$1 = (\theta')^2 = \alpha \cdot (\theta\alpha\theta) = \alpha \cdot \alpha^{\theta}$$

Namely $\alpha^\theta = \alpha^{-1}$. Therefore α_1, α_2 define the same real form if there exists an automorphism ψ of G such that

$$\alpha_1 = \psi\alpha_2(\psi^{-1})^\theta$$

This precisely corresponds to our Galois cohomology description.

Example. *Fix G real semisimple, $G = G_{\mathbb{C}}^\theta$. Fix a maximal torus T, and suppose we want to classify all maximal tori in G up to conjugacy. Indeed, Suppose T' is another maximal torus, we know there exists g such that $T'_{\mathbb{C}} = gT_{\mathbb{C}}g^{-1}$. Observe that $\theta(T'_{\mathbb{C}}) = T'_{\mathbb{C}}$, so*

$$gT_{\mathbb{C}}g^{-1} = \theta(g)\theta(T_{\mathbb{C}})\theta(g^{-1}) = \theta(g)T_{\mathbb{C}}\theta(g^{-1})$$

So we require $g^{-1}\theta(g) \in N_{G_{\mathbb{C}}}(T_{\mathbb{C}})$.

Example. *Let G be a real reductive group, and choose $g \in G$ be a regular semisimple element. Let $C = g^{G_{\mathbb{C}}}$ be a regular semisimple conjugacy class in $G_{\mathbb{C}}$, we want to know how does $C \cap G$ decompose into conjugacy classes in G (i.e. we want to find conjugacy classes in a given stable conjugacy class).*

Let $T_{\mathbb{C}} = \mathbb{Z}_{G_{\mathbb{C}}}(g)$ be a complex maximal torus. The answer is given by

$$\ker(H^1(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), T_{\mathbb{C}}) \to H^1(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), G_{\mathbb{C}}))$$

Moreover, in our case:

$$H^1(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), T_{\mathbb{C}}) = \{t \in T \mid \bar{t} = t^{-1}\}/(\bar{t}t^{-1}) \simeq \pi_0(T_{\mathbb{C}}^{\theta'})$$

for $\theta'(x) = \theta(x^{-1})$ (like for semisimple groups, $T_{\mathbb{C}}$ has unique involution θ_0 such that $T_{\mathbb{C}}^{\theta_0}$ is compact, and $\theta' = \theta_0 \circ (t \mapsto \bar{t}^{-1})$ the complex involution).

For example, if $G = K$ is compact adjoint, T also compact, $\theta' : t \mapsto t^{-1} \implies T^\sigma = (\mathbb{Z}/2)^r$ for $r = \dim T$. But the map to $H^1(\mathrm{Gal}, G)$ is injective.

5.4 Geometric Proof

We provides a geometric proof of the existence of Cartan involution.

Proof. (Existence for Complex Semisimple Groups). Start with a compact connected group K, $C(K)_{\mathrm{fin}} = \oplus V_\lambda \otimes V_\lambda^*$. Now if $G = K_{\mathbb{C}}$, then

$$\mathbb{C}[G] = C(K)_{\mathrm{fin}} = \oplus V_\lambda \otimes V_\lambda^*$$

Note V_λ has a K-invariant Hermitian form since V_λ is finite dimensional. Hence on $\mathrm{End}(V_\lambda)$ we get a $\mathbb{C}$-anti-linear involution: $M \mapsto \overline{M^t}$, namely the adjoint for the Hermitian form. Compose this with $V_\lambda \otimes V_\lambda^* \simeq V_\lambda^* \otimes V_\lambda$ coming from $\iota : g \mapsto g^{-1}$. We get an anti-complex-involution θ on $\mathbb{C}[G]$ such that

$$\theta(f)(g) = \overline{f(g)}, g \in K$$

We get an involution of G with $G^\theta = K$. □

Proof. (General Case). Let G be a real group. Let $K_c \subset G_{\mathbb{C}}$ be the maximal compact. We have an anti-involution θ on $G_{\mathbb{C}}$ such that $G_{\mathbb{C}}^{\theta} = G$. The key point is that we can choose θ_0 such that $K_c = G_{\mathbb{C}}^{\theta_0}$ and θ commutes with θ_0. Then $\sigma = \theta\theta_0$ restricts to a Cartan involution on G. First of all, for $g \in G$, we have $\theta\theta_0(g) = \theta_0\theta(g) = \theta_0(g)$. So θ preserves $\theta_0 g \implies \theta_0 g \in G \implies \theta\theta_0 g \in G$. Moreover, we check for $x \in \mathfrak{g}$, we know

$$-(x, \theta\theta_0 x) = -(x, \theta_0\theta x) = -(x, \theta_0 x) \geq 0$$

Now we need to find such a θ_0. Consider $X = G_{\mathbb{C}}/K_c$. It has an invariant metric coming from K_c-invaraint metric on $\mathfrak{g}_{\mathbb{C}}/\mathfrak{k}_c = i\mathfrak{k}_c$. The claim is that this metric nonpositive sectional curvature. (e.g. $\mathrm{SL}(2,\mathbb{C})/\mathrm{SU}(2) \simeq H^3$ hyperbolic 3-space. The result is also true for G/K, G real. For example $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2,\mathbb{R}) = \mathbb{H}$). Also X is simple connected.

But if X is simply connected with nonpositive sectional curvature, then every 2 points are connected by a unique geodesic segment (this is a CAT(0) space). But a compact group acting on a CAT(0) space has a fixed point, because any compact set has a unique circumcenter and it must be fixed. We see that every maximial compact subgroup in $G_{\mathbb{C}}$ is conjugate to K_c.

To find θ_0 commuting with θ, we note θ induces an involution on $X = \{$maximal compact subgroups in $G_{\mathbb{C}}\}$. Pick $x \in X$, pick a geodesic segment $[x, \theta(x)]$ has middle point y, then $\theta(y) = y$, i.e. θ preserves a maximal compact inside $G_{\mathbb{C}}$. By conjugating θ, we may assume θ preserves K_c. Now let θ_0 be such that $G_{\mathbb{C}}^{\theta_0} = K_c$

Then it is clear that $\theta\theta_0(g) = \theta_0\theta(g)$ for $g \in K_c$. But both sides are holomorphic involution, so they must agree on all of $G_{\mathbb{C}}$. Hence $\sigma|_G$ is a Cartan involution on G. Set $K = G^{\sigma}$, we see it's a maximal compact subgroup in G, by applying the fixed point principle to a compact subgroup in G. □

Remark. *The uniqueness can also be deduced by arguments of the same flavor.*

Remark. *(Helgason symmetric metric) We can define a canonical Riemmannian curvature tensor for the G-invariant metric on G/K: $R \in \Omega^2(\mathrm{End}(T_M))$. We write $T_{G/K}|_1 = \mathfrak{g}/\mathfrak{k}$, then:*

$$R(x, y) : z \mapsto ([x, y], z)$$

for $x, y, z \in \mathfrak{g}^{\theta,-}$, where $\mathfrak{g} = \mathfrak{g}^{\theta} \oplus \mathfrak{g}^{\theta-}$ and $\mathfrak{g}^{\theta} = \mathfrak{k}$.

5.5 Algebraic Proof

We now provide an algebraic proof for uniqueness, existence, and bijectivity.

Lemma 5.8. *Suppose G has a Cartan involution θ. Let σ be another involution, then there is $\sigma' = g\sigma g^{-1}$ for $g \in G, \theta\sigma' = \sigma'\theta$*

Proof. Denote $(-,-)$ the Killing form, $B_\theta(x, y) = -(x, \theta(y))$ positive definition. Set $s = \theta\sigma$. The observation is that s is self-adjoint for B_θ. Indeed:

$$-B_\theta(sx, y) = (\theta\sigma(x), \theta(y)) = (\sigma(x), y)$$

and

$$-B_\theta(x, sy) = (\theta(x), \theta(\sigma(y))) = (x, \sigma(y))$$

So the action of s on $\mathfrak{g}$ can be diagonalized with real eigenvalues. We write $\mathfrak{g} = \oplus \mathfrak{g}_{r_i}$. Also:

$$\theta s \theta^{-1} = s^{-1} = \sigma s \sigma^{-1}$$

Note s is an automorphism, this means $[\mathfrak{g}_{r_i}, \mathfrak{g}_{r_j}] \subset \mathfrak{g}_{r_i r_j}$.

Now set $\lambda = \log|s|$. We see that $\lambda|_{\mathfrak{g}_{r_i}} = \log|r_i|$, so it acts by derivation. But $\mathfrak{g}$ is semisimple, this shows the action λ is inner! Hence for $t \in \mathbb{R}$, we have $|s|^t = \exp(t\lambda) \in G$.

Now $\theta |s|^t \theta^{-1} = |s|^{-t}$, same for σ. We write $s = |s| \cdot \sigma'$, where σ' acts on each $\mathfrak{g}_{r_i}$ via the sign of r_i. So σ' commutes with θ, σ. Now consider:

$$|s|^{\frac{1}{2}} \sigma |s|^{-\frac{1}{2}} = |s|\sigma = s\sigma'\sigma = s\sigma\sigma' = \theta\sigma'$$

But then $\theta(\theta\sigma') = (\theta\sigma')\theta$, so $\theta\sigma'$ is the desired conjugation. □

Proof. (Algebraic Proof of the Theorem). Say $G = G_{\mathbb{C}}^{\sigma}$, and $G_{\mathbb{C}}$ has Cartan involution θ_c, which defines K_c the compact form. By lemma, we may assume $\sigma\theta_c = \theta_c\sigma$, so $\theta_c : G \to G$ and

$$(-,-)_{\mathfrak{g}} = \frac{1}{2}(-,-)_{\mathfrak{g}_{\mathbb{C}}}|_{\mathfrak{g}}$$

Hence $\theta_c|_G$ is a Cartan involution for G. This shows existence.

For uniqueness, suppose θ, θ' are two Cartan involutions, and we may assume they commute by the lemma. So either $\theta = \theta'$ or there exists $x \in \mathfrak{g}$ such that $\theta(x) = -\theta'(x)$, contradiciting the positivity of both forms.

We also show the bijectivity: $(G, \theta) \mapsto (G_{\mathbb{C}}, \theta_{\mathbb{C}})$. Given a complex involution σ of $G_{\mathbb{C}}$,we may assume σ commutes with Cartan involution θ, then $G = G_{\mathbb{C}}^{\sigma\theta}$ is the corresponding real form. This is the inverse map to $\theta \mapsto \theta_{\mathbb{C}}$. We must check this is well-defined: given two θ and θ' commuting with σ, $\sigma\theta$ and $\sigma\theta'$ must define the conjugate real form. But it is sufficient to see that θ, θ' are conjugate in $K_{\mathbb{C}} = G_{\mathbb{C}}^{\sigma}$: indeed, if $\theta = g\theta' g^{-1}$ for some $g \in K_{\mathbb{C}}$,then:

$$G^{\sigma\theta} = \{x \mid \sigma g \theta' g^{-1} x = x\} = \{x \mid g\sigma\theta' g^{-1} x = x\} = \{x \mid \sigma\theta(g^{-1}x) = g^{-1}x\} = g \cdot G^{\sigma\theta'}$$

But note both θ and θ' descends to a compact real form on $K_{\mathbb{C}}$, and all such forms are conjugate in $K_{\mathbb{C}}$. Moreover, we have freedom to conjugate by an element in $K_{\mathbb{C}}$, so without loss we may assume:$\theta|_{G^\sigma} = \theta'|_{G^\sigma}$. Hence it remains to prove the following:

Claim: θ is uniquely determined by $\theta|_{G^\sigma}$

Proof of Claim: WLOG, we may assume $\mathfrak{g}_{\mathbb{C}}$ is either almost simple, or is a sum of two simple algebras permuted by σ. Then we know $\mathfrak{g}^{\sigma,-}$ is a simple $\mathfrak{k}_{\mathbb{C}}$-module. Indeed, in the second case, $\mathfrak{g}^{\sigma,-} \simeq \mathfrak{g}^{\sigma}$ as $\mathfrak{g}^{\sigma}$-module; in the first case: if $V \subset \mathfrak{g}^{\sigma,-}$ is a $\mathfrak{g}^{\sigma}$-submodule, then $V \oplus \mathfrak{g}^{\sigma}$ is an ideal in $\mathfrak{g}$. Contradiction.

Now $\mathfrak{g} = \mathfrak{g}^{\sigma} \oplus \mathfrak{g}^{\sigma,-}$ is an orthgonal decomposition for B_θ, $B_\theta|_{\mathfrak{g}^{\sigma,-}}$ is a $G^\sigma = G^{\sigma'}$-invariant pairing. By Schur lemma, it's unique up to a scalar. By positivity, it's unique up to a real positive scalar r. So $\theta|_{\mathfrak{g}^{\sigma,-}} = r\theta'|_{\mathfrak{g}^{\sigma,-}}$. Hence $r^2 = 1 \implies r = 1$. □

6 Harish Chandra Admissibility Theorem

Theorem 6.1 (Harish-Chandra Admissibility Theorem)**.** *If G is a real semisimple group, K maximal compact. If V is an unitary representation, then for any $\rho \in \mathrm{Irr}(K)$*

$$\dim(V_\rho) \leq \dim(\rho)$$

In particular V is admissible.

The rest part of this section are devoted to prove this theorem.

6.1 Step 1: Schur Lemma and Density Theorem

Recall we have the Schur Lemma:

Theorem 6.2. *Let A be an associative algebra, V irreducible representation over $\mathbb{C}$. If $\dim_{\mathbb{C}} V < \infty$, then $A \to \mathrm{End}_{\mathbb{C}}(V)$ is onto.*

It has a natural generalization for infinite dimensional case:

Proposition 6.3. *Let A be an associative algebra, V irreducible representation over $\mathbb{C}$ not necessarily finite dimensional. Assume $\mathrm{End}_A V = \mathbb{C}$. Then given $v_1, \cdots, v_n \in V$ linearly independent, for any $w_1, \cdots, w_n \in V$ there exists $a \in A$, $a(v_i) = w_i$.*

Proof. Consider $\underline{v} = (v_1, \cdots, v_n) \in V^{\oplus n}$, need to check $A\underline{v} = V^{\oplus n}$. We write $V^{\oplus n} = \mathbb{C}^n \otimes V$. We know every submodule is of form $U \otimes V$ for some vector subspace $U \subset \mathbb{C}^n$. Now if $U \subsetneq \mathbb{C}^n$, we can find $0 \neq (a_1, \cdots, a_n) \in U^{\perp} \subset (\mathbb{C}^n)^*$, then $\sum a_i v_i = 0$ giving a contradiction. □

Here are the versions of Schur Lemma and Density Theorem for Hilbert space representations.

Theorem 6.4 (Schur Lemma)**.** *Let V be an irreducible Hilbert space representation of a group, and let e be a bounded self-adjoint endomorphism, then $e \in \mathbb{C}$.*

Proof. (Sketch). Use spectral theorem to present $V = L^2(X, \mu)$, then e becomse multiplication by a measurable function $\varphi : X \to \mathbb{R}$. We get an operator $f(\varphi)$ for $f \in C_c^{\infty}(\mathbb{R})$. If $e \notin \mathbb{C}$, we wish to construct $f(\varphi)$ such that $\ker(f(\varphi))$ is a proper closed subspace invariant under G. □

Theorem 6.5 (Density Theorem)**.** *If V is an irreducible Hilbert space representation of G, then for any linear independent $v_1, \cdots, v_n \in V$, and any $w_1, \cdots, w_n \in V$ and $\varepsilon > 0$, there exists $\varphi \in \mathrm{Meas}_c(G)$ such that*

$$\|\varphi(v_i) - w_i\| < \varepsilon$$

Proof. Again write $\underline{v} \in V^{\oplus n}$. It suffices to check

$$\overline{\mathrm{Meas}_c(X) \cdot \underline{v}} = V^{\oplus n}$$

Again all closed invaraint closed subspaces are $U \otimes V$ for $U \subset \mathbb{C}^n$ a vector subspaces. If $W \subset V^{\oplus n}$ is such a subspace, we get an orthogonal projection $\Pi_W : V^{\oplus n} \to W$, which is bounded and self adjoint endomorphism of $V^{\oplus n}$. But from Schur lemma, it follows that such endomorphism comes from a matrix in $\mathrm{Mat}_n(\mathbb{C})$, so is in fact onto. □

6.2 Step 2: Algebraic Finiteness

Proposition 6.6. *1. $K_{\mathbb{C}}$ has an open dense orbit on $G_{\mathbb{C}}/B_{\mathbb{C}}$*

2. $K_{\mathbb{C}}$ has finitely many orbits on $G_{\mathbb{C}}/B_{\mathbb{C}}$

Proof. We write $G = G_{\mathbb{C}}$ and $K = K_{\mathbb{C}}$. Let θ be the Cartan involution determining K, namely $K = G^{\theta}$. Let $X = G/B$. First of all, orbits are locally closed subvarieties. Hence it's enough to check there are finitely many of them. For this, we wll deduce from the finiteness of $B\backslash G/B = (X^2)/G \simeq W$. We write X^2_w as the orbit indexed by w.

Note θ acts on G, hence on $X = G/B$ since it maps a Borel subgroup to another Borel subgroup. Let $Z \subset X^2$ be the graph of θ. So if $\widetilde{\theta}(x, y) = (\theta(y), \theta(x))$, then $Z = (X^2)^{\widetilde{\theta}}$. Note $Z \simeq X$ as K-varieties, so the result follows from the following:

Lemma. *$Z \cap X^2_w$ is a finite union of K orbits.*

Proof. (of Lemma). It is enough to see if $(x, y) \in Z \cap X^2_w$ (so $y = \theta(x)$), then

$$\mathrm{Lie}(K) = \mathfrak{k} \twoheadrightarrow T_{(x,y)}Z \cap T_{(x,y)}X^2_w$$

is surjective. (This is a general phenomenon, suppose a group G acting on a space X, $\mathfrak{g}$ surjects onto T_xX for some point x, then $G \cdot x$ is open in X. But in Zariski topology, open orbits must be disjoint. The finiteness follows since X has finitely many connected components.) For $x \in X$, we set $\mathfrak{b}_x = \mathrm{Ad}_x(\mathfrak{b})$. So $T_xX = \mathfrak{g}/\mathfrak{b}_x, T_{(x,y)}X^2 = \mathfrak{g}/\mathfrak{b}_x \oplus \mathfrak{g}/\mathfrak{b}_y$. Hence

$$T_{(x,y)}Z = (\mathfrak{g}/\mathfrak{b}_x \oplus \mathfrak{g}/\mathfrak{b}_y)^{\widetilde{\theta}}, \quad T_{(x,y)}X^2_w = \mathrm{im}\ (\mathfrak{g} \to \mathfrak{g}/\mathfrak{b}_x \oplus \mathfrak{g}/\mathfrak{b}_y)$$

Now

$$\mathfrak{k} = \mathfrak{g}^{\theta} \to \mathrm{im}\ (\mathfrak{g} \to \mathfrak{g}/\mathfrak{b}_x \oplus \mathfrak{g}/\mathfrak{b}_y)^{\widetilde{\theta}}$$

is clearly surjective. □

□

Remark. *The set of orbits is disjoint union of subsets of W/W', indexed by conjugacy classes of tori in G.*

Example. *Let $G = \mathrm{SL}(3, \mathbb{R}), K = \mathrm{SO}(3, \mathbb{R})$. Note that:*

$$G_{\mathbb{C}}/B_{\mathbb{C}} \simeq \{(x, L) \mid x \in \mathbb{P}^2, L \in (\mathbb{P}^2)^*, x \in L\}$$

Consider the moduli space:

$$\mathcal{M} = \{(x, L, Q) \mid (x, L) \in G/B, Q \text{ conic in } \mathbb{P}^2\}$$

This has a natural left G-action. Now consider the natural surjective map:

$$G/B \to G \setminus \mathcal{M}$$

$$(x, L) \mapsto (x, L, Q_0)$$

for standard conic $Q_0 : x_0^2 + x_1^2 + x_2^2 = 0$, which naturally descends to

$$K \setminus G/B \to G \setminus \mathcal{M}$$

which is a bijection. The orbits are indexed by relative positions between a quadric with a "point on a line".

Proposition 6.7. *Suppose V is a finite dimensional irreducible representation of G. Then for every $\rho \in \mathrm{Irr}(K)$, we have*

$$\dim(V_\rho) \leq \dim \rho$$

Proof. Note a finite dimensional representation of G is algebraic, namely an algebraic representation of $G_{\mathbb{C}}$. In this dictionary, K-isotypic components are precisely $K_{\mathbb{C}}$-isotypic component, where $K_{\mathbb{C}} = G_{\mathbb{C}}^{\theta_{\mathbb{C}}}$ and $\theta_{\mathbb{C}}$ is complexification of Cartan involution.

The problem reduces to the case that V is a irreducible representation of complex reductive group $G_{\mathbb{C}}$, $K_{\mathbb{C}} = G_{\mathbb{C}}^{\theta_{\mathbb{C}}}$. Now by Borel-Weil $V = \Gamma(G/B, \mathcal{L}_\lambda)$ for some dominant weight λ. Pick $x \in G/B$ in the open orbit of $K_{\mathbb{C}}$, we have a map

$$K_{\mathbb{C}} \to G/B, \quad g \mapsto g(x)$$

Consider pullback under this map, this induces an injective map

$$\pi^* : \Gamma(\mathcal{L}_\lambda) \hookrightarrow \Gamma(\pi^* \mathcal{L}_\lambda) \simeq \mathbb{C}[K_{\mathbb{C}}]$$

But $\dim_K(V_\rho, \mathbb{C}[K_{\mathbb{C}}]) = \dim(\rho)$. □

6.3 Step 3: Reduction to the Finite Case

Lemma 6.8. *For any $\mu \in \mathrm{Meas}_c(G)$, there exists a finite dimensional irreducible reps V of G such that the action of μ on V is nonzero.*

Proof. Assume not, then we can find μ acts as zero on every finite dimensional irreducible reps. We may replace μ by $\mu * (f dg)$ for some $f \in C_c^0(G)$ which is smooth and acts as zero. So we may assume $\mu = f dg$ for $f \in C_c^0(G)$. Let $\mathbb{C}[G_{\mathbb{C}}]$ be the space of polynomial functions on $G_{\mathbb{C}}$, and it is a sum of finite dimensional irreducible representations of G (locally finiteness of algebraic representations).

By Stone Weierstrass, there exists a polynomial function $P \in \mathbb{C}[G_{\mathbb{C}}]$ such that $|P(g) - f(g)| < \varepsilon$ for all $g \in \mathrm{supp}(f)$. We take the finite dimensional representation generated by P. But then we see:

$$(f \cdot P)(h) = \int_G f(g) P(hg) dg$$

Hence $(f \cdot P)(1) > 0$, contradiction. □

We also need the following elementary linear algebra result:

Theorem 6.9 (Amitsur-Levitzki-Kolchin)**.** *For any $A_1, \cdots, A_{2n} \in \mathrm{Mat}_n(\mathbb{C})$ (or other coefficients), then*

$$\sum_{\sigma \in S_{2n}} (-1)^{|\sigma|} A_{\sigma(1)} \cdots A_{\sigma(2n)} = 0$$

Remark. *This identity fails for $\mathrm{Mat}_m(\mathbb{C})$ with $m > n$.*

Proof. (of HC Admissibility).Let V be an irreducible unitary representation of G. We check that $\dim(V_\rho) \leq \dim \rho = d_\rho \iff V_\rho \otimes \rho \subset V_{\text{fin}}$ has dimension $\leq d_\rho^2$.

Consider $\text{Meas}_c(G)^\rho = e_\rho * \text{Meas}_c(G) * e_\rho \subset \text{Meas}_c(G)$ subring, for $e_\rho = \chi(\rho)dg \in \text{Meas}_c(K)$ such that it acts by 1 in ρ and 0 in other representations.

By Stone-Weierstrass, we know for each $\mu \neq 0$, ther exists a finite dimensional irreducible representaiton of G such that μ acts nonzeroly on W. But if $\mu \in \text{Meas}_c(G)^\rho$, μ acts on ρ isotypic components, so it acts nontrivially on $W_\rho \otimes \rho$ the ρ isotypic component of W. Hence $\text{Meas}_c(G)^\rho$ acts faithfully on $\prod_{W \in \text{FIrr(G)}} W_\rho \otimes \rho$, for each component has dim $\leq d_\rho^2$.

Now suppose $\dim(V_\rho) > d_\rho$, then we can pick $v_1, \cdots, v_{d_\rho^2+1}$ linearly independent in $V_\rho \otimes \rho$. Let's find $M_i \in \text{Mat}_{d_\rho^2+1}$ such that

$$\sum (-1)^{|\sigma|} M_{\sigma(1)} \cdots M_{\sigma(2d_\rho^2)} \neq 0$$

By density theorem, can find $\mu_1, \cdots, \mu_{2d_\rho^2} \in \text{Meas}_c(G)$ to arbitrarily approximate M_i. By replacing μ_i by $e_\rho \mu_i e_\rho = \mu_i'$, we get that

$$\sum (-1)^{|\sigma|} \mu'_{\sigma(1)} \cdots \mu'_{\sigma(2d_\rho^2)} \neq 0$$

But this acts trivially on each $W_\rho \otimes \rho$ by AL, which forces this expression to be zero. Contradiction. □

Definition 6.10. Admissible topological representations V, W of G are called *infinitesimally equivalent* if $V_{\text{fin}} \simeq W_{\text{fin}}$ as $(\mathfrak{g}, K)$-modules.

Proposition 6.11. *If V, W are unitary Hilbert space representations(automatically admissible), V, W are infinitesimally equivalent iff they're equivalent.*

Proof. One direction is trivial. The converse direction follows from the fact that $V = \hat{V}_{\text{fin}}$ completion for topological vector space given by invariant norm. □

Remark. *This is not true in general. For example for* $\text{SL}(2, \mathbb{R})$*, we can use L^p with $p > 1$, C^k, or C^∞. They are all infinitesimally equivalent.*

Remark. *Given an irreducible $(\mathfrak{g}, K)$-module M, can it be realized as $M = V_{\text{fin}}$ for a canonically defined V? The answer is yes, this is called Casselman-Wallach globalization theorem.*

7 Harish Chandra Globalization Theorem

Let G be a real semisimple group. Fix a maximal compact K The goal is to prove the following:

Theorem 7.1. *If M is an irreducible unitary $(\mathfrak{g}, K)$-module, then $M = V_{\text{fin}}$ for a unique unitary (Hilbert space) representation of G. (We already know $V = \overline{M}$)*

Proof. Let $(-,-)$ be the invariant product on M. Write $M = \bigoplus M_\nu = \bigoplus V_\nu \otimes W_\nu$ the isotypic decomposition, for ν weight of $K_{\mathbb{C}}^\circ$.

Lemma. *For $x \in \mathfrak{g}$, $v \in M_\nu$, then:*

$$|x(v)|^2 \leq |v|^2 \cdot q(\nu) \cdot |x|^2$$

for q quadratic in ν, $|x|$ the norm for x in B_θ.

Proof. (For lemma) $|x(v)|^2 = (x(v), x(v)) = -(x^2 v, v)$. We will estimate this by estimating $\sum(x_i^2 v, v)$ for x_i orthonormal basis in $\mathfrak{g}$ for the form B_θ coming from the Cartan involution θ. Then we have $\sum x_i^2 = C_\mathfrak{g} - 2C_\mathfrak{k}^\mathfrak{g}$ for $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{k}^\perp$ and C stands for the Casimir. We know $C_\mathfrak{g}$ acts by a scalar on M by Schur lemma.
Claim: For a nondegenerate invariant form on a reductive Lie algebra $\mathfrak{k}$, the action of C on V_ν is $(\nu + \rho)^2 - \rho^2$ up to a scalar, for $\rho = \frac{1}{2}\sum_{\alpha>0} \alpha$ This implies the lemma. □

Now let $V = \overline{M}$ be the completion for norm topology. Now check that $x \in \mathfrak{g}$, $|x| < r$ for some r (independent of v), then

$$\sum \frac{x^n v}{n!}$$

converges in V for $v \in V$. It suffices to consider $v \in M_\nu$. Then it follows from the lemma and the observation: there exists a finite subset $\nu_1, \cdots, \nu_n$ such that:

$$x : M_\nu \to \bigoplus M_{\nu+\nu_i}$$

just by considering $\mathfrak{g} \bigoplus M_\nu$ as a K-module. In fact, for ν_i, can look at weights of K in $\mathfrak{g}$.

Remark. *We can understand this by Borel-Weil. Let V, V_λ be reps of a reductive group H, with V_λ simple. We write $V_\lambda = \Gamma(H/B, \mathcal{L}_\lambda)$, then $V \otimes V_\lambda = \Gamma(H/B, \mathcal{L}_\lambda \otimes V)$, where we view V as an equivariant bundle with fibre V. Now suppose $V|_B$ has a filtration with* $\mathrm{gr} = \bigoplus \mathbb{C}_\mu$, *it follows that*

$$\mathrm{gr}(\mathcal{L}_\lambda \otimes V) = \bigoplus \mathcal{L}_{\lambda+\mu}$$

Now if $v \in V_\mu$, $|x^n v| \leq \sum_{i_1,\cdots,i_d} |x|^n |v| q(\nu) q(\nu + \nu_{i_1}) \cdots q(\nu + \nu_{i_d}) \leq C(C+R)(C+2R)\cdots(C+(n-1)R)|x|^n \cdot |v|$, for $R = \max|\nu_i|, C = |\nu|$. So the above series converges for $|x| < R^{-1}$. The we defines an operator $\rho(g)$ on $\overline{M}$ for g in a neighborhood of 1. Then it follows from CH formula we see $\rho(g_1 g_2) = \rho(g_1)\rho(g_2)$ for g_1, g_2 in a neighborhood of 1, this gives an action of $\widetilde{G}$ (the universal cover) on V. Finally note that $\pi_1(G) = \pi_1(K^\circ)$, and V is a K-reps, this implies $\pi_1(G) = \pi_1(K^\circ)$ acts trivially, hence we get an action of G. □

Remark. *One sees $\pi_1(G) = \pi_1(K^\circ)$ via the following: in fact, we have polar decompsition: $G = K^\circ \cdot P$, and $P \simeq \mathbb{R}^N$.*

Corollary 7.2. *Any unitary $(\mathfrak{g}, K)$-module M is admissible.*

Remark. *This is true for any irreducible $(\mathfrak{g}, K)$-module and can be proved algebraically. The key tool is the following: we know $U(\mathfrak{g})$ acts on M, and in particular $Z(U(\mathfrak{g}))$ acts. So the quadratic Casimir element acts. Using Schur lemma, we see Z acts via $\chi : Z \to \mathbb{C}$. We call this χ an infinitesimal character.*

8 Some Lie Algebra Theory

8.1 Verma Module

Definition 8.1. Fix $\mathfrak{b} \subset \mathfrak{g}$, we write $\mathfrak{g} = \mathfrak{u}^- \oplus \mathfrak{t} \oplus \mathfrak{u}$, where $\mathfrak{b} = \mathfrak{u} \oplus \mathfrak{t}$. A Verma module:

$$M_\lambda = \mathrm{Ind}_{\mathfrak{b}}^{\mathfrak{g}} \mathbb{C}_\lambda = U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbb{C}_\lambda$$

Then M_λ is generated by a single element v_λ, and in our convention $U(\mathfrak{u})$ acts trivially. We get a basis from PBW theorem.

Remark.

$$\mathrm{Hom}_{\mathfrak{g}}(M_\lambda, M) \simeq \{v \in M \mid e_\alpha(v) = 0, hv = \langle \lambda, h \rangle v\}$$

These are called the singular vectors *of M.*

Lemma 8.2. *Let $\alpha^\vee$ be a simple coroot, and let $m = \langle \alpha^\vee, \lambda \rangle$. Then $f_\alpha^{m+1}(v_\lambda)$ is singular vector in M_λ of weight $\lambda - (m+1)\alpha$.*

Proof. The key observation is that $\lambda - (m+1)\alpha = s_\alpha(\lambda + \rho) - \rho$, the shifted action □

8.2 Center of Enveloping Algebra

Let $\mathfrak{g}$ be semisimple Lie algebra over $\mathbb{C}$, with Cartan $\mathfrak{t}$. For $z \in Z(U(\mathfrak{g}))$, then z acts on M_λ by a scalar $C_\lambda(z)$, since $\mathrm{End}(M_\lambda) = \mathbb{C}$.

Theorem 8.3 (Harish-Chandra Isomorphism)**.** *For $z \in Z(U(\mathfrak{g}))$, then $C_\lambda(z)$ depends on λ as a polynomial. Say $C_\lambda(z) = P_z(\lambda)$. Then:*

1. *$P_z(\lambda) = P_z(w(\lambda + \rho) - \rho)$ for $w \in W$*
2. *$z \mapsto P_z$ is an isomorphism $Z(U(\mathfrak{g})) \simeq \mathbb{C}[\mathfrak{t}^*]^{(W,\cdot)}$, where we take W to have the twist action on $\mathfrak{t}$.*

Example. *For $\mathfrak{g} = \mathfrak{sl}_2$, $P(n) = P(-n-2)$.*

Theorem 8.4 (Chevalley Restriction Theorem)**.** $\mathbb{C}[\mathfrak{g}]^G \simeq \mathbb{C}[\mathfrak{t}]^W$

Example. *Let $\mathfrak{g} = \mathfrak{gl}_n$, then $\mathbb{C}[\mathfrak{g}]^G$ is generated by $\mathrm{Tr}(x), \mathrm{Tr}(\wedge^2 x), \cdots, \det$. These are coefficient of characteristic polynomials of the matrix. These are elementary symmetric polynomials of eigenvalues.*

Remark. *We take the following convention: we denote $\mathbb{C}[V] = \mathrm{Sym}(V^*)$ to be algebraic functions on a vector space V.*

Proof. (of Chevalley Restriciton). Obviously the restriction $\mathbb{C}[\mathfrak{g}]^G \to \mathbb{C}[\mathfrak{t}]$ lands in $\mathbb{C}[t]^W$. The restriction map is injective: if P is an invariant polynomial, and $P|_{\mathfrak{t}} = 0$, then $P(x) = 0$ for every semisimple element in $\mathfrak{g}$. But then semisimple elements are dense, this implies $P = 0$.

To check surjectivity, $\mathbb{C}[\mathfrak{t}]$ is spanned by λ^n, for λ running through all dominant weights. So W averaging of λ^n spans $\mathbb{C}[\mathfrak{t}]^W$. We need to check these lie in the image of $\mathbb{C}[\mathfrak{g}]^G$. Take $P_{\lambda,n} \in \mathbb{C}[\mathfrak{g}]^G : x \mapsto \mathrm{Tr}(x^n, V_\lambda)$. Then $P_{\lambda,n}$ restricst to an element of form $\mathrm{Av}_W(\lambda^n) + \sum_{\mu<\lambda} a_\mu \mathrm{Av}_W(\mu) \in \mathbb{C}[\mathfrak{t}]^W$, we're done. □

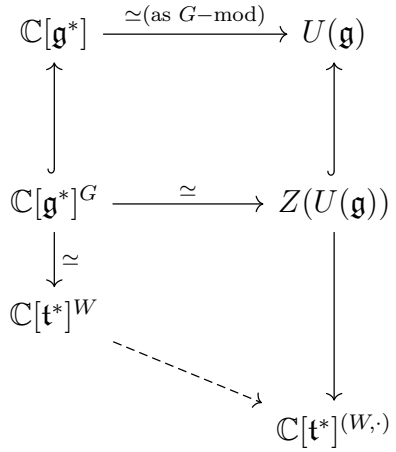
Proof. (of HC Isomorphism). We first check $P_z(\lambda)$ is a polynomial. Note $z \in U(\mathfrak{g})[0]$ the 0 weight space. Use PBW theorem, we fix a basis $f_\alpha h_I e_\beta$ for α, β positive roots. Write z as a linear combination of PBW monomials:

$$z = P(h) + (\text{terms containing some } e_\beta)$$

but the later term kill $v_[\lambda]$. So $P_z(\lambda) = P(h)(\lambda)$ for $P(h) \in \mathbb{C}[\mathfrak{t}^*]$.

We now check $P_z(\lambda)$ is invariant under $\cdot$ action. Indeed, from exercise we see for λ dominant, we have a nonzero map $M_{w\cdot\lambda} \to M_\lambda$. Therefore we see $P_z(\lambda) = P_z(w \cdot \lambda)$.

To check it's an isomorphism, we use Chevalley restriction + PBW.



One only need to check that the dashed bottom arrow is an isomorphism. In fact, it sends a homogeneous degree d polynomial P to ($P+$ smaller degree terms). □

We'll see that $Z(U(\mathfrak{g}))$ is a free algebra.

Theorem 8.5. *(Chevalley-Sheppard-Todd Theorem) If W acts on V a vector space, and W a finite group is generated by complex reflections. Then:*

1. *$\mathbb{C}[V]$ is a free module over $\mathbb{C}[V]^W$*
2. *$\mathbb{C}[V]^W$ is a free commutative algebra.*

Remark. *The converse is also true. If W acts on V and $\mathbb{C}[V]^W$ is a polynomial algebra, then W is generated by complex reflections.*

Corollary 8.6. $\mathrm{Hom}_G(V_\lambda, \mathbb{C}[\mathfrak{g}])$ *finitely generated over* $\mathbb{C}[\mathfrak{g}]^G$.

Proof. We claim that for every representation V, the restriction map $(V \otimes \mathbb{C}[\mathfrak{g}])^G \to V \otimes \mathbb{C}[\mathfrak{t}]$ is injective. Indeed suppose I have $u = v_1 \otimes f_1 + \cdots + v_n \otimes f_n \in (V \otimes \mathbb{C}[\mathfrak{g}])^G$ is sent to 0. Then $f_i|_\mathfrak{t} = 0$ for all i. Now if $x \in \mathfrak{g}$ is semisimple, then we can find $g \in G$ such that $g^{-1}xg \in \mathfrak{t}$. But then:

$$u(x) = (gu)(x) = \sum gv_i f_i(g^{-1}xg) = 0$$

by assumption. This implies $f_1(x) = \cdots = f_n(x) = 0$. So f_i vanishes on all semisimple elements, hence $f_i = 0 \implies u = 0$.

In particular, we consider the following embedding

$$(V_\lambda^* \otimes \mathbb{C}[\mathfrak{g}])^G \to V_\lambda^* \otimes \mathbb{C}[\mathfrak{t}]$$

But the target is finitely generated over $\mathbb{C}[\mathfrak{t}]$, hence over $\mathbb{C}[\mathfrak{t}]^W$. □

8.3 Admissibility of Irreducible $(\mathfrak{g}, K)$-modules

Recall the following setup. Let G be a simply connected complex semisimple Lie group, and θ be a holomorphic involution on G. Let $K = G^\theta$, this is automatically connected. Let $\mathfrak{k}$ be the Lie algebra of K, and $\mathfrak{p} = \mathfrak{k}^\perp$. Note θ becomes a linear involution on $\mathfrak{g}$ and $\mathfrak{k}$ and $\mathfrak{p}$ are just 1 and -1 -eigenspaces. Note that K acts on $\mathfrak{p}$.

Example. *We let $G = H \times H$ and θ be the involution exchanging the two copies of H. Then indeed $K = G^\theta = H \subset H \times H$ the diagonal copy. Hence we have:*

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{h} = \{(x, x) \mid x \in \mathfrak{h}\} \oplus \{(x, -x) \mid x \in \mathfrak{h}\}$$

Proposition 8.7. *Let $x \in \mathfrak{p}$ be generic (semisimple element with minimal dimension of centralizer). Then:*

1. *$Z_\mathfrak{g}(x) \cap \mathfrak{p} = \mathfrak{a}$ is abelian semisimple. So $\mathfrak{a} \subset \mathfrak{t}$ for some Cartan $\mathfrak{t}$.*
2. *$K \cdot \mathfrak{a}$ contains a (Zariski) dense open in $\mathfrak{p}$.*

Proof. If $\mathfrak{p} = 0$ we have nothing to prove. So we may assume $\mathfrak{p} \neq 0$. We first observe that $\mathfrak{p}$ has some semisimple element. Indeed suppose not, then by Jordan decomposition $\mathfrak{p}$ is nilpotent, hence Killing form vanishes on $\mathfrak{p}$. But this contradicts the fact that $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ is an orthogonal decomposition. Hence if we pick $x \in \mathfrak{p}$ generic, it's nonzero.

Set $\mathfrak{h} = \mathfrak{g}_x = Z_\mathfrak{g}(x)$. We know $\mathfrak{h}$ is reductive (centralizer of any semisimple element is reductive), and we can write $\mathfrak{h} = \mathfrak{h}' \oplus \mathfrak{h}_0$ with $\mathfrak{h}'$ semisimple and $\mathfrak{h}_0$ abelian. Now $\theta(x) = -x$, and θ preserves $\mathfrak{h}$ and $\mathfrak{h}'$. Indeed if $y \in \mathfrak{h}$, then $[\theta(y), x] = -[\theta(y), \theta(x)] = -\theta([y, x]) = 0$. Moreover $\mathfrak{h}' = [\mathfrak{h}, \mathfrak{h}]$.

We now claim that $\theta|_{\mathfrak{h}'} = \mathrm{id}$. If so, then $\mathfrak{a} = \mathfrak{h}_0 \cap \mathfrak{p} \subset \mathfrak{t}$ consisting of semisimple elements. Indeed, one sees that $\mathfrak{h}_0$ consists of semisimple elements. Indeed, since $\mathfrak{h}$ is reductive, all its semisimple elements are in fact semisimple in $\mathfrak{g}$, in particular every element in $\mathfrak{h}_0$.

Now let's prove the claim. If not, can find y semisimple in $\mathfrak{h}' \cap \mathfrak{p}$. Then $x + ty \in \mathfrak{p}$ semsimple for all but finitely many t, and in this case $\mathfrak{g}_{x+ty} = \mathfrak{g}_x \cap \mathfrak{g}_y = Z_\mathfrak{h}(y) \subsetneq \mathfrak{h}$ has smaller dimension. Contradiction.

The second part is a calculation on the tangent space. It suffices to see that the action map $a : K \times \mathfrak{a} \to \mathfrak{p}$ has surjective differential at some point. For example we look at $(1, x) \in K \times \mathfrak{a}$. Then $da_{1,x} : \mathfrak{k} \oplus \mathfrak{a} \to \mathfrak{p}, (y, b) \mapsto [y, x] + b$. Note that x is semisimple, so $\mathfrak{g} = \mathfrak{g}_x \oplus \mathrm{ad}_x(\mathfrak{g}) \implies \mathfrak{p} = \mathfrak{a} \oplus \mathrm{ad}_x(\mathfrak{k})$ by looking at -1-eigenspace of θ. This checks the surjectivity. □

Note we have a restriction map $\mathbb{C}[\mathfrak{g}]^G \to \mathbb{C}[\mathfrak{p}]^K$, hence we can view $\mathbb{C}[\mathfrak{p}]$ as a $\mathbb{C}[\mathfrak{g}]^G$-module. Hence for every K-reps V, $\mathrm{Hom}_K(V, \mathbb{C}[\mathfrak{p}])$ is also a $\mathbb{C}[\mathfrak{g}]^G$-module. In fact, we have:

Corollary 8.8. $\mathrm{Hom}_K(V, \mathbb{C}[\mathfrak{p}])$ *is finitely generated over* $\mathbb{C}[\mathfrak{g}]^G$

Proof. $\mathrm{Hom}_K(V, \mathbb{C}[\mathfrak{p}])$ are precisely K-invariant polynomial maps $\mathfrak{p} \to V^*$. By the above Proposition, such a map is determined when restricting to $\mathfrak{a}$. So:

$$\mathrm{Hom}_K(V, \mathbb{C}[\mathfrak{p}]) \hookrightarrow \mathbb{C}[\mathfrak{a}] \otimes V^*$$

$\mathfrak{a} \subset \mathfrak{t}$. Now $\mathbb{C}[\mathfrak{a}]$ is finitely generated over $\mathbb{C}[\mathfrak{t}]^W = \mathbb{C}[\mathfrak{g}]^G$. □

Corollary 8.9. *Every irreducible* $(\mathfrak{g}, K)$*-module is admissible.*

Proof. Let M be an irreducible $(\mathfrak{g}, K)$-module (we assume K connected, and M irreducible as a $\mathfrak{g}$-module). Pick $v \in V_\nu \subset M_\nu$, where V_ν is an irreducible K-submodule and M_μ is the K-isotypic component. Note we have a surjection: $U(\mathfrak{g}) \otimes v \twoheadrightarrow M$, this gives us an surjection:

$$\mathrm{Ind}_{\mathfrak{k}}^{\mathfrak{g}}(V_\nu) \twoheadrightarrow M$$

By Schur lemma $Z = Z(U(\mathfrak{g}))$ acts via quotient by maximal ideal, so we have a surjection:

$$\mathbb{C}_\chi \otimes_Z \mathrm{Ind}_{\mathfrak{k}}^{\mathfrak{g}}(V_\nu) \to M$$

Now $\mathrm{Ind}_{\mathfrak{k}}^{\mathfrak{g}} V_\nu$ has a filtration with $\mathrm{gr} = \mathbb{C}[\mathfrak{p}] \otimes V_\nu$. Action of $\mathrm{gr}Z$ is identified with action of $\mathbb{C}[\mathfrak{g}]^G$ on $\mathbb{C}[\mathfrak{p}] \otimes V_\nu$. So we have a surjection:

$$\mathbb{C} \otimes_{\mathbb{C}[\mathfrak{g}]^G} (\mathbb{C}[\mathfrak{p}] \otimes V_\nu) \to \mathrm{gr}(\mathbb{C}_\chi \otimes_Z \mathrm{Ind}_{\mathfrak{k}}^{\mathfrak{g}}(V_\nu))$$

Hence $\mathrm{Hom}_K(V_\mu, \mathbb{C} \otimes_{\mathbb{C}[\mathfrak{g}]^G} (\mathbb{C}[\mathfrak{p}] \otimes V_\nu)) = \mathrm{Hom}_K(V_\mu \otimes V_\nu^*, \mathbb{C} \otimes_{\mathbb{C}[\mathfrak{g}]^G} \mathbb{C}[\mathfrak{p}])$ is finite dimensional by the first part. □

Remark. *Here we use* $\mathrm{Coh}^K(\mathfrak{p}^*)$ *to study* $(\mathfrak{g}, K)$*-module; we use* $\mathrm{Coh}^G(\mathfrak{g}^*)$ *to study Harish-Chandra bimodules:* $(\mathfrak{g} \oplus \mathfrak{g}, G)$*-modules.*

For example, for a finitely generated $(\mathfrak{g}, K)$-module M, we can define $\mathrm{supp}(M) \subset \mathfrak{p}^*$. For an appropriate situation, $\mathrm{gr}(M)$ is a finitely generated K-equivariant module over $\mathbb{C}[\mathfrak{p}^*]$. $\dim(\mathrm{supp}(M))$ is called the Gelfand-Kirillov dimension of M. If M is irreducible, $\mathrm{supp}(M) \subset \mathfrak{p} \cap \mathcal{N}$.

8.4 Proof of CST Theorem

Proof. (of first part). Use Demazure operators. Let $s = s_\alpha \in W$ be a (complex) reflection, then $V^s = \ker(\alpha)$. Note if $P \in \mathbb{C}[V]$, then $s(P) - P$ vanishes on $\alpha = 0$. So $D_s(P) := \frac{s(P)-P}{\alpha} \in \mathbb{C}[V]$. Note that $P \in \mathbb{C}[V]^W \iff D_s(P) = 0$ for all s reflections.

Consider $A = \mathbb{C}[V]/(\mathbb{C}[V]_+^W)$ a finite dimensional graded vector space. Let $P_1, \cdots, P_l$ be a basis for this vector space. They generate $\mathbb{C}[V]$ over $\mathbb{C}[V]^W$ by graded Nakayama Lemma. Need to check there are no relations. Suppose we have a shortest relation:

$$\sum Q_i P_i = 0, \quad Q_i \in \mathbb{C}[V]^W$$

Lemma. *For each P_i, we can find a $D = \sum a_I D_{s_{i_1}} \cdots D_{s_{i_n}}$ such that $D(P_i) = 1$ constant, and $D(P_j) = 0$ if $\deg P_j \leq \deg P_i$ and $j \neq i$.*

The key step of the lemma is that for any homogeneous $P \in \mathbb{C}[t]$ such that $0 \neq \overline{P} \in A$, there exists $D = D_{s_1} D_{s_2} \cdots D_{s_n}$ with $D(P) = 1$. This follows from $A^W = \mathbb{C} \cdot 1$. If P is such that $\overline{P}$ is not W-invariant, such that there exists a reflection s such that $s(\overline{P}) \neq \overline{P} \implies \overline{D_s(P)} \neq 0$, but then $\deg(s(P)) = \deg(P) - 1$. Therefore, there exists a pairing

$$\mathrm{Dem}_d \times A_d \to \mathbb{C}$$

We showed the above pairing is perfect. Hence we can embed $A_d \hookrightarrow \mathrm{Dem}_d^*$ and $\mathrm{Dem}_d \twoheadrightarrow A_d^*$. This proves the lemma.

Now pick P_i of maximal degree, and find D as in the lemma, we apply D and see $Q_i = 0$, contradiction. □

The second part follows from the following general property:

Proposition 8.10 (Serre criterion)**.** *If $A = \oplus_{n \geq 0} A_n$ graded commutative algebra, $A_0 = k$. Then A is free and finitely generated (i.e. a polynomial algebra) iff there exists a finite free (equivalently projective) resolution for k as an A-module.*

We see this concludes the proof by considering a free resolution of $\mathbb{C}$ as a $\mathbb{C}[V]$-module, and restrict to $\mathbb{C}[V]^W$.

Remark. *More generally, for a A commutative finitely generated over $k = \overline{k}$. Then* Spec A *is smooth iff for any maximal idela $\mathfrak{m}$, $A/\mathfrak{m}$ has a finite projective resolution.*

Proof. (of Serre criterion). If $A = k[x_1, \cdots, x_n] = k[V]$. We can take the Koszul resolution. For the converse, if $\dim(A) < \infty, A \neq k$, then there is no finite resolution: indeed, set

$$P_A(t) = \sum \dim(A_i) t^i$$

and

$$Q(t) = \sum_{i,d} (-1)^i m_{i,d} t^d, \quad m_{i,d} = \dim V_i[d]$$

Then $Q(t)P(t) = 1$, and $P(t)$ is a polynomial, so $Q(t)$ is not a polynomial. Contradiction.

Now if $\dim A = \infty$, there exists a not a zero divisor and $a \notin I^2$, where $I = \oplus_{i>0} A_i$. Set $B = A/(a)$, we check there exits a finite free resolution for k over A then the same holds for B. Then B is free by induction, this will imply A is free.

Indeed, start with a free resolution for k over A:

$$0 \to F_n \to \cdots \to F_1 \to F_0$$

Apply $\otimes_A B$, we get a complex of freee B-module, with 1-dim homology in degree 0 and equals 1. We claim it contains a direct summand which is a resolution for k. (Think of A as a deformation of B. A/a^2 is a first order deformation of k. Then k admits first order deformation $\implies \mathrm{Ext}^2(k,k)$ vanishes). □

Remark. *$C[V]^W$ is free of rank $|W|$. This can be seen by passing to fraction fields and use Galois theory. Or more geometrically: we can think of the embedding $\mathbb{C}[V]^W \hookrightarrow \mathbb{C}[V]$ as the projection map $V \to V//W$. On the level of $\mathbb{C}$-points, this map is $|W|$-to-1.*

This implies for $A = \mathbb{C}[V]/\mathbb{C}[V]^W_+$ the coinvariant ring, we have $\dim A = |W|$. In fact:

$$A \simeq H^*(G/B)$$

and $\dim A_n = \#\{w \in W \mid \ell(w) = n\}$.

Corollary 8.11. *$\mathbb{C}[\mathfrak{g}]^G$ is a free algebra.*

8.5 Kostant Slice

Recall if $x \in \mathfrak{g}$ is regular semisimple, then $\mathfrak{g}_x$ the centralizer is a torus $\mathfrak{t}$. We say $\operatorname{rank}(\mathfrak{g}) = \dim(\mathfrak{g}_x) = r$. We also know $\mathfrak{g}^{\mathrm{rs}}$ is a dense open subset. It follows that $\dim(\mathfrak{g}_x) \geq r$ for all $x \in \mathfrak{g}$.

Definition 8.12. We say $x \in \mathfrak{g}$ is regular, if $\dim \mathfrak{g}_x = r$.

Remark. *Regular semisimple elements form a Zariski open set.*

Example. *For $\mathfrak{g} = \mathfrak{sl}_n$, x is regular iff it only has one Jordan block for each eigenvalue.*

Proposition 8.13. *For a semisimple Lie algebra $\mathfrak{g}$, the element $e = \sum_{\alpha \text{ simple}} e_\alpha$ is regular*

Proof. For α a simple root, set $f_\alpha \in \mathfrak{g}_{-\alpha}$ be such that $[e_\alpha, f_\alpha] = \alpha^\vee$. We now choose $h = 2\rho^\vee$, where $\rho^\vee$ the fundamental coweight; and $f = \sum_{i=1}^r m_i f_{\alpha_i}$, where $2\rho^\vee = \sum_{i=1}^r m_i \alpha_i^\vee$. One can check that (e, f, h) forms a $\mathfrak{sl}_2$-triple, called the principal $\mathfrak{sl}_2$-triple.

Now $\dim(\mathfrak{g}_e)$ equals the number of summands in the decomposition of $\mathfrak{g}$ as a sum of irreducible $\mathfrak{sl}_2$-modules. But note that $\mathfrak{g}$ as a principal $\mathfrak{sl}_2$-representation has only even weights. So the number equals $\dim(\mathfrak{g}_0) = \dim(\mathfrak{g}_h) = r$. □

Lemma 8.14. *Let $P_1, \cdots, P_r$ be free homogeneous generators of $\mathbb{C}[\mathfrak{g}]^G \simeq \mathbb{C}[\mathfrak{t}]^W$. Consider:*

$$dP_i : \mathfrak{g} \to \mathfrak{g}^* \simeq \mathfrak{g}$$

These are G-equivariant polynomial maps. We have $dP_i(x) \in \mathfrak{z}(\mathfrak{g}_x)$, and $dP_i(x)$ span $\mathfrak{g}_x$ if x is regular.

Example. *Let $\mathfrak{g} = \mathfrak{sl}_n$, $P_i : x \to \operatorname{Tr}(x^{i+1})$ for $i = 1, 3, \cdots, n-1$. Then $dP_i(x) = (i+1)x^i$.*

Proof. Firstly, we see for each $y \in \mathfrak{g}_x$, we have:

$$[y, dP_i(x)] = dP_i([y, x]) = 0 \implies dP_i(x) \in \mathfrak{z}(\mathfrak{g}_x)$$

On $\mathfrak{g}^{\mathrm{reg}} \subset \mathfrak{g}$, we have a vector subbundle $\underline{\mathfrak{g}_x} \subset \underline{\mathfrak{g}}$ of rank r. Then dP_i are r sections. We want to check they give a basis at every fibre, i.e. they trivialize the bundle. We check it:

1. On the open part $\mathfrak{g}^{\mathrm{rs}}$. Let $x \in \mathfrak{g}^{\mathrm{rs}}$, $\mathfrak{g}_x = \mathfrak{t}$ a maximal torus. Let $\pi : \mathfrak{t} \to \mathfrak{t}/W \simeq \mathbb{C}^r$, it sends $x \mapsto \overline{x}$. Then it suffices to check

$$d\pi : T^*_{\overline{x}}(\mathfrak{t}/W) \xrightarrow{\sim} T^*_x(\mathfrak{t}) \iff \forall \xi \in \mathfrak{t}^*, \exists P \in \mathbb{C}[\mathfrak{t}]^W, dP|_x = \xi$$

   This is true: start from any P_0 such that $dP_0|_x = \xi$, $dP_0|_{w(x)} = 0$ for $1 \neq w \in W$. And consider W-averaging of P_0.

2. Fix a Borel subalgebra $\mathfrak{b}$. Note a generic non-semisimple element has form $x = s + n$ for $\alpha(s) = 0$ for a unique positive root α. But then s acts on $\mathfrak{g}_\alpha$ as $\alpha(s)$, this means $n \in \mathfrak{g}_\alpha$. This means we need to check surjectivity for these x.

   We know $\mathfrak{g}_x = \mathfrak{t}' \oplus \mathbb{C} \cdot n \subset \mathfrak{g}_s = \mathfrak{t}' \oplus \mathfrak{sl}(2)_\alpha$. Set $V = n + \mathfrak{t}' \oplus \mathbb{C} \cdot f_\alpha$. There is a natural map $\pi : V \to \mathfrak{t}/W$. We need to check the map:

$$T^*_{\pi(x)}(\mathfrak{t}/W) \to T^*_x V = T^*_x(n + \mathfrak{t}' \oplus \mathbb{C} \cdot f_\alpha) = (\mathfrak{t}' \oplus \mathbb{C} \cdot f_\alpha)^* \simeq \mathfrak{t}' \oplus \mathbb{C} \cdot e_\alpha$$

   is an isomorphism. But by Kostant slice in the case of $\mathfrak{sl}(2)$, we know:

$$V \simeq \mathfrak{t}' \oplus \mathbb{C} \cdot h_\alpha / \langle s_\alpha \rangle \simeq \mathfrak{t}/\langle s_\alpha \rangle$$

   So indeed $\pi : V \simeq \mathfrak{t}/\langle s_\alpha \rangle \to \mathfrak{t}/W$, which induces isomorphism on cotangent spaces as in the first case.

This suffices because the locus in $\mathfrak{g}^{\mathrm{reg}}$ where dP_i don't span is the zero set of a section of the line bundle $\wedge^r \underline{\mathfrak{g}_x}$, such a set can't have codim ≥ 2. □

This lemma is equivalent to say $\mathfrak{g}_x \to \mathfrak{t}/W$ has nondegenerate differnetial at x, for x regular.

Theorem 8.15 (Kostant Slice)**.** *Let $S = (f + \mathfrak{g}_e)$. Then $S \simeq \mathfrak{t}/W$. In other words, $\mathbb{C}[\mathfrak{g}]^G \xrightarrow{\sim} \mathbb{C}[S]$ via restriction map. Moreover, we have:*

$$(f + \mathfrak{b})/U \simeq S$$

in fact $U \times S \simeq f + \mathfrak{b}$ is the trivial torsor.

Remark. *$\mathfrak{b}$ can be recovered from e as $\mathfrak{b} = \mathfrak{g}_{\geq 0}$. For e nilpotent operator on a finite dimensional vector space V, there is a canonical filtration on V. By JNF, there exists an $\mathfrak{sl}_2$-action on V and set $V_{\geq i} = \oplus_{j \geq i} V_j$, where V_j is the j-eigenspace of h. V_j depends on the choice of h, but for another choice h', there exists $u \in U = G_{>0}$ such that $h' = uhu^{-1}$, so they have the same filtration. In fact, this is the unique filtration such that*

1. *$eV_{\geq i} \to V_{\geq i+2}$*
2. *On V/eV, it's given by $(V/eV)_{\geq -i} = \mathrm{im}\ (\ker(e^{i+1}))$*

Explicitly: $V_{\geq i} = \oplus_{m-n=i-1} \ker(e^n) \cap \mathrm{im}\ (e^m)$.

Proof. We look at $\mathbb{C}[\mathfrak{t}]^W \to \mathbb{C}[S]$. To see it's injective, it suffices to see $G \cdot S$ contains a dense open subset in $\mathfrak{g}$. It's enough to check

$$T(G \times S) \to T(\mathfrak{g}) = \mathfrak{g} \iff \mathfrak{g} \oplus \mathfrak{g}_e \to \mathfrak{g}$$

mapping $(x, s) \mapsto [f, x] + s$. But this follows from the representation theory of $\mathfrak{sl}_2$.

The surjectivity follows from Lemma by graded Nakayma. We wish to give a grading on $\mathbb{C}[S]$, which is equivalent to proved a $\mathbb{C}^*$-action. For $s \in f + \mathfrak{g}_e$, we define:

$$t \cdot s = \mathrm{Ad}\begin{pmatrix} t^{\frac{1}{2}} & \\ & t^{-\frac{1}{2}} \end{pmatrix}(ts)$$

In fact, the action of SL(2) on $\mathfrak{g}$ factors through PGL(2). So formally we should write:

$$t \cdot s = \mathrm{Ad}\begin{pmatrix} t & \\ & 1 \end{pmatrix}(ts)$$

In this grading $\mathbb{C}[S]$ is a polynomial algebra with generators in degrees $m_i + 1$, where $2m_i$ are the highest weight of $\mathfrak{sl}(2)$-representation appearing in the decomposition of $\mathfrak{g}$ acted by $\{e, f, h\}$. Now I have $\mathbb{C}[\mathfrak{t}]^W \subset \mathbb{C}[S]$ as graded algebras, and both are polynomial algebras. Now $d(S \to \mathfrak{t}/W)$ is an isomorphism, this implies $\mathbb{C}[S]/\mathbb{C}[\mathfrak{t}]^W_+$ is one-dimensional, hence $\mathbb{C}[S] = \mathbb{C}[\mathfrak{t}]^W$.

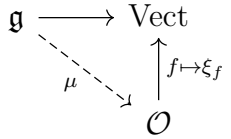
For the second part, we need to show $U \times S \simeq f + \mathfrak{b}$. First note that $\mathfrak{u} = \mathfrak{g}_{>0}$, and we write $\mathfrak{g} = \oplus \mathfrak{g}_{2i}$ for $f \in \mathfrak{g}_{-2}$. We know $[f, \mathfrak{u}] \subset \mathfrak{g}_{\geq 0}$. Hence $u(f) \in f + \mathfrak{b}, \forall u \in U$. We now need to check that for all $x \in \mathfrak{b}$, there exists $u \in U$ such that $u(f + x) = f + \mathfrak{g}_e$. The key point is to write

$$\mathfrak{b} = [f, \mathfrak{u}] + \mathfrak{g}_e$$

this follows from $\mathfrak{sl}_2$-reps theory and $\mathfrak{z}_{\mathfrak{g}}(e, f, h) = 0$. We now check by induction in d that there exists $u \in U$ such that $u(f+x) \bmod \mathfrak{g}_{>d}$ is in the image of $f+\mathfrak{g}_e$. Assume by induction $x \bmod \mathfrak{g}_{\geq d}$ is in the image of $\mathfrak{g}_e$. Write $x_d = x^\circ + [f, y]$ for $x^\circ \in \mathfrak{g}_e$ and $y \in \mathfrak{g}_{d-2}$. Let's consider $\exp(-y)(f + x)$. This proves $U \times S \to f + \mathfrak{b}$ is onto. But also $[f + x, y] \neq 0$ for $0 \neq y \in U \implies \mathrm{Stab}_U(z) = 1$ for $z \in S$. Hence this is in fact a bijection. It's also algebraic by construction. □

Remark. *This is a special case of* **Hamiltonian reduction**. *Let X be a variety, a Poisson structure comes from a section π of $\bigwedge^2 T_X$: $\{f, g\} = \langle \pi, df \wedge dg \rangle$ with the property that $[\pi, \pi] = 0$ (the Schouten bracket).*

Let G acts on X preserving π. A moment map is a G-equivariant $\mu : X \to \mathfrak{g}^*$ making the following diagram commutes:



the map $\mathcal{O} \to$ Vect is given by $\pi(df, -) = \xi_f(-)$. This is in fact a Lie algebra homomorphism. Suppose $N = X/G$ is a principal G-bundle. Then $T^*N = \mu^{-1}(0)/G$, which is the Hamiltonian reduction.

In our case $X = \mathfrak{g}^*$ is a Poisson variety such that $f_x \in \mathbb{C}[\mathfrak{g}^*]$ linear function corresponding to $x \in \mathfrak{g}$, then $\{f_x, f_y\} = f_{[x,y]}$. This restricts to a symplectic structure on each G-orbit in $\mathfrak{g}^*$. Explicitly, after we identifying $\mathfrak{g} \simeq \mathfrak{g}^*$, an orbit looks like $\mathfrak{g}/[\mathfrak{g}, x]$. The symplectic form is given by $\omega(y, z) = ([y, z], x)$.

Corollary 8.16. $\deg P_i = m_i + 1$.

Corollary 8.17. $\mathbb{C}[\mathfrak{g}]_+^G \subset \mathbb{C}[\mathfrak{g}]$ *is a radical ideal with zero set $\mathcal{N}$, the set of nilpotent elements.*

Proof. Set theoretically, for $x \in \mathfrak{g}$, x nilpotent iff $P(x) = 0$ for all $P \in \mathbb{C}[\mathfrak{g}]_+^G$. Use Jordan decomposition, we write $x = x_s + x_n$. We can find Borel subalgebra $\mathfrak{b} = \mathfrak{t} \oplus \mathfrak{u}$ such that $x_s \in \mathfrak{t}, x_n \in \mathfrak{u}$. One sees

$$P(x_s + x_n) = P(x_s)$$

because consider conjugation, say by $\rho^\vee(t) \in G_{\text{ad}}$, this sends e_α to $t^{\langle \rho^\vee, \alpha \rangle} e_\alpha$. But then we can write $x_n = \sum_{\alpha > 0} c_\alpha e_\alpha$. Then:

$$P(x) = P(\text{Ad}(\rho^\vee(t))(x)) = P(x_s + \sum c_\alpha t^{\langle \rho^\vee, \alpha \rangle} e_\alpha) \to P(x_s)$$

as $t \to 0$. Hence $P(x) = P(x_s)$. Now use Chevalley restriciton theorem, we know: x nilpotent $\iff x_s = 0 \iff P(x_s) = 0, \forall P \in \mathbb{C}[\mathfrak{t}]_+^W \iff P(x) = 0, \forall P \in \mathbb{C}[\mathfrak{g}]_+^G$.

Recall we have:

$$\chi : \mathfrak{g} \to \mathbb{A}^r$$

given by invariant polynomials $P_1, \cdots, P_r$. The nilpotent cone is defined to be

$$\mathcal{N} = \chi^{-1}(0) = \text{Spec } (\mathbb{C}[\mathfrak{g}]/\mathbb{C}[\mathfrak{g}]_+^G)$$

We need to see $\mathcal{N}$ is reduced. In fact, first note $\dim(\mathcal{N}) \geq \dim(\mathfrak{g}) - r$ since $\mathcal{N}$ is a fibre. On ther other hand, $\mathbb{C}[\mathfrak{g}]_+^G$ is generated by r elements and thus has height $\leq$ r. This shows $\dim(\mathcal{N}) = \dim(\mathfrak{g}) - r$. Therefore $P_1, \cdots, P_r$ form a regular sequence and hence $\mathcal{N}$ is Cohen-Macaulay.

Another input is the following. Let e be a regular nilpotent element, we've seen $d\chi_e$ has full rank and therefore e is a smooth point of $\mathcal{N}$. Therefore $\mathcal{N}^{\text{reg}} \subset \mathcal{N}$ is smooth, hence $\mathcal{N}$ is generically reduced. A classical theorem of Serre states that CM + generic reducedness implies reducedness. □

Remark. *It is conjectured that scheme-theoretic preimage under $\mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$, $(x, y) \mapsto [x, y]$ is reduced.*

8.6 Generalizations of Kostant Slice

Let $e \in \mathfrak{g}$ be any nilpotent. We can complete this into a $\mathfrak{sl}_2$-tripe (e, f, h) by Jacobson Morozov. Now set $S_f = f + \mathfrak{g}_e$ is again a slice to the orbit $G(f)$, called *Slowdowy Slice*. By a slice we mean:

$$T_f \mathfrak{g} = T_f(G(f)) \oplus T_f S_f \iff \mathfrak{g} = [\mathfrak{g}, f] \oplus \mathfrak{g}_e$$

We still have a map $S_f \to \mathfrak{t}/W$, with the fibre of 0 equals $S_f^0 = S_f \cap \mathcal{N}$. This is a singular variety (symplectic singularity).

Example. *Let $\mathfrak{g}$ be simple. Consider the G-action on $\mathcal{N}$, which has finitely many orbits. There exists a unique open orbit (consisting of regular nilpotents); there also exists a unique orbit of codimension 2 (subregular orbit). For example, for $\mathfrak{g} = \mathfrak{sl}(n)$, the subregular orbit corresponds to Jordan type $(n-1, 1)$.*

Now if f is subregular, then $\dim(S_f^0) = 2$. This is a rational surface singularity, namely it is isomorphic to $\mathbb{A}^2/\Gamma$ for $\Gamma \subset \mathrm{SL}(2)$ a finite subgroup.

Recall we have McKay correspondence

$$\begin{array}{ccc} \{\text{finite subgroups of } \mathrm{SL}_2\} & \xleftrightarrow{\Gamma \mapsto \mathbb{A}^2/\Gamma} & \{\text{rational surface singularities}\} \\ \uparrow & \nearrow_{\mathfrak{g} \mapsto S_f^0} & \uparrow \\ \{\text{simply-laced Lie algebra}\} & \longleftrightarrow & \{\text{simply-laced Dynkin diagram}\} \end{array}$$

For $\Gamma \subset \mathrm{SL}(2)$, we wish to produce a Dynkin diagram. The vertices are nontrivial irreducible representations of Γ. Two vertices v_ρ and v_ψ are connected if $\mathrm{Hom}(\psi, \rho \otimes \mathbb{C}^2) \neq 0$.

Example. *For $\Gamma = \mathbb{Z}/n\mathbb{Z}$, this produces A_n. For Γ the icosahedron group, this produces E_8.*

Let S be rational surface singularity, it has a unique minimal resolution $\pi : \widetilde{S} \to S$. Choose $x_0 = f \in S$, then $\pi^{-1}(x_0)$ is the union of projective lines. This defines a graph with vertices being projective lines, and two vertices v_{ℓ_1}, v_{ℓ_2} care connected if $\ell_1 \cap \ell_2 \neq \emptyset$.

1. $\widetilde{S}$ can also be recovered from the geometry of $\mathfrak{g}$. Consider the Springer resolution:

$$\begin{array}{ccc} \widetilde{\mathcal{N}} = T^*(G/B) = \{(a, \mathfrak{b}) \mid a \in \mathrm{rad}(\mathfrak{b})\} & \hookrightarrow & \mathfrak{g} \times G/B \\ & & \downarrow p \\ S & \hookrightarrow & \mathcal{N} \end{array}$$

   One can check that $p : T^*(G/B) \to \mathcal{N} \subset \mathfrak{g}^*$ is the moment map. Then for $S = S_f^0 \subset \mathcal{N}$, we have $\widetilde{S} = p^{-1}(S_f^0)$.

   Example. *For $\mathfrak{g} = \mathfrak{sl}(n)$, e of type $(1, n-1)$, then $p^{-1}(e)$ is consecutive gluing of $\mathbb{P}^1$.*

2. To a non simply laced diagram D, one can assign unfolding: a simply laced diagram with an automorphism. For example for type G_2, we associate D_3 with a S_3-action.

   When $\mathfrak{g}$ is not simply-laced, $S_f^0 \simeq S$ coming from the unfolding (of the dual group), the automorphism coming from the action of $\pi_0(Z_G(f))$ on lines in $p^{-1}(f)$.

We have $S = f + \mathfrak{g}_e \simeq (f + \mathfrak{b})/U = \mathfrak{g}^* /\!/_\psi U$ the Hamiltonian reduction. This picture also generalizes. Again, can define $\mathfrak{g}_{\geq 0} = \mathfrak{p}$ the parabolic, $\mathfrak{g}_{>0} = \mathfrak{u}$ its radical. Let's assume that e is *even*, namely h only has even eigenvalues in $\mathfrak{g}$ (say in $\mathfrak{sl}(n)$: all block sizes are of the same parity). Then the linear functional $\psi_f : x \mapsto (f, x)$ is a character ($\psi_f|_{[\mathfrak{u},\mathfrak{u}]} = 0$). Hence we have $S \xrightarrow{\sim} \mathfrak{g}^* /\!/_\psi U$.

Otherwise, we need to modify U. On $\mathfrak{g}_1$ we get a skew form $\omega(x, y) = (f, [x, y])$. It turns out that it's nondegenrate. Now we can pick a Lagrangian $L \subset \mathfrak{g}_1$, and we redefine $\mathfrak{u} \coloneqq L \oplus \mathfrak{g}_{\geq 2}$. Now $\psi_f|_{[\mathfrak{u},\mathfrak{u}]} = 0$. In this case, we have $S \xrightarrow{\sim} \mathfrak{g}^* /\!/_{\psi_f} U$.

Exercise. $\dim(\mathfrak{g}) - \dim(S) = 2\dim(\mathfrak{u})$.

Another generalization is Kostant Rallis slice. Let θ be an involution acting on $\mathfrak{g}$. Write $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$. Choose a generic nilpotent in $\mathfrak{p}$. Complete it as (e, f, h) for $e, f \in \mathfrak{p}$ and $h \in \mathfrak{k}$. We then have

$$f + (\mathfrak{g}_e \cap \mathfrak{p}) \xrightarrow{\sim} \mathfrak{p}/\mathfrak{k} \simeq \mathfrak{a}/W^{\mathfrak{a}}$$

8.7 Whittaker Module

Consider the triangular decomposition $\mathfrak{g} = \mathfrak{u}_- \oplus \mathfrak{t} \oplus \mathfrak{u}_+$. Let's fix a character ψ of $\mathfrak{u}$

Definition 8.18. The **universal Whittaker module** for $(\mathfrak{g}, \psi)$ is

$$\mathcal{W}_\psi = U(\mathfrak{g}) \otimes_{U(\mathfrak{u})} \mathbb{C}_\psi$$

The category of Whittaker modules is

$$\mathrm{Wh}_\psi(\mathfrak{g}) = \{M \in U(\mathfrak{g})\text{-Mod} \mid x - \psi(x) \text{ acts locally nilpotent for all } x \in \mathfrak{u}\}$$

Theorem 8.19 (Kostant)**.** *1. $Z(U(\mathfrak{g})) \simeq \mathrm{End}(\mathcal{W})$ and $\mathrm{Ext}^{>0}_{\mathfrak{g}}(\mathcal{W}, \mathcal{W}) = 0$.*

2. The category of Whittaker module is equivalent to $Z(U(\mathfrak{g}))$ modules via:

$$M \mapsto \mathrm{Hom}_{\mathfrak{g}}(\mathcal{W}, M)$$

In particular, this category is abelian.

9 Representation of $\mathbf{SL}(2, \mathbb{R})$

9.1 Classification of Irreducible $(\mathfrak{g}, K)$-modules

Let $G_{\mathbb{R}} = \mathrm{SL}(2, \mathbb{R})$, $K_{\mathbb{R}} = \mathrm{SO}(2, \mathbb{R})$, $\mathfrak{g} = \mathfrak{sl}(2, \mathbb{C})$, $K = K_{\mathbb{C}} = \mathrm{SO}(2, \mathbb{C}) = \mathbb{C}^*$. We wish to classify irreducible $(\mathfrak{g}, K)$-modules.

Given an irreducible $(\mathfrak{g}, K)$-module V, we write $V = \oplus V_n$ such that K acts on V_n by $t \mapsto t^n$. Now choose basis (h, e, f) for $\mathfrak{g}$:

$$h = i\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, e = \frac{1}{2}\begin{pmatrix} i & 1 \\ 1 & -i \end{pmatrix}, f = \frac{1}{2}\begin{pmatrix} -i & 1 \\ 1 & i \end{pmatrix}$$

complex conjugation acts by $h \to -h$, $e \to f$, $f \to e$.

Now $e : V_n \to V_{n+2}$, $f : V_n \to V_{n-2}$. So if V irreducible, it can have either even weights or odd weights. Let $C = \frac{1}{2}(ef + fe + \frac{h^2+1}{2}) = fe + \frac{(h+1)^2}{4} = ef + \frac{(h-1)^2}{4}$ be the Casimir. Then C acts by scalar c for $c \in \mathbb{C}$.

We see $\dim(V_n) \leq 1$ for all n. Indeed using PBW, any $x \in U(\mathfrak{g})$ of weight 0 is a polynomial in fe and h. Now if $v \in V_n$, we see $U(\mathfrak{g})v \cap V_n = (U(\mathfrak{g})_0 v) \cap V_n = \mathbb{C}v$, as $hv = nv, (fe)v = (C - \frac{(h+1)^2}{4})v$.

Clearly if $V_n = 0 \implies V = V_{<n} \oplus V_{>n}$, so one of them is zero. Hence $\{n | V_n \neq 0\}$ is a segment, or a ray, or a line.

1. For the segment case, these are finite dimensional representations.

2. For the ray case, these are Verma modules for $\mathfrak{sl}(2)$ of highest weight m for $m < 0$ (for Borel spanned by h, e), and Verma modules of lowest weight m for $m > 0$ (here we choose a different Borel spanned by h, f). These are called L_m. For $m \neq \pm 1$, these are called discrete series representations; for $m = \pm 1$, these are called limits of discrete series.

3. For the line case, fix $v_n \in V_n$ such that $v_{n-2} = fv_n$. We can compute $ev_n = efv_{n-2} = (c - \frac{(n+1)^2}{4})v_{n+2}$. Easy to see this defines a module, and it's irreducible if $c \neq \frac{(n+1)^2}{4}$ for all n. In this case, set $c = \frac{s^2}{4}$. We compute $ev_n = \frac{(s-n-1)(s+n+1)}{4}v_{n+2}$ for $s \neq 2\mathbb{Z}$ or $2\mathbb{Z}+1$. Let's rescale v_n and get w_n. We have formula:

$$ew_n = \frac{1}{2}(s-1-n)w_{n+2}, \quad fw_n = \frac{1}{2}(s-1+n)w_{n-2}$$

Theorem 9.1. *Irreducible $(\mathfrak{g}, K)$-module are*

1. *V_n: finite dimensional representations*

2. *L_m, for $m \in \mathbb{Z}$ and $m \neq 0$. These are called discrete series for $m > 1$. $L^{\pm}_{\pm 1}$ are called limis of* **discrete series**.

3. *P^+_s for $s \notin 2\mathbb{Z}+1$, P^-_s for $s \notin 2\mathbb{Z}$. These are called* **principal series**. *Moreover, $P^{\pm}_s = P^{\pm}_{-s}$*

9.2 Unitary Representations

Recall $\overline{h} = -h, \overline{e} = f, \overline{f} = e$. Look for Hermitian form $(-,-)$ on an irreducible module M such that $(xv, w) + (v, \overline{x}w) = 0$. Hence we need $(hv, w) = (v, hw), (ev, w) = (v, -fw), (fv, w) = (v, -ew)$.

Recall $C = fe = \frac{(h+1)^2}{4}$ actes by $c = \frac{s^2}{4}$. This forces $(w_n, w_m) = 0$ for $m \neq n$. We only need to describe (w_n, w_n). Note $(ew_n, ew_n) = (-few_n, w_n) = (\frac{(n+1)^2}{4} - c)(w_n, w_n)$. This shows: M is unitary $\iff \frac{(n+1)^2}{4} - c > 0, \forall n$ such that $w_n \neq 0, ew_n \neq 0$.

For L^+_m, $c = \frac{(m-1)^2}{4}$, we have $\frac{(n+1)^2}{4} - c > 0$ for $n \geq m$. We can also make similar calculation for L^-_{-m}.

For principal series, one sees we either have $s \in i\mathbb{R}$; or P^+_s with $|s| < 1$

Proposition 9.2. *The following $(\mathfrak{g}, K)$-modules are are unitary modules:*

1. *Trivial representations.*

2. *$L^{\pm}_{\pm m}$ for $m \geq 1$.*

3. *$P^{\pm}_s$ for $s \in i\mathbb{R}$. These are called* **unitary principal series**.

4. *P^+_s for $|s| < 1$. These are called* **complementary series**.

9.3 Realizations

For P_s^+, recall we have realization:

$$P_s^+ = \{f \in C^\infty(\mathbb{R}^2\backslash\{0\}) \mid f(tx) = |t|^{s-1}f(x)\}$$

and these can be identified as $\frac{1-s}{2}$-densities on $\mathbb{RP}^1$. Hence for $s \in i\mathbb{R}$ this is unitary. Indeed, choose a parametrization:

$$S^1 \to \mathbb{RP}^1, \quad \theta \mapsto x(\theta)$$

Then (after completion) P_s^+ then has realization $\{f(\theta)(d\theta)^{\frac{1-s}{2}} \mid f \in L^2(S^1, d\theta)\}$. Let us define:

$$(f(\theta)|d\theta|^{\frac{1-s}{2}}, g(\theta)|d\theta|^{\frac{1-s}{2}}) = \int_{\mathbb{R}} f(\theta)\overline{g(\theta)}|d\theta|$$

Then obiously G acts unitarily by change of variables.

Discrete series L_m^+ is realized as $m/2$-densities on unit disc $|z|< 1$. Note $\mathrm{SL}(2,\mathbb{R})$ acts on the disc, preserving the metric $\mu = \frac{du d\overline{u}}{(1-|u|^2)^2}$. If $\omega_1 = f(du)^{m/2}, \omega_2 = g(du)^{m/2}$. We can define:

$$(\omega_1, \omega_2) = \int_D \omega_1\overline{\omega_2}/\mu^{m/2-1}$$

For complementary series, we still realize them as $\frac{1-s}{2}$-densities, as before. We can define:

$$(f, g) = \int_{\mathbb{R}^2} f(x)\overline{g(y)}\frac{dxdy}{|x-y|^{1+s}} = \int_{\mathbb{R}^2} f(x)\overline{g(y)}(dxdy)^{\frac{1-s}{2}} \cdot (\frac{dxdy}{|x-y|^2})^{\frac{1+s}{2}}$$

The point is that the form $\frac{dxdy}{|x-y|^2}$ is invariant when transforming both coordinates under the action of $\mathrm{PSL}(2,\mathbb{R})$. Hence the above form is actually invariant under the G-action.

10 Remaining Goals

Definition 10.1. An irreducible representation of G is of **discrete series** if some matrix coefficient $m_{v,w} : g \mapsto (gv, w) \in L^2(G)$, with $v \in V, w \in V^*$. Note we can take v, w be K-finite.

Remark. *In this case, $x \mapsto m_{x,w}$ embeds V into $L^2(G)$, thus V is unitary.*

For admissible representations, we can define its characters which is a distribution on the group.

Theorem 10.2. *It's well-defined, and its restriction to G^{rs} is an analytic function. Moreover, the distribution is given by a locally L^1-functions.*

We will discuss classifications and character formula.

For classification purpose, we will see the building blocks are discrete series: there is Langlands classification lemma of all starting from discrete series. There will be an algebraic

geometric classification using $\mathcal{D}$-modules and localization theory: this is classified using $K_{\mathbb{C}}$-orbits on $G_{\mathbb{C}}/B_{\mathbb{C}}$ and extra data. Alternatively, we can use the special case of local Langlands conjecture. For F local field,

$$\mathrm{Irr}(G(F)) \leftrightarrow \mathrm{Hom}(\mathrm{Gal}(F), {}^L G) \text{ with extra data}$$

Let's discuss character formula. For principal series (induction of finite dimensional representations), the character is essentially given by Frobenius formula for standard representations. A basic question is to express irreducible characters as linear combinations of standard ones. For $G_{\mathbb{R}} = H(\mathbb{C})$ a complex group, this is equivalent to Kazhdan-Lusztig conjectures. More generally, we need to turn to Lusztig-Vogan generalization.

Fact: an irreducible $(\mathfrak{g}, K)$-module has a realization in Hilbert space, where $g \in G$ acts by bounded continuous, but not necessarily unitary operator.

Example. *Consider $P_s^+ = \{f(x)dx^{\frac{1-s}{2}}\}$ for $G = \mathrm{SL}(2,\mathbb{R})$, where we fix some non-vanishing K-invariant dx identified for different s as vector spaces. We get $(L^2, (-,-))$ on P_s^+.*

11 Characters of Representations

11.1 Some Functional Analysis

Definition 11.1. Let V be a Hilbert space, and let $T \in \mathrm{End}(V)$. We say T is **compact** if for every bounded subset $X \subset V$, we have $\overline{T(X)}$ compact. We say T is **Hilbert Schmidt** if

$$\|T\|^2 = \sum \|Te_i\|^2 < \infty$$

for some choice of orthonormal basis e_i. We say T is of **trace class** if the series

$$\sum_i (|T|(e_i), e_i)$$

absolutely converges, where $|T| = (T^\dagger \cdot T)^{\frac{1}{2}}$. In this case, we can define the **trace** of T:

$$\mathrm{Tr}(T, V) \coloneqq \sum (T(e_i), e_i)$$

Remark. *Trace class $\implies$ Hilbert-Schmidt $\implies$ Compact $\implies$ Bounded.*

Proposition 11.2.

1. *Let v_i be an orthonormal basis of V, and let S be an endomorphism of the algebraic span of v_i so that the sum $\sum_i \|Sv_i\|^2$ converges. Then S (uniquely) extends to a HS operator.*
2. *If S is HS and S' is bounded, then $S \circ S'$ is HS.*
3. *If S_1, S_2 are HS, then $S_1 \circ S_2$ is trace class and*
$$\mathrm{Tr}(S_1 \circ S_2) = (S_1, S_2^\dagger)_{\mathrm{HS}}$$
4. *if S is trace class and S' bounded, then $S \circ S'$ and $S' \circ S$ are trace class, and $\mathrm{Tr}(S \circ S') = \mathrm{Tr}(S' \circ S)$.*

11.2 Definition of Character

Let V be a Hilbert space representation.

Lemma 11.3. *Let $G_{\mathbb{R}}$ be a real Lie group with maximal compact K. Then we have polar decomposition:*

$$\mathfrak{g}_{\mathbb{R}} = \mathfrak{k}_{\mathbb{R}} \oplus \mathfrak{p}_{\mathbb{R}}$$

Let $G = G_{\mathbb{C}}, \mathfrak{g} = \mathfrak{g}_{\mathbb{C}}, \mathfrak{k} = \mathfrak{k}_{\mathbb{C}}, \mathfrak{p} = \mathfrak{p}_{\mathbb{C}}$ be all complexified. Let $\mathcal{N} \subset \mathfrak{g}$ be the nilpotent cone.

1. *G acts on $\mathcal{N}$ with finitely many orbits.*
2. *$K_{\mathbb{C}}$ acts on $\mathcal{N} \cap \mathfrak{p}$ with finitely many orbits.*

Proof. 1. Pick $\mathfrak{g} \subset \mathfrak{sl}(n)$ and $G \subset \mathrm{SL}(n, \mathbb{C})$. Write $\mathfrak{sl}(n) = \mathfrak{g} \oplus \mathfrak{c}$. Then $\mathcal{N} = \mathcal{N}_{\mathfrak{sl}(n)} \cap \mathfrak{g}$. But by JNF, it's enough to see if $\mathcal{O}$ is an $\mathrm{SL}(n, \mathbb{C})$ orbit on $\mathcal{N}_{\mathfrak{sl}(n)}$, then $\mathcal{O} \cap \mathfrak{g}$ is finite union of G orbits. But this follows from every G orbit in $\mathcal{O} \cap \mathfrak{g}$ is open, but this is another calculation on the tangent space. Indeed, for any $x \in \mathcal{O} \cap \mathfrak{g}$:

$$T_x(G \cdot x) = [x, \mathfrak{g}] = ([x, \mathfrak{g}] \oplus [x, \mathfrak{c}]) \cap \mathfrak{g} = [x, \mathfrak{sl}(n)] \cap \mathfrak{g} = T_x(\mathcal{O}) \cap \mathfrak{g}$$

2. This is the same calculation by replacing $\mathrm{SL}(n, \mathbb{C})$ by G and G by $K_{\mathbb{C}}$. Check for G orbit $\mathcal{O} \subset \mathfrak{g}$, $\mathcal{O} \cap \mathfrak{p}$ is a finite union of K-orbits. Indeed, for $x \in \mathfrak{p}$, we still write $[x, \mathfrak{g}] = [x, \mathfrak{k}] \oplus [x, \mathfrak{p}]$.

□

Remark. *In fact G orbits in $\mathfrak{g} = \mathfrak{g}^*$ are symplectic, and $\mathcal{O} \cap \mathfrak{p}$ is a Lagrangian.*

Proposition 11.4. *Let M be an irreducible $(\mathfrak{g}, K)$-module, write $M = \oplus W_\nu \otimes V_\nu$, then*

$$\dim(W_\nu) \leq C \cdot \dim(V_\nu)$$

for some constant C.

Remark. *We showed this for unitary representation before.*

Proof. Pick $V_\nu \subset M$. Then $U(\mathfrak{g}) \otimes V_\nu \twoheadrightarrow M$. Now write $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$, we therefore know:

$$\mathrm{Sym}(\mathfrak{p}) \otimes V_\nu \twoheadrightarrow M$$

Passing to associated graded for PBW filtration, we get a finitely generated module $\overline{M}$ for $\mathbb{C}[\mathfrak{g}^*] = \mathbb{C}[\mathfrak{g}]$. Since this module is K-equivariant, we know

$$\mathrm{supp}(\overline{M}) \subset \mathfrak{p} \cap \mathcal{N}$$

Check for any N a K-equivariant $\mathrm{Sym}(\mathfrak{p})$-module as above, $[V_\nu : N] \leq C \dim V_\nu$. WLOG, N is a torsion free module on the closure of a K orbit $\mathcal{O} \subset \mathcal{N} \cap \mathfrak{p}$. Then $N \subset$ sections of an equivariant vector bundle on an orbit $\subset \mathbb{C}[K]^r$ for r the rank of the bundle. This proves the claim. □

Proposition 11.5. *For $f \in C_c^\infty(G)$, fdg acts by trace class operator on V.*

Proof. Recall we have $C_{\mathfrak{k}}$ a Casimir operator for K. We know

$$V_{\text{fin}} = \bigoplus_{\nu} W_\nu \otimes V_\nu$$

as K-isotypical decomposition. We know $\dim(W_\nu) \leq C \cdot \dim(V_\nu) \sim O(|\nu|^d)$, where d is the number of positive roots. But $C_{\mathfrak{k}}$ acts on V_ν by a scalar $c_\nu \geq 0$ approximately $|\nu|^2$. Note $(1 + C_{\mathfrak{k}})^{-1} \in \text{End}_{\mathbb{C}}(V_{\text{fin}})$.

Lemma. *$(1 + C_{\mathfrak{k}})^{-N}$ is HS for large N*

Indeed, this follows from the convergence of

$$\sum_{\nu \in \Lambda_K^+} \frac{\dim(V_\nu)^2}{1 + |\nu|^{2N}}$$

Using HS times bounded is again HS, we write

$$f = (1 + C_{\mathfrak{k}})^{-N}((1 + C_{\mathfrak{k}})^{-N}(1 + C_{\mathfrak{k}})^{2N} f)$$

Note $(1 + C_{\mathfrak{k}})^{2N} f$ is bounded, $(1 + C_{\mathfrak{k}})^{-N}$ is Hilbert Schmidt, it follows f is trace class. $\square$

Recall

$$\text{Dist}(G) \coloneqq C_c^\infty(G)^*$$

and fix a Haar measure dg of G. We thus define $\chi_V \in \text{Dist}(G)$ via:

$$(\chi_V, f) = \text{Tr}(f dg, V)$$

Remark. *$\chi_V = \chi_{V'} \iff V_{\text{fin}} \simeq V'_{\text{fin}}$ as $(\mathfrak{g}, K)$-modules, because $K \times K$ finite functions are dense in $C_c^\infty(G)$.*

Theorem 11.6. *$\chi_V|_{G^{\text{rs}}}$ is an analytic function.*

We have two immediate properties of χ_V:

1. χ_V is an invariant (under conjugation) distributions.
2. Suppose V has central character μ'. We have a map $U(\mathfrak{g}) \to \text{Diff}(G)$ coming from the left translation action of G on itself. Note $L \in \text{Diff}(G)$ acts on $T \in \text{Dist}(G)$ via:

   $$(LT, \varphi) = (T, L^* \varphi)$$

   where $L \mapsto L^*$ is the canonical anti-involution. Now we compute for $z \in Z(U\mathfrak{g})$:

   $$(z \cdot \chi_V, \varphi) = (\chi_V, z^* \varphi) = \text{Tr}(z^* \varphi dg, V) = \mu'(z^*)\text{Tr}(\varphi dg, V) = (\mu'(z^*) \cdot \chi_V, \varphi)$$

   But we can find μ such that $\mu(z) = \mu'(z^*)$, hence: $z \cdot \chi_V = \mu(z) \cdot \chi_V$ for some character $\mu : Z(U\mathfrak{g}) \to \mathbb{C}$.

Hence to prove the above theorem, we only need to show:

Theorem 11.7. *Any invariant distribution Θ satisfying: $z \cdot \Theta = \mu(z) \cdot \Theta$ for some character μ is given by integration against a locally L_1 function. Moreover, this function is C^∞ on the regular semisimple locus.*

In the process of the proof we will show:

Theorem 11.8. *The space of functions on the regular semisimple locus that are invariant and satisfying $z \cdot f = \mu(z) \cdot f$ for a given character μ is finite dimensional.*

Corollary 11.9. *The set of irreducible $(\mathfrak{g}, K)$-modules with a given character $Z(\mathfrak{g})$ is finite.*

11.3 Proof of Regularity of Characters

Let G be a complex reductive group whose derived group is simply connected. Let $\lambda : \mathfrak{t} \to \mathbb{C}$ be a character, which integrates to $e^\lambda : T \to \mathbb{C}$. The key identity relating these: for $H \in \mathfrak{t}$ we have

$$e^\lambda(\exp(H)) = \exp(\lambda(H))$$

In particular, when view H as a differential operator, we can compute: for $t \in T$

$$H(e^\lambda)(t) = \frac{d}{ds}\Big|_{s=0} e^\lambda(\exp(sH)t) = \frac{d}{ds}\Big|_{s=0} \exp(s\lambda(H))e^\lambda(t) = \lambda(H)e^\lambda(t)$$

Let

$$T^{\mathrm{reg}} = T \cap G^{\mathrm{rs}} = \{t \in T \mid \alpha(t) \neq 0, \forall \alpha\}$$

Let $\Delta = e^\rho \cdot \prod_\alpha (1 - e^{-\alpha})$ be a function on T. Note Δ is W-anti-invariant. Note also that Δ is invertible on T^{reg}. Hence we can define an automorphism on $\mathrm{Diff}(T^{\mathrm{reg}})$ given by:

$$L \mapsto \Delta^{-1} \cdot L \cdot \Delta$$

Note this preserves W-invariant elements of $\mathrm{Diff}(T^{\mathrm{reg}})$.

Recall we have the normalized Harish-Chandra isomorphism $\varphi^n : Z(\mathfrak{g}) \to \mathrm{Sym}(\mathfrak{t})^W$.

Proposition 11.10. *The following diagram commutes:*

$$\begin{array}{ccccc}
Z(U\mathfrak{g}) & \longrightarrow & \mathrm{Diff}(G)^G & \longrightarrow & \mathrm{Diff}(G^{\mathrm{rs}})^G \\
\downarrow{\scriptstyle \varphi^n} & & & & \downarrow \\
\mathrm{Sym}(\mathfrak{t})^W & & & & \mathrm{Diff}(T^{\mathrm{reg}})^W \\
\downarrow & & & & \uparrow{\scriptstyle \mathrm{Ad}_{\Delta^{-1}}} \\
\mathrm{Diff}(T)^W & & \longrightarrow & & \mathrm{Diff}(T^{\mathrm{reg}})^W
\end{array}$$

Proof. One needs to check for every $f \in \mathcal{O}(T^{\mathrm{reg}})^W$ and every $z \in Z(U\mathfrak{g})$, the action of z on f via the two circuits coincide. Note we can take f to be the character of an irreducible representation of highest weight λ. The Weyl character formula says:

$$f = \Delta^{-1} \cdot \Big(\sum_{w \in W} (-1)^{\ell(w)} e^{w(\lambda+\rho)}\Big)$$

Ther upper circuit says:

$$z(f) = (\lambda, \varphi(z)) \cdot f = (\lambda + \rho, \varphi^n(z)) \cdot f$$

where φ is the ordinary HC map. The lower circuit says:

$$\begin{aligned} z(f) &= \mathrm{Ad}_{\Delta^{-1}}(\varphi^n(z))(f) \\ &= \Delta^{-1} \cdot \varphi^n(z) \cdot (\sum_{w \in W} (-1)^{\ell(w)} e^{w(\lambda+\rho)}) \\ &= \Delta^{-1} \cdot (\sum_{w \in W} (w(\lambda+\rho), \varphi^n(z))(-1)^{\ell(w)} e^{w(\lambda+\rho)}) \\ &= \Delta^{-1}(\lambda + \rho, \varphi^n(z)) \cdot \Delta \cdot f \\ &= (\lambda + \rho, \varphi^n(z)) \cdot f \end{aligned}$$

Hence the diagram indeed commutes. □

Now let Θ generates a $\mathcal{D}(G^{\mathrm{rs}}/\mathrm{Ad}_G)$-module M. Note M is a quotient of $\mathcal{D}/(z - \mu(z) : z \in Z(U\mathfrak{g}))$. To check analycitiy of Θ is equivalent to check that $M = \mathcal{D}/(z - \mu(z) : z \in Z(U\mathfrak{g}))$ is $\mathcal{O}$-coherent. Note we can rewrite the previous diagram as:

$$\begin{array}{ccccc} Z(U\mathfrak{g}) & \longrightarrow & \mathcal{D}(G^{\mathrm{rs}}/\mathrm{Ad}_G) & \longrightarrow & \mathcal{O}(T^*(G^{\mathrm{rs}}/\mathrm{Ad}_G)) \\ & & & & \downarrow \wr \\ \downarrow \varphi^n & & & & \mathcal{O}(T^*(T^{\mathrm{reg}}/W)) \\ & & & & \downarrow \text{locally} \\ \mathbb{C}[\mathfrak{t}^*]^W & & \longrightarrow & & \mathcal{O}(T^{\mathrm{reg}}/W) \otimes \mathbb{C}[\mathfrak{t}^*] \end{array}$$

embedding to the second component

up to a twist of Δ^{-1}

Therefore locally on $T^{\mathrm{reg}}/W \simeq G/\mathrm{Ad}_G$

$$\mathrm{gr}M \simeq (\mathcal{O} \otimes \mathbb{C}[\mathfrak{t}^*])/(\overline{z} : z \in Z(U\mathfrak{g}))$$

But twisting by Δ is an isomorphism, hence

$$\mathrm{gr}(M) \simeq (\mathcal{O} \otimes \mathbb{C}[\mathfrak{t}^*])/(\mathbb{C}[\mathfrak{t}^*]^W)$$

But $\mathbb{C}[\mathfrak{t}^*]$ is finite over $\mathbb{C}[\mathfrak{t}^*]^W$, this shows $\mathrm{gr}(M)$ is finitely generated over $\mathcal{O}$, hence M is $\mathcal{O}$-coherent.

11.4 Characters for $\mathbf{SL}_2$

For $\mathrm{SL}(2, \mathbb{R})$, we can compute $\chi_V(g)$ for g compact. Suppose

$$g = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \in K = \mathrm{SO}(2)$$

and set $z = e^{i\theta}$.

For spherical principal series $V = P_s^+$, note that as K-representation this is just $L^2(S^1)$, and:

$$V^{\text{fin}} = \bigoplus_{n \in 2\mathbb{Z}} \mathbb{C} \cdot e^{in\theta}$$

Now we can compute:

$$\chi_V(g) = \sum_{n \in 2\mathbb{Z}} \text{Tr}(g, \mathbb{C} \cdot e^{in\theta}) = \sum_{n \in 2\mathbb{Z}} z^n$$

this is 0 for $z \neq \pm 1$. So the character is $c \cdot (\delta_1 + \delta_{-1})$ for some constant c. (One can also see this by the fact that for $g \neq \pm 1$, it has no fixed points on $\mathbb{RP}^1$).

For (limits of) discrete series $V = L_m^+$ for $m > 0$, we have character $z^m + z^{m+2} + \cdots = \frac{z^m}{1-z^2}$ for $z \neq \pm 1$. Indeed, take a function $\varphi \in C^\infty(S^1)$, $\varphi(\pm 1) = 0$. On this, χ_V acts by a trace class operator, whose trace is

$$\int_{\lambda \in S^1} \varphi(z) \frac{z^m}{1 - z^{-2}} dz$$

by Fourier series theory. Similarly, we find that on L_{-m}^-, we have character $\frac{z^{-m}}{1-z^{-2}}$. This calculates the character value on noncompact elements.

Now if $g \in G^{\text{rs}}$ is noncompact, then $\chi_{L_m}(g) = \chi_{L_{-m}}(g)$. To see this, note that we have an automorphism of $\text{SL}_2(\mathbb{R})$ given by conjugation by $g = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \in \text{GL}_2(\mathbb{R})$. This automorphism changes L_m^+ to L_{-m}^-, but fixes the split semisimple elements $g = \begin{pmatrix} a & \\ & a^{-1} \end{pmatrix}$. Hence the result follows.

Thus we will know all values if we computed characters on principal series:

$$\chi_{L_m^+} + \chi_{L_{-m}^-} + \chi_{V_{m-2}} = \chi_{\text{PS}}$$

which will be treated systematically in the next section.

11.5 Characters for Principal Series

Setup

For simplicity let's assume G be a split real reductive group, and fix (T, B) split maximal torus and Borel. Given a character $\nu : T \to C^\times$, it determines a G-equivariant line bundle $\mathcal{L}_\nu = G \times^{B,\nu} \mathbb{C}$, whose global section is a sphereical principal series representaiton via parabolic induction:

$$V_\nu = \{f : G \to \mathbb{C} \mid f(gb) = \nu(b)^{-1} f(g)\}$$

We wish to compute its character.

A Digression on Volume Form

Let's discuss some general methods of producing volume forms. Let $\pi : P \to X$ be a smooth fibre budnle, there is a compact pushforward map given by integration on fibres:

$$\pi_! : \mathcal{M}_{c,\pi}(P) \to \mathcal{M}(X)$$

where $\mathcal{M}_{c,\pi}(P)$ denotes measures that are compactly supported on fibres.

The relative tangent sequence:

$$0 \to T_{P/X} \to T_P \to \pi^* T_X \to 0$$

gives $\det T_P^* \simeq \det T_{P/X}^* \otimes \pi^* \det T_X^*$

Fix a volume form $\alpha \in \Gamma(X, \det T_X^*)$, then every top form $\beta \in \Gamma(P, \det T_P^*)$ uniquely decomposes as $\beta = \alpha \otimes \gamma$ for $\gamma \in \Gamma(P, \det T_{P/X}^*)$.

In this case, given a function $f \in C_{c,\pi}(P)$ compactly supported along fibres, the measure $\pi_! f\beta$ is given by the top form whose value at $x \in X$ is

$$(\int_{P_x} f d\gamma|_{P_x})\alpha_x$$

Application to Flag Variety

The cotangent bundle T^*G/B is G-equivariant, so is determined by the representation $(\mathfrak{g}/\mathfrak{b})^* = \mathfrak{u}$ of B. Taking top exterior power we see the canonical bundle is $\mathcal{L}_{\delta^{-1}}$, where $\delta : \mathfrak{b}/[\mathfrak{b},\mathfrak{b}] = \mathfrak{t} \to \mathbb{C}$, sending $t \mapsto \prod_{\alpha>0} \alpha(t)$. Therefore, we have identifications:

$$V_{\delta^{-1}} = \Gamma(G/B, \det T_{G/B}^*) = \mathcal{M}(G/B)$$

We wish to specify a measure on G/B. Let's look at the tangent sequence at point g:

$$0 \to \mathrm{Ad}_g \mathfrak{b} \to \mathfrak{g} \to \mathfrak{g}/\mathrm{Ad}_g \mathfrak{b} \to 0$$

corresponding to cotangent sequence:

$$0 \to \mathrm{Ad}_g((\mathfrak{g}/\mathfrak{b})^*) \to \mathfrak{g}^* \to \mathrm{Ad}_g \mathfrak{b}^* \to 0$$

Hence we have a canonical isomorphism:

$$\det(\mathfrak{g}^*) \simeq \mathrm{Ad}_g(\det((\mathfrak{g}/\mathfrak{b})^*)) \otimes \mathrm{Ad}_g(\det(\mathfrak{b}^*))$$

Let's fix a left-invariant Haar measure μ_G and μ_B on G and B, they corresponds $\omega \in \det \mathfrak{g}^*$ and $\omega_1 \in \det \mathfrak{b}^*$. Hence there exists unique $\omega_2 \in \mathfrak{g}/\mathfrak{b}$ such that $\omega = \omega_1 \otimes \omega_2$. Suppose then ω_2 corresponds to measure $\mu_{G/B}$.

This gives us a way to compute the integration on G. Indeed we have:

$$\mathcal{M}_c(G) \xrightarrow{\pi_!} \mathcal{M}_c(G/B) \xrightarrow{\int_{G/B}} \mathbb{C}$$

We can explicitly compute $\pi_!(f\mu_G)$:

$$\pi_!(f\mu_G) = (\int_{b\in B} f(gb) d\mu_B) d\mu_{G/B}$$

Setup of Integral Kernel

Recall the goal is the following: we have a measure $f\mu_G \in \mathcal{M}_c(G)$, this can be viewed as a trace class operator in $\mathrm{Aut}(V_\nu)$, we wish to compute its trace.

The general method of integral kernel works as following. Suppose I have an operator $T : L^2(X) \to L^2(Y)$. The integral kernel its a function $K(x, y)$ on $X \times Y$ such that

$$(Tf)(y) = \int_X K(x, y)f(x)dx$$

In the discrete case this is nothing but writing out the matrix for the linear operator. Now if $X = Y$, and T is trace class, then we have:

$$\mathrm{Tr}(T) = \int_X K(x, x)dx$$

But here our Hilbert space is $\Gamma(G/B, \mathcal{L}_\nu)$, so we need a more general setup.

Indeed, a more geometric way to view integral kernel is the following. Consider the diagram

$$\begin{array}{ccccc} & & X \times Y & & \\ & \overset{\mathrm{pr}_X}{\swarrow} & & \overset{\mathrm{pr}_Y}{\searrow} & \\ X & & & & Y \end{array}$$

The integral kernel for a map $T : \mathcal{O}(X) \to \mathcal{O}(Y)$ is a section $K \in \mathcal{M}_c(X) \boxtimes \mathcal{O}(Y)$ such that

$$Tf = \mathrm{pr}_{Y*}(K \cdot \mathrm{pr}_X^* f)$$

Hence more generally, the integral kernel for a map $T : \Gamma(X, \mathcal{L}_X) \to \Gamma(Y, \mathcal{L}_Y)$ is a section $K \in \Gamma(X \times Y, (\mathcal{L}_X^{-1} \otimes \Omega_X^{\mathrm{top}}) \boxtimes \mathcal{L}_Y)$ such that for $s \in \Gamma(X, \mathcal{L}_X)$:

$$Ts = \mathrm{pr}_{Y*}(K \otimes \mathrm{pr}_X^* s)$$

In particular, if we take $X = Y$, then the integral kernel is a section of the line bundle $(\mathcal{L}^{-1} \otimes \Omega^{\mathrm{top}}) \boxtimes \mathcal{L}$, which restricts to $\mathcal{O}_X$ on the diagonal. Hence we can do integration on the diagonal and computes the trace.

Compute the Integral Kernel

Recall we wish to compute the integral kernel for: $f\mu_G$. Let $\varphi \in V_\nu$, we wish to write

$$((f\mu_G) \cdot \varphi)(g) = \int_{h \in G} f(h)\varphi(h^{-1}g)d\mu_G = \int_{G/B} K(x, g)\varphi(x)d\mu_{G/B}$$

where K must have the transformation law:

$$K(xb, y) = \nu(b)K(x, y) = \nu(b)^{-1}K(x, yb)$$

This hints us to do the change of variable $z = h^{-1}g$, so that

$$\begin{aligned}\int_{h\in G} f(h)\varphi(h^{-1}g)d\mu_G &= \int_{z\in G} f(gz^{-1})\varphi(z)d\mu_G \\ &= \int_{z\in G/B}(\int_B f(gb^{-1}z^{-1})\varphi(zb)d\mu_B)d\mu_{G/B} \\ &= \int_{z\in G/B}(\int_B \nu(b)^{-1}f(gb^{-1}z^{-1})d\mu_B)\varphi(z)d\mu_{G/B}\end{aligned}$$

Hence:

$$K(x,g) = \int_B \nu(b)^{-1}f(gb^{-1}x^{-1})d\mu_B \implies K(g,g) = \int_B \nu(b)^{-1}f(gb^{-1}g^{-1})d\mu_B$$

Namely:

$$\mathrm{Tr}(f\mu_G, V_\nu) = \int_{g\in G/B}(\int_{b\in B} \nu(b)^{-1}f(gb^{-1}g^{-1})d\mu_B)d\mu_{G/B}$$

Grothendieck Springer Resolution

In this section we denote B for any Borel (rather than a fixed Borel). Consider the following diagram:

$$\begin{array}{ccc} & \widetilde{G} = \{(g,B) \mid g \in B\} & \\ & \swarrow \quad \downarrow \pi & \\ T & & G \end{array}$$

Over $\mathbb{C}$, generically (on regular semisimple elements) it's $|W|$: 1. Indeed, for $\mathrm{SL}(n,\mathbb{C})$, this correspondist to ordering of eigenvalues. In gereal, given $g \in T$, we require $g \in B$ implies $T \subset B$. Hence there are $|W|$ choices. Since we're working with a split real group, the same thing happens over $\mathbb{R}$.

Note that we have a fibre bundle:

$$\widetilde{G} \to G/B, \quad (g,B) \mapsto B$$

Fix g, B, we have a natural map

$$B \to \widetilde{G} \to G/B$$

Look at tangent space at (g,B) we have a short exact sequence:

$$0 \to T_gB \to T_{(g,B)}\widetilde{G} \to T_BG/B \to 0$$

Let's compare this with the tangent sequence from the fibre bundle $G \to G/B$ at g:

$$0 \to T_gB \to T_gG \to T_BG/B \to 0$$

Put them together we get:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & T_gB & \longrightarrow & T_{(g,B)}\widetilde{G} & \longrightarrow & T_BG/B \simeq \mathfrak{g}/\mathfrak{b} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow & & \downarrow{\scriptstyle \mathrm{Ad}_{g^{-1}}-1} & & \\ 0 & \longrightarrow & T_gB & \longrightarrow & T_g(G) & \longrightarrow & T_BG/B \simeq \mathfrak{g}/\mathfrak{b} & \longrightarrow & 0 \end{array}$$

Therefore we see $\widetilde{G}$ has trivial canonical bundle. Let $h \in G$ be such that $\mathrm{Ad}_h(\mathfrak{b}_0) = \mathfrak{b}$ (for a preferred Borel $\mathfrak{b}_0$), then the top form at point (g, B) is

$$\det(\mathrm{Ad}_{g^{-1}} - 1 \mid \mathfrak{g}/\mathfrak{b}) \cdot \mathrm{Ad}_h(\omega_1) \otimes \mathrm{Ad}_h(\omega_2)$$

This corresponds to the pullback measure of μ_G on $\widetilde{G}$. We still denote it as μ_G.

Let us denote $\mathcal{B}$ for the flag variety (no preferred choice of Borel). Again we wish to decompose the top form along the fibre bundle $\widetilde{G} \to \mathcal{B}$. Recall we have a preferred top form corresponding to $\mu_{\mathcal{B}}$, whose value at B is just $\mathrm{Ad}_h(\omega_2)$. So the volume form along the fibre B is just:

$$\det(\mathrm{Ad}_{g^{-1}} - 1 \mid \mathfrak{g}/\mathfrak{b}) \cdot \mathrm{Ad}_h(\omega_1)$$

Hence for any function $\Phi \in C_{c,\pi}(\widetilde{G})$, we have:

$$\begin{aligned} \int_{(g,hB_0)\in\widetilde{G}} \Phi(g,h) d\mu_G &= \int_{h\in G/B_0} \Big(\int_{g\in hBh^{-1}} \Phi(g,h) \det(\mathrm{Ad}_{g^{-1}} - 1 \mid \mathfrak{g}/\mathrm{Ad}_h(\mathfrak{b}_0)) \mathrm{Ad}_h(\omega_1)) d\mu_{\mathcal{B}} \\ &= \int_{h\in G/B_0} \Big(\int_{b\in B} \Phi(hbh^{-1}, h) \det(\mathrm{Ad}_{b^{-1}} - 1 \mid \mathfrak{g}/\mathfrak{b}_0) d\mu_{B_0} \Big) d\mu_{G/B_0} \end{aligned}$$

Putting Everything Together

Drop the subscript 0 in the above section, we have formula:

$$\int_{(h,gB)\in\widetilde{G}} \Phi(h,g) d\mu_G = \int_{g\in G/B} \Big(\int_{b\in B} \Phi(gbg^{-1}, g) \det(\mathrm{Ad}_{b^{-1}} - 1 \mid \mathfrak{g}/\mathfrak{b}) d\mu_B \Big) d\mu_{G/B}$$

Therefore if we take:

$$\Phi(h,g) = \nu(g^{-1}h^{-1}g) f(h^{-1}) \det(\mathrm{Ad}_{g^{-1}h^{-1}g} - 1 \mid \mathfrak{g}/\mathfrak{b})^{-1}$$

then we have:

$$\begin{aligned} \mathrm{Tr}(f\mu_G, V_\nu) &= \int_{(h,gB)\in\widetilde{G}} \nu(g^{-1}h^{-1}g) f(h^{-1}) \det(\mathrm{Ad}_{g^{-1}h^{-1}g} - 1 \mid \mathfrak{g}/\mathfrak{b})^{-1} d\mu_G \\ &= \int_{(h,gB)\in\widetilde{G}} \nu(g^{-1}hg) f(h) \det(\mathrm{Ad}_{g^{-1}hg} - 1 \mid \mathfrak{g}/\mathfrak{b})^{-1} d\mu_G \\ &= \int_{h\in G} \sum_{g^{-1}hg\in B} \nu(g^{-1}hg) \det(\mathrm{Ad}_{g^{-1}hg-1} - 1 \mid \mathfrak{g}/\mathfrak{b})^{-1} f(h) d\mu_G \end{aligned}$$

Therefore, if we take f to be supported on the maximal torus T, and take $t \in T^{\mathrm{reg}}$, we see:

$$\chi_{V_\nu}(t) = \sum_{w\in W} \frac{\nu(w(t))}{\det(\mathrm{Ad}_{w(t)} - 1 \mid \mathfrak{g}/\mathfrak{b})} = \sum_{w\in W} \frac{e^{w\cdot\nu}}{\prod_{\alpha>0} |1 - e^{-w(\alpha)}|}(t)$$

The trick is the following, recall $\rho = \sum_{\alpha>0} \frac{1}{2}\alpha$, so that:

$$\prod_{\alpha>0}|1-e^{-w(\alpha)}| = \prod_{\alpha>0, w(\alpha)<0} e^{\alpha} \prod_{\alpha>0}|1-e^{-\alpha}| = e^{\rho - w\rho}\prod_{\alpha>0}|1-e^{-\alpha}| = e^{-w\rho}\prod_{\alpha>0}|e^{\alpha/2}-e^{-\alpha/2}|$$

Hence:

$$\chi_{V_\nu} = \sum_{w\in W} \frac{e^{w(\nu-\rho)}}{\prod_{\alpha>0}|e^{\alpha/2}-e^{-\alpha/2}|}$$

In fact, in light of the notion of unitary induction (and the above formula), it is very common to denote V_ν as the character induced from the character $\nu \otimes \rho$. Under this convention:

Proposition 11.11. *Let V_ν be a spherical principal series representation, then $\chi_V(g) = 0$ if $g \in G^{\mathrm{rs}}$ is not diagonalizable. Otherwise if $g \in T$ for some maximal split torus T:*

$$\chi_V(g) = \sum_{w\in W} \frac{e^{w(\nu)}(g)}{\prod_{\alpha\in\Pi^+}|e^{\alpha/2}(g)-e^{-\alpha/2}(g)|}$$

Remark. *$V_\nu = \Gamma(\mathcal{L}_\nu \otimes \text{half densities})$. If ν is unitary, $|\nu(g)| = 1$ for all $g \in T$, then V_ν is unitary. Indeed:*

$$\langle f_1, f_2\rangle = \int_{x\in G/B} f_1(x)\overline{f_2}(x)$$

is an invariant inner product. Note $f_1\overline{f_2}$ is a measure on G/B.

More generally, consider $P \subset G$ parabolic subgroup, with Levi L. Let σ be a unitary representation of L, and $\rho_P = \frac{1}{2}\sum_{\alpha\in\Phi(\mathfrak{u})}\alpha$ then:

$$i_P^G(\sigma) \coloneqq \Gamma(\mathcal{F}_\sigma \otimes \text{half measures}) = \{f : G/U \to V_\sigma \mid f(xl) = e^{-\rho}(l)\sigma(l)^{-1}f(x)\}$$

In fact $i_P^G(\rho)$ is unitary if ρ is unitary. This is called **normalized unitary induction**.

Example. *For $G = \mathrm{SL}(2,\mathbb{R})$, and $g = \begin{pmatrix} a & 0 \\ 0 & a^{-1}\end{pmatrix}$, $\nu(g) = |a|^s$, we have:*

$$\chi_\nu(g) = \frac{|a|^s + |a|^{-s}}{|a-a^{-1}|}$$

12 A Study on Discrete Series

12.1 More Structure Theory for Real Groups

Let's first discuss some general setup. Let G be a real group, with maximal compact K. Let us fix a maximal split torus A: $(\mathbb{R}^\times)^r \to G$ with Lie algebra $\mathfrak{a}$. This is a maximal $\mathbb{R}$-diagonalizable subalgebra of $\mathfrak{g}$, and $\mathfrak{a} \subset \mathfrak{k}^\perp$. It's a fact that any two maximal $\mathbb{R}$-diagnolizable subalgebra of $\mathfrak{g}$ are conjugate.

Now set $\mathfrak{l} = \mathfrak{z}_\mathfrak{g}(\mathfrak{a})$ be the Lie algebra of a Levi subgroup L.

Proposition 12.1. $\mathfrak{l} \simeq \mathfrak{a} \oplus \mathfrak{l} \cap \mathfrak{k}$.

Corollary 12.2. *1. The map $A \times (K \cap L) \to L$ is the polar decomposition for L.*

2. The map $A \times (K \cap Z_L) \to Z_L$ is the polar decomposition for Z_L.

3. The group L is compact modulo its center.

Now let's study the adjoint action of $\mathfrak{a}$ on $\mathfrak{g}$:

$$\mathfrak{g} = \mathfrak{l} \bigoplus \oplus_\alpha \mathfrak{g}_\alpha$$

One can show that these α form a root system. One chosoe an positive system on this root system and set:

$$\mathfrak{p} \coloneqq \mathfrak{l} \bigoplus \oplus_{\alpha>0} \mathfrak{g}_\alpha$$

be the Lie algebra of a parabolic subgroup P of G. Now if we set $\mathfrak{u} = \oplus_{\alpha>0}\mathfrak{g}_\alpha$ be the nilpotent part, we can write $\mathfrak{p} = \mathfrak{l} \oplus \mathfrak{u}$.

Fix a parabolic $\mathfrak{p}$ with Levi $\mathfrak{l}$, with $\mathfrak{p}$ minimal real parabolic. Then the semisimple part of $\mathfrak{l}$ is compact (noncompact groups contains a split torus). Now write $\mathfrak{p} = \mathfrak{l} \oplus \mathfrak{u}$. Denote $\mathfrak{a}_+ = \{h \in \mathfrak{a} \mid \alpha(h) > 0, \forall \alpha > 0\}$.

Now we fix a Cartan decomposition:

$$G = KA^-K$$

where $A^- = \{\exp(-x) \mid x \in \mathfrak{a}_+\}$

Example. *Let $G = \mathrm{SL}(n, \mathbb{R})$, $K = \mathrm{SO}(n, \mathbb{R})$. Then*

$$G/K = \{\text{positive definite quadratic forms}\}$$

Then K acts on the space of positive definite quadratic forms. From linear algebra we see:

$$K\backslash G/K \simeq A^-$$

For $\mathrm{SL}(2, \mathbb{R})$, we can see this geometrically using the action on $\mathbb{H} \simeq \mathrm{SL}(2, \mathbb{R})/\mathrm{SO}(2, \mathbb{R})$ preserving the hyperbolic metric.

Remark. *Recall we have the* **unique** *polar decomposition $G = PK$, where $P = \exp(\mathfrak{k}^\perp)$. To see the uniqueness, if θ is the Cartan involution, then $P = \{p \mid \theta(p) = p^{-1}\} = \{\sqrt{g\theta(g^{-1})}\}$. Note the later expression makes sense since $g\theta(g^{-1})$ has positive eigenvalues. Now For A a maximal split torus, one can show:*

$$\mathfrak{a}/W(\mathfrak{a}, K) \to \mathfrak{k}^\perp/K$$

is an isomorphism, where $W(\mathfrak{a}, K) = N_K(\mathfrak{a})/Z_K(\mathfrak{a})$. This in particular shows that:

$$K \cdot \mathfrak{a} \twoheadrightarrow \mathfrak{k}^\perp \implies KA^+K^{-1} = P$$

Then the Cartan decomposition follows.

Remark. *Can choose $G \subset \mathrm{SL}(n, \mathbb{C})$ compatible with Cartan involution. Then: $K = \mathrm{SU}(n) \cap G$, $P = P_{\mathrm{SL(n,C)}} \cap G$.*

12.2 Asymptotics of Matrix Coefficients for Discrete Series

Let's fix some notations. Choose a maximal split torus A, with $\mathrm{Lie}(A) = \mathfrak{a}$. Let $L = Z_G(A)$ be the Levi, and $\mathfrak{z}_{\mathfrak{g}}(\mathfrak{a}) \coloneqq \mathfrak{l} = \mathfrak{a} \oplus \mathfrak{m}$. Note that $\theta|_{\mathfrak{m}} = \mathrm{id}$. So $L = A \cdot M$. Now if P is parabolic with Levi L (minimal real parabolic), we have decomposition $P = MAN$ for N unipotent radical.

Remark. *Clearly G split iff $\dim \mathfrak{a} = \mathrm{rank}\ \mathfrak{g}$. In this case $\mathfrak{m} = 0$ and P is just the Borel. In fact, if P is the Borel $\iff$ $\mathfrak{a}$ contains a regular element $\iff$ G is quasi-split.*

Remark. *G is an inner twist of a unique quasi-split form.*

Example. *For example, if $G = \mathrm{SU}(n,n)$, $G_{\mathbb{C}} = \mathrm{SL}(2n,\mathbb{C})$. This is a quasi-split form.*

Definition 12.3. Let M be an irreducible $(\mathfrak{g}, K)$-module. **Exponents** (as elements in $\mathfrak{a}^*$) are the eigenvalues of A on $M/\mathfrak{u}^d M$ for some d.

Lemma 12.4. *The $(\mathfrak{g}, K)$-module M is finitely generated over $\mathfrak{u}$. So $M/\mathfrak{u}^d M$ is finite dimensional.*

Proof. Everything is complexified. Look at $\mathrm{gr}(M) \in \mathrm{Coh}^K(\mathfrak{k}^\perp \cap \mathcal{N})$. We can choose such a filtration to be compatible with PBW filtration and K-invariant, moreover it's a good filtration (namely $\mathrm{gr}(M)$ is finitely generated over $\mathbb{C}[\mathfrak{g}]$). Need to see the map:

$$\mathfrak{k}^\perp \cap \mathcal{N} \xrightarrow{\mathrm{pr}} \mathfrak{g}/\mathfrak{p}$$

is finite. Since $\mathfrak{k}^\perp \cap \mathcal{N}$ is conical, it's enough to see $\mathrm{pr}^{-1}(0) = 0$. Hence need to study $\mathcal{N} \cap \mathfrak{k}^\perp \cap \mathfrak{p}$.

Note $\mathfrak{a} \subset \mathfrak{k}^\perp$, so $\theta|_{\mathfrak{a}} = -1$. Hence $\theta : \mathfrak{p} \to \mathfrak{p}^-$ is the opposite parabolic. Now:

$$\mathfrak{p} \cap \mathfrak{k}^\perp \subset \mathfrak{p} \cap \theta(\mathfrak{p}) = \mathfrak{l} = \mathfrak{a} \oplus \mathfrak{m}$$

It follows $\mathfrak{p} \cap \mathfrak{k}^\perp = \mathfrak{a}$. □

Now let V be an admissible representation, with $M = V_{\mathrm{fin}}$. The goal is to find a criterion for detecting whether V is discrete series. To do this, let's fix a decomposition:

$$\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{m} \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$$

We set $\rho = \frac{1}{2}\sum_{\alpha \in \Phi^+} \alpha \in \mathfrak{a}^*$. Namely, for $a \in A$, we have $e^{2\rho}(a) = \det(\mathrm{Ad}(a) \mid \mathfrak{u})$.

Theorem 12.5. *A representation V is discrete series iff all exponents λ satisfies $\lambda > \rho$.*

Remark. *We say $\lambda_1, \lambda_2 \in \mathfrak{a}^*$ satisfies $\lambda_1 \geq \lambda_2$ iff $\lambda_1 - \lambda_2$ is a sum of positive roots.*

Example. *Let $G = \mathrm{SL}(2,\mathbb{R})$ and $V = L_m^+$. We wish to check that the discrete series L_m^+ satisfy the growth condition. Recall we pick*

$$h = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, e = \frac{1}{2}\begin{pmatrix} i & 1 \\ 1 & -i \end{pmatrix}, f = \frac{1}{2}\begin{pmatrix} -i & 1 \\ 1 & i \end{pmatrix}$$

Then we can realize:

$$L_m^+ = U(\mathfrak{g}_{\mathbb{C}}) \otimes_{U(\mathfrak{b})} \mathbb{C}_m, \text{where } \mathfrak{b} = \langle h, f \rangle \subset \mathfrak{sl}(2, \mathbb{C}) = \mathfrak{g}_{\mathbb{C}}$$

Hence a generator for $\mathfrak{a} = \mathbb{R} \cdot a, a = -2ie - h$, $\mathfrak{u} = \mathbb{R} \cdot (e + f - ih)$. *Now that* $V/\mathfrak{u}V$ *is generated by* v_m *of* h*-weight* m, *and* a *acts by*

$$av_m = (-2ie - h)v_m = (-2i(ih - f) - h)v_m = hv_m = mv_m$$

We see we require $m > 1 = \rho$ *to be the discrete series.*

Now let's prove the theorem. The proof splits into two steps. First we need to estimate matrix coefficient by the exponents. Let's take v, v^* be K-finite (co)vector. We wish to approximate $m_{v^*,v}(g)$, where we will take $g = \exp(a)$ for a regular anti-dominant element $a \in \mathfrak{a}$. We will prove:

Theorem 12.6. *For every real number* R *there exists a finite collection of exponents* λ_n *of* $\mathfrak{a}$ *and polynomial function* p_n *on* $\mathfrak{a}$ *such that the function*

$$m_{v^*,v}(\exp(a)) - \sum_n \exp(\lambda_n(a)) \cdot p_n(a)$$

tends to zero faster than $\exp(R \cdot \max_\alpha(\alpha(a)))$.

Proof. Let's assume V is a Hilbert representation of G. Since the operators $T_{\exp(a)}$ are bounded, we can find a real number R' such that

$$\|T_{\exp(a)}\| < \exp(-R' \cdot \rho(a))$$

for $a \in \mathfrak{a}$ anti-dominant. Let m be such that $m \cdot \max_\alpha(\alpha(a)) < \rho(a)$ and choose integer k such that $k > R + m \cdot R'$. Let's consider the quotient:

$$\pi : M \to M/\mathfrak{u}^k M$$

This is a finite dimensional representation of Lie algebra $\mathfrak{a}$, so it integrates to a representation of A.

Let $v_1, \cdots, v_n$ be vectors in M that project to a basis for $M/\mathfrak{u}^k M$. Then there exists polynomial functions p_n on $\mathfrak{a}$ such that the action of A on $\pi(v)$ is given by

$$\exp(a) \cdot \pi(v) = \sum_n \exp(\lambda_n(a)) \cdot p_n(a) \cdot \pi(v_n)$$

We will show that:

$$m_{v^*,v}(\exp(a)) - \sum_n \exp(\lambda_n(a)) \cdot p_n(a) \cdot \langle v^*, v_n \rangle \tag{1}$$

decreases faster than $\exp(R \cdot \max_\alpha \alpha(a))$. We need the following easy lemma:

Lemma. *The natural map* $i : M/\mathfrak{u}^k M \to V_{\text{sm}}/\mathfrak{u}^k V_{\text{sm}}$ *is* A*-equivariant.*

From this lemma, we see that

$$\exp(a) \cdot v = \sum_n \exp(\lambda_n(a)) \cdot p_n(a) \cdot v_n + v'$$

for some $v' \in \mathfrak{u}^k V_{\mathrm{sm}}$. Therefore 1 becomes: $m_{v^*,v'}$. But we can write v' as a sum of vectors of form $\xi_1 \cdots \xi_k \cdot v''$ with $\xi_i \in \mathfrak{u}_{\alpha_i}$ and $v'' \in V_{\mathrm{sm}}$. Without loss we may assume v' is a monomial, it follows that:

$$m_{v^*,v'}(\exp(a)) = \prod_i \exp(\alpha_i(a)) \cdot m_{v^*,v''}(\exp(a))$$

But by assumption we know the second term increases slower than $\exp(-R' \cdot \rho(a))$ hence slower than $\exp(-mR' \cdot \max_\alpha(\alpha(a)))$, while the first term decreases faster than $\exp(k \cdot \max_\alpha(\alpha(a)))$. □

Let $\lambda \in \mathfrak{a}$. We say another character $\lambda' \geq \lambda$ iff:

$$\Re(\lambda') - \lambda \in \mathrm{Span}_{\mathbb{R}_{\geq 0}}(\text{positive roots})$$

Proposition 12.7. *Let V be an admissible representation, and let M be the $(\mathfrak{g}, K)$-module of K-finite vectors. Given $\lambda \in \mathfrak{a}^*$, TFAE:*

1. *All λ_n appears in the above theorem are $\geq \lambda$*
2. *All eigenvalues of $M/\mathfrak{u}M$ are $\geq \lambda$*
3. *All exponents are $\geq \lambda$*

We skip the proof of the above proposition. We are now ready to prove the original theorem. For simplicity, let's do the case for $\mathrm{SL}_2(\mathbb{R})$.

Proof. Consider the map

$$K \times \exp(t \cdot a \mid t > R) \times K \to G$$

This is an open embedding with compact complement. Let $f(g)$ be a matrix coefficient of a $(\mathfrak{g}, K)$-module. We need to study the integrability of $|f(g)|^2$ times the pullback of the invariant top form under this map. Since f is K-finite on both sides, this question of integrability only depends on the further restriction of $f(g)$ to $g = \exp(t \cdot a)$ with $t > R$. The result follows by the next geometric assertion. □

Lemma 12.8. *The pullback of the invariant top form on G to $\mathbb{R}$ along the exponential map $t \mapsto \exp(t \cdot a)$ behaves like $\exp(-2t \cdot \rho(a))$ times a function that tends to 1 at $t \to \infty$.*

Proof. We identify the tangent space to G at the point $\exp(t \cdot a)$ with $\mathfrak{g}$ via right translation. Therefore we need to compute the differential:

$$\mathfrak{k} \oplus \mathfrak{a} \oplus \mathfrak{k} \to \mathfrak{g}$$

at point $t \cdot a$. Then it's easy to see this map is given by:

$$(x, y, z) \mapsto x + y + \mathrm{Ad}(\exp(-t \cdot a))(z)$$

We wish to compute its determinant. Let's now complexify everything. Now if θ is the Cartan involution on G, then if we write $\mathfrak{g} = \mathfrak{u}^- \oplus \mathfrak{a} \oplus \mathfrak{u}^+$, we must have

$$\mathfrak{k} = \{x + \theta(x) \mid x \in \mathfrak{u}^+\}$$

But then note that for $u \in \mathfrak{u}^-$

$$\mathrm{Ad}(\exp(-t \cdot a))(u) \to 0$$

as $a \to \infty$. Hence the determinant is approximately:

$$\det \mathrm{Ad}(\exp(-t \cdot a))|_{\mathfrak{u}^+} = \exp(-2t \cdot \rho(a))$$

as required. □

12.3 Langlands Classification

In this section, we list several results without proofs.

Definition 12.9. A representation is *tempered* if matrix coefficents $m_{v^*,v}$ lie in $L^{2+\varepsilon}$ for all $\varepsilon > 0$

Proposition 12.10. *A representation V is tempered iff all exponents λ satisifes $\lambda \geq \rho$.*

Proposition 12.11. *Unitary induction sends tempered to tempered. Every tempered representation is a summand in a unitary induction of discrete series.*

One can classify all admissible representation (modulo infinitesimal equivalence) in terms of tempered representations. This is called Langlands Classification.

Theorem 12.12 (Langlands Classification)**.** *For every admissible irreducible representation V, there exists a unique up to conjugacy of the following data:*

1. *P a parabolic in G*
2. *A tempered representation ρ of L*
3. *A (real-valued) character $\lambda : L \to \mathbb{R}_{>0}$, which is dominant for L.*

such that V is an irreducible quotient of $i_P^G(\rho \otimes e^\lambda)$. This is called the **Langlands quotient**.

Remark. *In the theorem used a condition on eigenvalues of $\mathfrak{a}$ on $M/\mathfrak{u}^n M$. But it's actually enough to take $n = 1$. Indeed a weight $\mathfrak{u}^i M/\mathfrak{u}^{i+1} M$ must have form $\lambda = \alpha + \lambda'$ for α a root of A in $\mathfrak{u}$, λ' a weight in $\mathfrak{u}^{i-1} M/\mathfrak{u}^i M$. We can see this from the action map:*

$$\mathfrak{u} \otimes \mathrm{gr}_i M \to \mathrm{gr}_{i+1} M$$

The module $r_P^G = M/\mathfrak{u}M$ is a $(\mathfrak{l}, K \cap L)$ Harich-Chandra module. One can show that r_P^G is left adjoint to parabolic induction.

Theorem 12.13 (Casselman Submodule Theorem)**.** *Every irreducible $(\mathfrak{g}, K)$-module can be realized as a submodule of some $i_{P_0}^G(\mathcal{L})$ for some $(\mathfrak{m}, K_M)$-module $\mathcal{L}$.*

Let Λ be the weight lattice, $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} = \mathfrak{t}_{\mathbb{R}}^*$. There are two cones inside $\mathfrak{t}_{\mathbb{R}}^*$:

- $\mathfrak{t}_{\mathbb{R}}^{*+} \coloneqq \Lambda^+ \otimes_{\mathbb{Z}} \mathbb{R}$, for Λ^+ dominant weights
- $\mathfrak{t}_{\mathbb{R}}^{*-} \coloneqq R^- \otimes_{\mathbb{Z}} \mathbb{R}$ for $R = \sum_{n_i \leq 0} n_i \alpha_i$, α_i positive roots.

Lemma 12.14 (Langlands Lemma)**.** *Every $\lambda \in \mathfrak{t}_{\mathbb{R}}^*$ can be written uniquely as $\lambda_1 + \lambda_2$ for $\lambda_1 \in \mathfrak{t}_{\mathbb{R}}^{*+}$, $\lambda_2 \in \mathfrak{t}_{\mathbb{R}}^{*-}$ and $(\lambda_1, \lambda_2) = 0$.*

Proof. (Of Langlands Classification for split groups, sketched). Consider all exponents of M, pick the minimal one in the partial order. Let $\Re\lambda = \lambda_1 - \lambda_2$. Pick P maximal such that λ_1 is a character of L, namely:

$$\mathfrak{l} = \mathfrak{a} \oplus \bigoplus_{(\alpha, \lambda_1) = 0} \mathfrak{g}_\alpha$$

Claim: P, λ_1 satisfies the claim in the theorem.

To recover the tempered representation of L, we take $r_P^G(M)_{\lambda_1}$, the part where the center of L acts via λ_1. Check this is $e^{\lambda_1} \otimes$ some tempered representation of L. □

13 Localization Theory

13.1 $\mathcal{D}$-modules

Let $\mathfrak{g}$ be a complex reductive Lie algebra. A $\mathfrak{g}$-module with a fixed infinitesimal character is a module of $U_\lambda = U\mathfrak{g} \otimes_{Z(U\mathfrak{g})} \lambda$ for some $\lambda \in \mathfrak{t}^*/W$. Then $\mathrm{gr}(U_\lambda) = \mathbb{C}[\mathcal{N}]$. Note that $\mathcal{N} \subset \mathfrak{g}^*$ is a Poisson variety.

Given $\varphi, \psi \in \mathbb{C}[\mathcal{N}]$ of degree d_1, d_2, we can pick lift $\widetilde{\varphi}, \widetilde{\psi} \in U_\lambda$, and:

$$\{\varphi, \psi\} = [\widetilde{\varphi}, \widetilde{\psi}] (\mathrm{mod}\ U_\lambda^{\leq d_1 + d_2 - 1}) \in \mathbb{C}[\mathcal{N}]_{d_1 + d_2 - 1}$$

This is saying: U_λ is a quantization of $(\mathbb{C}[\mathcal{N}], \{-,-\})$. Note also that $\mathcal{N}$ is singular, with Poisson leaves of various dimensions.

Consider the Springer resolution:

$$\pi : T^*(G/B) = \widetilde{\mathcal{N}} \to \mathcal{N}$$

This is a resolution of singularity: it's proper and isomorphism on the regular nilpotent orbit. Moreover $\widetilde{\mathcal{N}}$ is symplectic, and π is a Poisson map.

The symplectic variety $\widetilde{\mathcal{N}}$, or more generally, $X = T^*M$ for M smooth algebraic variety, it has a specific quantization: the sheaf of algebraic differential operators on M, namely the sheaf $\mathcal{D}_M$. We wish to study $\mathcal{D}$-modules.

Let X be a smooth algebraic variety, and let M be a $\mathcal{D}_X$-module. Recall we can define the notion of singular support of M, which is well-defined as a closed subset of T^*X. The most important class of $\mathcal{D}$-modules is holonomic ones.

Definition 13.1. A $\mathcal{D}_X$-module M is holonomic if $\dim(\mathrm{SS}(M)) = \dim X$.

Theorem 13.2 (Classification of Irreducible Holonomic Modules)**.** *Irreducible holonomic $\mathcal{D}$-modules on X are in bijection with the data $(Z, \mathcal{E}, \nabla)$, for Z is a smooth locally closed subvariety, and $(\mathcal{E}, \nabla)$ a vector bundle with flat connection on Z, modulo the equivalence relation:*

$$(Y, \mathcal{E}) \sim (Y', \mathcal{E}|_{Y'})$$

if $Y' \subset Y$ is open dense.

Consider a variety X together with an action of an affine algebraic group G: $\rho : G \times X \to X$. Recall the definition of G-equivariant quasicoherent sheaves:

Definition 13.3. A G-equivariant quasicoherent sheaf on X is a quasicoherent sheaf $\mathcal{F}$ on X with an isomorphism:

$$\varphi : \rho^* \mathcal{F} \to \mathrm{pr}_2^* \mathcal{F}$$

satisfying the simplicial cocycle condition.

Definition 13.4. A *weakly G-equivariant $\mathcal{D}$-module* on X is a $\mathcal{D}_X$-module $\mathcal{F}$ with a G-equivariant structure φ which is $\mathcal{D}_X$-linear.

Let $\mathcal{F}$ be a weakly G-equivariant $\mathcal{D}_X$-module. Then $\mathcal{F}$ carries two different actions of the Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$.

The first one comes from the $\mathcal{D}$-module structure.:

$$\mathfrak{g} \to \mathrm{Vect}(X) \to \mathcal{D}(X)$$

Call this action $a_{\mathcal{D}\text{-mod}}$. Note this is independent of the equivariant structure of the sheaf.

On the other hand, note that G acts on the space $G \times X$ by left translation. This induces:

$$\mathfrak{g} \to \mathrm{Vect}(G \times X) \subset \mathcal{D}(G \times X)$$

Now $\mathcal{D}(G \times X)$ acts on $\mathcal{O}_G \boxtimes \mathcal{F} = \mathrm{pr}^* \mathcal{F} \simeq \rho^* \mathcal{F}$, via φ. Restricting to $1 \times X$, this gives another $\mathfrak{g}$-action on $\mathcal{F}$. Call this action a_{equiv}.

Definition 13.5. A G-equivariant $\mathcal{D}_X$-module is a weakly equivariant $\mathcal{D}_X$-module such that $a_{\mathcal{D}\text{-mod}} = a_{\mathrm{equiv}}$.

This is precisely saying the structure isomorphism φ is $\mathcal{D}_{G \times X}$-linear. Indeed:

$$\varphi : \rho^* \mathcal{F} \to \mathcal{O}_G \boxtimes \mathcal{F}$$

For $f \in \mathcal{F}(X)$, viewed as a function on $G \times X$ with constant value along the G-direction, we see that φ being $\mathcal{D}_G$-linear is saying that for all $z \in \mathfrak{g}$:

$$\varphi(a_{\mathcal{D}\text{-mod}}(z)(f)) = \varphi(z \cdot f) = z \cdot \varphi(f)$$

since the $\mathfrak{g}$-action on $\rho^* \mathcal{F}$ is precisely the $\mathcal{D}$-module action. Hence its equivalent to:

$$a_{\mathcal{D}-\mathrm{mod}}(z)(f) = \varphi^{-1}(z \cdot \varphi(f)) \coloneqq a_{\mathrm{equiv}}(z)(f)$$

This proves the claim.

Remark. *Suppose G is connected, then every $\mathcal{D}$-module has at most one G-equivariant structure. This is because the $\mathfrak{g}$-action on $\mathcal{F}$ is determined by the Lie algebra homomorphism $\mathfrak{g} \to \mathcal{D}(X)$, and this can be integrated to a G-equivariant structure in at most one way. Moreover, in this case any $\mathcal{D}_X$-linear map between G-equivariant $\mathcal{D}_X$-module is automatically $\mathfrak{g}$-linear (since $\mathfrak{g}$ action factors through $\mathcal{D}(X)$), so is automatically G-linear. Hence: for connected G, the category of G-equivariant $\mathcal{D}_X$-module is a full subcategory of the category of $\mathcal{D}_X$-modules. So G-equivariance for $\mathcal{D}_X$-module is a property, not a structure!*

13.2 Beilinson-Bernstein Localization

Theorem 13.6 (Beilinson-Bernstein)**.** *The global section functor:*

$$\Gamma : \mathcal{D}\text{-mod}(G/B) \to U_0\text{-mod}$$

is a equivalence of categories.

We will prove this by proving:

1. Flag varieties are $\mathcal{D}$-affine.
2. $U_0 \simeq \Gamma(\mathcal{D}_{G/B})$. In fact, this map deforms $\pi^* : \mathbb{C}[\mathcal{N}] \to \mathbb{C}[\widetilde{\mathcal{N}}]$.

Example. *Consider $G = \mathrm{SL}(2)$, then $G/B = \mathbb{P}^1$. Let's consider the $\mathcal{D}$-module δ_0. It has a basis*

$$v = t^{-1}, t^{-2}, \cdots$$

Note that $t\partial_t$ acts on v with eigenvalue -1, and $t^2\partial_t, t^3\partial_t, \cdots$ kill it. We set $e = t^2\partial_t$, $h = 2t\partial_t, f = -\partial_t$. Therefore we get a vector killed by e, and $hv = -2v$. We're thus getting M_{-2}, an irreducible Verma. Note that M_{-2} indeed has central character 0.

Before the proof, let us first see some applications. Let $K = K_{\mathbb{C}}$.

Corollary 13.7 (of Localization Theorem)**.** *We have an equivalence of categories:*

$$(\mathfrak{g}, K)\text{-mod}_0 \simeq \mathcal{D}_K\text{-mod}(G/B)$$

This can be made even more explicit. Note $K_{\mathbb{C}}$ acts on G/B with finitely many orbits. We have the following general fact:

Proposition 13.8. *Let a group H act on X with finitely many orbits. Then irreducible H-equivariant $\mathcal{D}$-modules are classifeid by the pair $(\mathcal{O}, \psi)$, for $\mathcal{O}$ an orbit, and ψ an irreducible representation of $\pi_0(\mathrm{stab}_H(x))$ for some $x \in \mathcal{O}$.*

To understand representations with other central characters, we need to introduce the notion of *twisted differential operator* (TDO).

Let X be smooth over $\mathbb{C}$, and $L \to X$ a line bundle. We will define $\mathcal{D}_L$ a sheaf of differential operators acting on sections of L.

Let's choose a trivialization $L|_{V_i} \simeq \mathcal{O}$, $f_{ij} \in \mathcal{O}^\times(V_i \cap V_j)$ the transition function. Then one each V_i we identify $\mathcal{D}_L|_{V_i} \simeq \mathcal{D}|_{V_i}$, on each V_{ij} we have twist:

$$\alpha_{ij} : \mathcal{D}|_{V_{ij}} \to \mathcal{D}|_{V_{ij}}, \quad d \mapsto d + f_{ij}^{-1} df_{ij}$$

Note that this can be rewritten as:

$$\alpha_{ij}|_{\mathcal{O}} = \mathrm{id}$$

$$v \in \mathrm{Vect} \mapsto v + \langle v, \frac{df_{ij}}{f_{ij}} \rangle$$

This can be easily checked. Indeed, given a function $\varphi \in \mathcal{O}(V_j)$ and $v \in \mathrm{Vect}$, we must have:

$$v(f_{ij}^{-1}\varphi) = f_{ij}^{-1}\alpha_{ij}(v)(\varphi) \iff \alpha_{ij}(v)(\varphi) = d\varphi + f_{ij}^{-1}v(f_{ij}) \cdot \varphi$$

The second form can therefore easily be generalized by:

Exercise. If ω is a closed 1-form, then

$$f \mapsto f, \quad v \mapsto v + \langle v, \omega \rangle$$

is an automorphism of $\mathcal{D}$

Therefore, we can define TDO using any $h \in H^1(\Omega^1_{\mathrm{cl}})$. In particular, for a line bundle L, can define $\mathcal{D}_{L^c}$ given by cocycle:

$$v \mapsto v + c\langle v, \frac{df_{ij}}{f_{ij}} \rangle$$

Remark. *For a space X, there exists an equivalence of categories*

$$\mathcal{D}\text{-mod}(X) \simeq \mathcal{D}_L\text{-mod}(X)$$

by $M \mapsto M \otimes_{\mathcal{O}} L$. More generally, $\mathcal{D}_{L^c}$-mod$(X) \simeq \mathcal{D}_{L^{c+1}}$-mod$(X)$.

Let's find a concrete realization of the category $\mathcal{D}_{L^c}$-mod(X). Consider $\widetilde{X}$ the punctured total space for L, and $\pi : \widetilde{X} \to X$ a principal $\mathbb{G}_m$-bundle, so that $X = \widetilde{X}/\mathbb{C}^\times$. So

$$\mathcal{D}\text{-mod}(X) \simeq \mathbb{C}^\times\text{-equivariant } \mathcal{D}\text{-mod}(\widetilde{X})$$

with $M \mapsto \pi^* M$, and the inverse map $\widetilde{M} \mapsto (\pi_* \widetilde{M})^{\mathbb{G}_m} = (\pi_* \widetilde{M})_0$.

This motivates the following nice description:

$$\mathcal{D}_{L^c}\text{-mod}(X) \simeq \text{weakly } \mathbb{C}^\times\text{-equivariant } \mathcal{D}\text{-mod}(\widetilde{X}) \text{ with log monodromy } c$$

RHS means the action of $e = t\partial_t \in \mathrm{Lie}(\mathbb{C}^\times)$ on M disagrees by:

$$a_{\mathcal{D}\text{-mod}}(e) = a_{\mathrm{equiv}}(e) + c$$

The inverse map is $\widetilde{M} \to (\pi_* \widetilde{M})_c = (\pi_* \widetilde{M})^{\mathbb{G}_m}$, where the subscript c denotes the eigenvalue of e.

In particular let's set $X = G/B$, $\widetilde{X} = G/U \to G/B$ a principal T-bundle. We write:

$$\Lambda = X^*(T) \quad T = \mathbb{C}^\times \otimes \Lambda^\vee$$

So $\Lambda \otimes \mathbb{C} = \mathfrak{t}^*$. For $\lambda = c_1\lambda_1 + \cdots c_n\lambda_n \in \mathfrak{t}^*$, where $\lambda_1, \cdots, \lambda_n$ are simple roots, we get a TDO ring:

$$\mathcal{D}_\lambda \coloneqq \mathcal{D}_{L_{\lambda_1}^{c_1} \otimes \cdots \otimes L_{\lambda_n}^{c_n}}$$

Theorem 13.9. *1. For all λ, $\Gamma(\mathcal{D}_\lambda) \simeq U_{\overline{\lambda}}$, where $\overline{\lambda} = \lambda \bmod W$. In particular, we have well-defined map:*

$$\Gamma_\lambda : \mathcal{D}_\lambda\text{-mod} \to U_{\overline{\lambda}}\text{-mod}$$

2. Now assume $\langle \lambda + \rho, \alpha^\vee \rangle \notin \mathbb{Z}_{<0}$ for all $\alpha^\vee$ positive coroots, then Γ_λ is exact.

3. Now assume $\langle \lambda + \rho, \alpha^\vee \rangle \notin \mathbb{Z}_{\leq 0}$ for all $\alpha^\vee$ positive coroots, then Γ_λ is an equivalence.

Remark. *For every regular $\overline{\lambda}$, there exists λ for which the above theorem applies. For all $\overline{\lambda}$, there exists λ for which Γ_λ is exact.*

Let's explain how to deduce (3) from (1) and (2). We use the following general result:

Proposition 13.10. *Let $\mathcal{A}$ be an abelian category with projective generator P. Suppose in addition:*

$$\mathrm{Hom}_{\mathcal{A}}(P, \bigoplus M_i) \simeq \bigoplus \mathrm{Hom}_{\mathcal{A}}(P, M_i)$$

that $\mathrm{Hom}_{\mathcal{A}}(P)$ commutes with all direct sums. Then we have an equivalence of category:

$$\mathrm{Hom}_{\mathcal{A}}(P, -) : \mathcal{A} \xrightarrow{\sim} \mathrm{End}_{\mathcal{A}}(P)^{\mathrm{op}}\text{-mod}$$

Now take $\mathcal{A} = D_\lambda$-mod and $P = \mathcal{D}_\lambda$. Then the first part tells me $\mathrm{End}_{\mathcal{A}}(P)^{\mathrm{op}} \simeq U_{\overline{\lambda}}$. The second part tells me that $\mathcal{D}_\lambda$ is projective. Now the extra input is: if $\langle \lambda + \rho, \alpha^\vee \notin \mathbb{Z}_{\leq 0} \rangle$, then Γ_λ sends nonzero object to nonzero object (i.e. $\mathcal{D}_\lambda$ is a generator). Once I can see this then the third part follows from the general proposition.

Corollary 13.11 (Translation Functor)**.** *Let $\overline{\lambda}, \overline{\mu}$ be regular central characters, with λ, μ dominant. If $\nu = \mu - \lambda \in \Lambda$-integral, then:*

$$U_{\overline{\lambda}}\text{-mod} \simeq U_{\overline{\mu}}\text{-mod}$$

In particular, if $\lambda \in \Lambda^+$, then U_0-mod $\simeq U_\lambda$-mod.

If λ regular, μ not, $\lambda, \mu \in \Lambda^+$, we have:

$$U_{\overline{\lambda}}\text{-mod} \simeq \mathcal{D}_\lambda\text{-mod} \simeq \mathcal{D}_\mu\text{-mod} \to U_{\overline{\mu}}\text{-mod}$$

This is not an equivalence, but it's exact. This is called "translation to the wall".

Corollary 13.12.

$$(\mathfrak{g}, K)\text{-mod}_\lambda \simeq K\text{-equivariant } \mathcal{D}_\lambda\text{-mod}$$

Fix $B = T \cdot N$. Let's talk about connections to the theory of category $\mathcal{O}$.

Definition 13.13. Fix a decomposition:

$$\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{t} \oplus \mathfrak{n}^+$$

The category $\mathcal{O}$, denoted as $\mathcal{O}^{\mathrm{BGG}}$ consists of $U(\mathfrak{g})$-modules such that:

1. It is finitely generated over $U(\mathfrak{g})$.

2. $\mathfrak{t}$-acts diagonally so that it has weight decomposition.

3. It is locally $U(\mathfrak{n}^+)$-finite.

Fix a central character χ, we denote $\mathcal{O}_\chi^{\mathrm{BGG}}$ as the subcategory of $\mathcal{O}^{\mathrm{BGG}}$ where for each $z \in Z(U\mathfrak{g})$, $z - \chi(z)$ acts locally nilpotently.

It is a well known fact that:

$$\mathcal{O}^{\mathrm{BGG}} = \bigoplus_\chi \mathcal{O}_\chi^{\mathrm{BGG}}$$

We will instead consider two different versions of it. We define $\mathcal{O}_\chi$ to be the category of $U(\mathfrak{g})$-modules satisfying:

1. It is finitely generated over $U(\mathfrak{g})$.

2. $\mathfrak{t}$-acts locally finitely (not necessarily semisimply).

3. It is locally $U(\mathfrak{n}^+)$-finite.

4. It has central character χ.

Note that we weaken the second condition by strengthen the fourth condition. Moreover, we define $\mathcal{O}'_\chi \subset \mathcal{O}_\chi$ by replacing the second condition by our original one: $\mathfrak{t}$ acts semisimply. So that $\mathcal{O}'_\chi \subset \mathcal{O}_\chi^{\mathrm{BGG}}$. But anyway we have:

$$\mathrm{Irr}(\mathcal{O}_\chi) \simeq \mathrm{Irr}(\mathcal{O}_\chi^{\mathrm{BGG}}) \simeq \mathrm{Irr}(\mathcal{O}'_\chi)$$

Then we have:

Corollary 13.14. *Let λ be a regular dominant weight. We have:*

$$N\text{-equivarint } \mathcal{D}_\lambda\text{-mod} \simeq \mathcal{O}_{\overline{\lambda}}$$

$$B\text{-equivariant } \mathcal{D}_\lambda\text{-mod} \simeq \mathcal{O}'_\lambda$$

As an application, starting from $G_{\mathbb{R}}$ complex group, we have the diagonal copy $G \subset G \times G$. We have:

$$G\text{-equivariant } \mathcal{D}\text{-mod}(G/B \times G/B) \simeq B\text{-equivariant } \mathcal{D}\text{-mod}(G/B) \simeq \mathcal{O}'_0$$

Corollary 13.15. *$\mathcal{O}'_0$ is equivalent to the category of Harish-Chandra bimodules:*

$$\mathcal{O}'_0 \simeq (\mathfrak{g} \oplus \mathfrak{g}, G)\text{-mod}_{0,0}$$

Corollary 13.16.

$$\mathrm{Irr}(\mathfrak{g}, K)\text{-mod}_0 \simeq \{(\mathcal{O}, \psi) \mid \mathcal{O} \text{ is a } K\text{-orbit on } G/B \ , \psi \in \mathrm{Irr}(\pi_0(\mathrm{stab}(x)))\}$$

The right hand side is called an equivariant local system. In fact:

$$\pi_0(\mathrm{stab}(x)) = \pi_0(T_x^\theta) \simeq (\mathbb{Z}/2)^m$$

Corollary 13.17.

$$\mathrm{Irr}(\mathcal{O}_0) \simeq B\backslash G/B = W$$

We wish to understand characters of irreducible representations for category $\mathcal{O}$. We can think these as formal characters. If $M \in \mathcal{O}_0$, we have:

$$M = \bigoplus M[\lambda]$$

then

$$\mathrm{ch}(M) = \sum \dim M[\lambda] e^{\lambda}$$

In all these cases, we have an easy formula for character of costandard objects, via the $*$-extension of equivariant bundle with flat connection from a K or B-orbit. For $\mathcal{O}_0$ we get the **dual Verma modules**:

$$\mathrm{ch}(M_\lambda) = e^{\lambda} \cdot \prod_{\alpha>0} \frac{1}{1-e^{-\alpha}}$$

Consider Grothendieck group $K_0(\mathcal{O}_0)$ or $(K_0((\mathfrak{g}, K)\text{-mod}_0))$. It has two bases:

1. Costandard modules.
2. Irreducible modules.

We wish to describe the transformation matrix. For $\mathcal{O}_0$, we have:

$$\mathbb{Z}[W] \simeq K_0(\mathcal{O}_0), w \mapsto M_{ww_0 \cdot 0}$$

These classes of irreducibles give Kazhdan-Lusztig basis / canonical basis.

13.3 Proof of Localization Theorem

Lemma 13.18. *$\mathbb{P}^n$ is $\mathcal{D}$-affine. Namely:*

$$\mathcal{D}\text{-mod}(\mathbb{P}^n) \simeq \Gamma(\mathcal{D}_{\mathbb{P}^n})\text{-mod}$$

Proof. This amounts to check:

1. Γ is exact on $\mathcal{D}$-mod$(\mathbb{P}^n)$.
2. $\Gamma(M) \neq 0$ if $M \neq 0$.

Let $M \in \mathcal{D}$-mod$(\mathbb{P}^n)$, and let $\pi : \mathbb{A}^{n+1}\backslash\{0\} \to \mathbb{P}^n$. Set $\widetilde{M} = \pi^*(M)$. Then we know:

$$\Gamma(\widetilde{M}) = \bigoplus_{i\in\mathbb{Z}} \Gamma(M(i))$$

Note if $\mathrm{Lie}(\mathbb{C}^\times) = \langle e \rangle$ for the Euler vector field $e = \sum x_i \partial_i$, we know e acts on i-th summand by scalar i. Let's denote $\Gamma(\widetilde{M})_i := \Gamma(M(i))$.

We need the basic fact for project space that if $0 \neq \mathcal{F} \in \mathsf{QCoh}(\mathbb{P}^n)$, then $\Gamma(\mathcal{F}(i)) \neq 0$ for $i \gg 0$.

Now that if $\Gamma(\widetilde{M})_i = 0$ and $i \neq -1$, then $\Gamma(\widetilde{M})_{i+1} = 0$. This is because, for $v \in \Gamma(\widetilde{M})_{i+1}$, we have:

$$(i+1)v = \sum x_i(\partial_i v)$$

with each $\partial_i v \in \Gamma(\widetilde{M})_i$. Thus is $M \neq 0$, then $\Gamma(\widetilde{M})_i \neq 0$ for $i \geq 0$. This proves the second part.

For the first part, we know for $\mathcal{F} \in \mathrm{Coh}(\mathbb{P}^n)$, we know $H^{>0}(\mathcal{F}(i)) = 0$ for $i \gg 0$. Now if $\mathcal{F} \to \mathcal{F}' \to 0$ of quasi-coherent sheaves, and let $s \in \Gamma(\mathcal{F}')$. For $i \gg 0$, we have

$$s(\Gamma(\mathcal{O}(i))) \subseteq \mathrm{Im}(\Gamma(\mathcal{F}(i)))$$

Now $e = \sum x_i\partial_i = \sum \partial_i x_i - (n+1)$, and $M \to M' \to 0$. Let s be a section of M'. If $v \in \Gamma(\widetilde{M})_i$, we have:

$$(i + (n+1))v = \sum \partial_i x_i v$$

So if $x_i v \in \mathrm{Im}(\Gamma(\widetilde{M}))$ for all i, then so does v unless $i = (-n-1)$. This proves the first part. □

Remark. *For $\mathcal{D}_c$-mod$(\mathbb{P}^n)$ with $c \in \mathbb{C}$, this proves Γ is exact except for*

$$c = -n-1, -n-2, \cdots$$

$\Gamma(M) \neq 0$ for $M \neq 0$ except for

$$c = -1, -2, \cdots$$

We wish to generalize this argument to arbitrary flag varieties. We need to use an analogue of

$$\mathbb{A}^{n+1} \backslash \{0\} \to \mathbb{P}^n$$

Namely:

$$G/U \to G/B$$

Note that G/U is quasi affine, namely:

$$G/U \hookrightarrow \overline{G/U} = \mathrm{Spec}\ (\mathbb{C}[G/U])$$

A serious problem is that $\mathbb{C}[G/U]$ is not smooth except for $G = \mathrm{SL}(2)$. But we have a ring isomorphism:

$$\mathbb{C}[G/U] = \bigoplus_{\lambda \in \Lambda^+} V_\lambda$$

with multiplication of RHS $V_\lambda \otimes V_\mu \to V_{\lambda+\mu}$.

A Key imput here is the Fourier transform of $\mathcal{D}$-modules:

$$\mathrm{FT} : \mathcal{D}(V) \to \mathcal{D}(V^*)$$

Namely, we can identify:

$$\mathcal{D}(V) = \mathbb{C}[V, V^*]/([v, f] = f(v))$$
$$\mathcal{D}(V^*) = \mathbb{C}[V^*, V]/([f, v] = f(v))$$

Then:

$$\mathrm{FT} : v \mapsto v, f \mapsto -f$$

More generally, given a vector bundle $\mathcal{V} \to S$, we have Fourier transform:

$$\mathrm{FT} : \mathcal{D}(\mathcal{V}) \to \mathcal{D}(\mathcal{V}^*)$$

Lemma 13.19. *After choosing a trivialization of* $\det(\mathcal{V})$*, we have a canonical isomorphism:*

$$\mathcal{D}(\mathcal{V}) \simeq \mathcal{D}(\mathcal{V}^*)$$

Let $D = \Gamma(\mathcal{D}_{G/U})$. If M is a $\mathcal{D}$-mod on G/B, corresponds to $\widetilde{M}$ on G/U. Then:

$$\Gamma(\widetilde{M}) = \bigoplus_{\lambda} \Gamma(M \otimes \mathcal{O}(\lambda))$$

Note that we have a $\mathbb{P}^1$-fibration: $G/B \to G/P_\alpha$ for some simple root α. The similar picture for G/U is

$$G/U \to G/P'_\alpha$$

where $P'_\alpha = [P_\alpha, P_\alpha] = \langle U, \mathrm{SL}(2)_\alpha \rangle$. This is an $(\mathbb{A}^2 \backslash \{0\})$-fibration. We complete this into a $\mathbb{A}^2$-bundle:

$$(\overline{G/U})_\alpha \to G/P'_\alpha$$

Then in fact this is a symplectic rank 2 vector bundle. So:

$$D = \Gamma(\mathcal{D}_{G/U}) = \Gamma(\mathcal{D}_{\overline{(G/U)_\alpha}})$$

Let Φ_{s_α} be the Fourier transform for this vector bundle. One check that

Lemma 13.20. $D^G = U(\mathfrak{t})$

Proof. $D \subset \mathrm{End}(\mathbb{C}[G/U]) = \mathrm{End}(\bigoplus V_\lambda)$. If $d \in D^G$, then d acts on V_λ by a constant c_λ. The map $\lambda \to c_\lambda$ must be a polynomial since d is a differential operator (check this!) So it must comes from $U(\mathfrak{t})$. □

For $V_\lambda \subset \mathbb{C}[G/U]$, we know $V_\lambda^* \simeq V_{\lambda'}$. Choose a dual basis f_i, f^i for $V_\lambda, V_{\lambda'}$. Consider the following generalization of Euler vector field:

$$e_\lambda = \sum f_i \Phi_{w_0}(f^i) \in U(\mathfrak{t})$$

One can compute:

$$e_\lambda = \prod_\alpha \prod_{i=1}^{\langle \alpha^\vee, \lambda \rangle} (\alpha^\vee + \langle \alpha^\vee, \rho \rangle - i)$$

Now we can run a similar argument. If $M \neq 0$, then $\Gamma(\widetilde{M})_\lambda \neq 0$ for $\lambda \in \Lambda^+$. Pick $0 \neq v \in \Gamma(\widetilde{M})_\lambda$. Note tht $e_{\lambda'} v \neq 0$, so

$$\Phi_{w_0}(f^i) v \neq 0 \in \Gamma(\widetilde{M})_0$$

for some i.

Remark. *We proved*

$$\mathcal{D}\text{-mod} \xrightarrow{\Gamma} U_0\text{-mod}$$

The inverse is given by

$$\mathcal{L} : M \to \mathcal{D} \otimes_{U_0} M$$

For example, for a $\mathcal{D}$-module $\mathcal{M}$ on G/B, its fibre at $x \in G/B$ is defined as:

$$\mathcal{M}_x \coloneqq \mathbb{C}_x \otimes_{\mathcal{O}} \mathcal{M} \simeq \delta_x \otimes_{\mathcal{D}} \mathcal{M}$$

where $\delta_x = \mathbb{C}_x \otimes_{\mathcal{O}} \mathcal{D}$. We call $\Gamma(\delta_x)$ a Verma module.

For $\mathcal{M} = \mathcal{L}(M)$, we compute:

$$\mathcal{L}(M)_x = \delta_x \otimes_{\mathcal{D}} \mathcal{D} \otimes_U M = \text{Verma} \otimes_U M = (M_{n_x})_{-2\rho}$$

This appeared in the definition of exponents. This leads to to Beilinson-Bernstein's proof of Casselman submodule theorem.

13.4 Applications

Theorem 13.21 (Localization of Discrete Series)**.** *Assume* rank $K =$ rank G. *Let* $0 \neq L \in (\mathfrak{g}, K)$-mod *be a discrete series. Then infinitesimal character of L is regular integral λ, then localization gives:*

$$\mathcal{L}_\lambda(L) = \mathcal{D}_\lambda \otimes_{U_\lambda} L$$

Then the support $\operatorname{supp}(\mathcal{L}_\lambda(L))$ *is a closed K-orbit on G/B. In fact* $\operatorname{gr}(\mathcal{L}_\lambda(L))$ *is a line bundle on the conormal.*

Let's discuss a little bit of Kazhdan-Lusztig theory. Recall we saw:

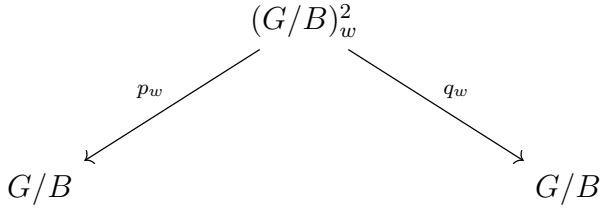
$$\mathcal{D}\text{-mod}_U(G/B) \simeq \mathcal{O}_0$$

and we can identify $K^0(\mathcal{O}_0) \simeq \mathbb{Z}[W]$. We wish to describe classes of irreducibels as canonical basis. In fact W acts naturally on K^0, coming from the action of braid group on the derived category

$$D^b(\mathcal{O}_0) = D^b(\mathcal{D}\text{-mod}_U(G/B))$$

It can be describe:

1. geometrically:



So that $\widetilde{s_\alpha}$ acts by $q_{s_\alpha *} p^*_{s_\alpha}$, hence $\widetilde{w}$ acts by $q_{w*} p^*_w$. Here we are using the presentation:

$$B = \langle \widetilde{w} \mid \widetilde{uv} = \widetilde{u}\widetilde{v} \text{ if } \ell(uv) = \ell(v) + \ell(v) \rangle$$

2. Tensoring with Harish-Chandra bimodules.

3. Via derived localization.

The definition of canonical basis uses the q-deformation of W-action on $K^0(\mathcal{O}_0)$. For example $\mathbb{Z}[W]$ deforms to be the Hecke algebra H_q.

A Classification of Real Forms

Let $\mathfrak{g}$ be a semisimple Lie algebra. We know the group $\mathrm{Aut}(\mathfrak{g})$ is a Lie group with Lie algebra $\mathfrak{g}$. We denote its identity component at the *the adjoint group* G_{ad}.

Now let $\mathfrak{h} \subset \mathfrak{g}$ be a Cartan subalgebra, and $H \subset G_{\mathrm{ad}}$ the corresponding connected Lie subgroup. This group acts by 1 on $\mathfrak{h}$ and by $\exp(\alpha(x)), x \in \mathfrak{h}$ on $\mathfrak{g}_\alpha$. The exponential map induces:

$$\mathfrak{h}/2\pi i P^\vee \simeq H$$

We call H the maximal torus of G_{ad} corresponding to $\mathfrak{h}$.

Proposition A.1. $N(H)/H \simeq N(\mathfrak{h})/\mathfrak{h} \simeq W$

Now let D be the Dynkin diagram of $\mathfrak{g}$. We denote $\mathrm{Aut}(D)$ the diagram automorphism of D. We then have an isomorphism:

$$\xi : \mathrm{Aut}(D) \ltimes G_{\mathrm{ad}} \to \mathrm{Aut}(\mathfrak{g})$$

Definition A.2. Over an arbitary field k of characteristic 0. The Lie algebras defined via Serre relations are called **split**.

Question: Let $\mathfrak{g}$ be a semisimple Lie algebra of a field K of characteristic 0, which splits over a Galois extension L of K. Can we calssify such $\mathfrak{g}$?

Let $\Gamma = \mathrm{Gal}(L/K)$ be the Galois group of L over K. We can recover $\mathfrak{g}$ as the subalgebra of invariants $\mathfrak{g}_L^\Gamma$. Note this action is *twisted-linear*: it is additive and $g(\lambda x) = g(\lambda)g(x)$ for $x \in \mathfrak{g}_L, \lambda \in L$.

Let's take $\rho_0(g)(\lambda x) = g(\lambda)x$ for $x \in \mathfrak{g}$ (this defines the split form on K). Now for any other twisted-linear action $\rho(g)$, we can write $\rho(g) = \eta(g)\rho_0(g)$ for some $\eta : \Gamma \to \mathrm{Aut}(\mathfrak{g}_L)$. We need ρ to be a homomorphisma:

$$\eta(gh)\rho_0(gh) = \eta(g)\rho_0(g)\eta(h)\rho_0(h)$$

which is equivalent to:

$$\eta(gh) = \eta(g)g(\eta(h))$$

where $g(a) = \rho_0(g)a\rho_0(g)^{-1}$ for $a \in \mathrm{Aut}(\mathfrak{g}_L)$. This is saying η is a 1-cocycle. We denote the Lie algebra attached to the cocycle by g_η.

Now let's discuss when $\mathfrak{g}_{\eta_1} \simeq \mathfrak{g}_{\eta_2}$. Indeed, this happens when there exists $a \in \mathrm{Aut}(\mathfrak{g}_L)$ such that $\rho_1(g)a = a\rho_2(g) \iff \eta_1(g) = a\eta_2(g)g(a)^{-1}$. Two 1-cocyles related in this way are called cohomologous. Hence:

Theorem A.3. *Semisimple Lie algbera $\mathfrak{g}$ over K which split over a Galois extension L of K are classified by:*

$$H^1(\Gamma, \mathrm{Aut}(\mathfrak{g}_L))$$

Let's now apply the general theory to the case $K = \mathbb{R}, L = \mathbb{C}$. Then $\Gamma = \mathbb{Z}/2$ is generated by the complex conjugation $s \mapsto \overline{s}$, and $\mathrm{Aut}(\mathfrak{g}_L) = \mathrm{Aut}(D) \ltimes G_{\mathrm{ad}}$. Note the cocycle η is determined by $s = \eta(-1) \in \mathrm{Aut}(D) \ltimes G_{\mathrm{ad}}$. The cocycle condition reads $s\overline{s} = 1$. We need to quotient out The relations $s \mapsto as\overline{a}^{-1}$.

Theorem A.4. *Real semisimple Lie algberas whose complexification is $\mathfrak{g}$ are classified by $s \in \mathrm{Aut}(D) \ltimes G_{\mathrm{ad}}$ such that $s\overline{s} = 1$ modulo $s \mapsto as\overline{a}^{-1}$ for $a \in \mathrm{Aut}(\mathfrak{g})$.*

We denote the real form of $\mathfrak{g}$ corresponding to s by $\mathfrak{g}_{(s)} = \{x \in \mathfrak{g} \mid \overline{x} = s(x)\}$.

Definition A.5.

1. A real form $\mathfrak{g}_{(s)}$ of complex simple Lie algbera $\mathfrak{g}$ is said to be inner to $\mathfrak{g}_{(s')}$ is $s' = gs$ up to equivalence, where $g \in G_{\mathrm{ad}}$. An inner form is a form inner to the split form.

2. We call $\mathfrak{g}_{(s)}$ to be quasi-split if $s = s_0 \in \mathrm{Aut}(D)$ modulo equivalence.

Definition A.6. The *compact real form* is determined by the automorphism:

$$\tau(h_j) = -h_j, \quad \tau(e_j) = -f_j, \quad \tau(f_j) = -e_j$$

Let's denote $\mathfrak{g}_{(\tau)} = \mathfrak{g}^c$

Proposition A.7. *The Killing form of $\mathfrak{g}^c$ is negative definite. As a consequence, the associated adjoint group is compact.*

Consider a complex simple Lie algebra $\mathfrak{g} = \mathfrak{g}^c \otimes_{\mathbb{R}} \mathbb{C}$. We have the compact antilinear involution $\omega = \sigma_\tau$ whose set of fixed points is $\mathfrak{g}^c$. For another real structure, we can write $\sigma = \omega \circ g$ for $g \in \mathrm{Aut}(\mathfrak{g})$ such that $\omega(g)g = 1$

B Exercises

Definition B.1. Fix a principal $\mathfrak{sl}(2)$-triple for $\mathfrak{g}$. Define $\psi : \mathfrak{u} \to \mathbb{C}$ by $\psi(x) = (f, x)$. A $\mathfrak{g}$-module M is called **nondegenerate** iff $M \otimes_{U(\mathfrak{u})} \mathbb{C}_\psi \neq 0$.

Exercise. Let M be an irreducible Harish-Chandra bimodule. Then M is nondegenerate iff $\mathrm{supp}(M) = \mathcal{N}$ has the full support.

We need the following basic facts:

1. For a finite length HC bimodule M, M is finitely generated as a left (or right) $U(\mathfrak{g})$-module, and $\mathrm{supp}(M)$ is well defined (same for left and right actions and independent of good filtration we chose). Moreover, $\mathrm{supp}(M) \subset \mathcal{N}$, and is G-invariant.

2. The nilpotent cone $\mathcal{N}$ is Cohen-Macaulay.

3. The theorem of **Miracle Flatness**: Let $f : X \to Y$ be a morphism of integral schemes of finite type over a field. Suppose further X Cohen-Macaulay and Y regular. Suppose for each closed point $y \in Y$, we have $f^{-1}(y)$ is equidimensional and $\dim X = \dim Y + \dim f^{-1}(y)$, then $f : X \to Y$ is flat. In particular, if I have a vector bundle E on X, then f_*E is flat over Y, namely: $\mathrm{Tor}_i^{\mathcal{O}_Y}(f_*E, k_y) = 0$ for each $y \in Y$.

4. The grading of $U(\mathfrak{sl}_2)$. Let $\mathfrak{g}$ be a semisimple Lie algebra with principal $\mathfrak{sl}_2$-tripel (e, f, h). We can view $U(\mathfrak{sl}_2)$ as a graded algebra by letting $\deg e = -1, \deg f = 1, \deg h = 0$. Therefore, we can view $\mathfrak{g}$ as a graded-module over $U(\mathfrak{sl}_2)$: for simple roots α,
$$\deg e_\alpha = -1, \deg f_\alpha = 1, \deg(\mathfrak{t}) = 0.$$
In addition, suppose we have a $U(\mathfrak{g})$-module M, one can put a compatible pair of grading and filtration on M. Indeed, we can use the $\mathfrak{g}$-action to put a grading on M, and choose a homogeneous set of generators of M w.r.t the above grading. These generators define a compatible filtration on M.

5. Note we have identifications:
$$\begin{array}{ccc} \mathbb{C}[\mathfrak{g}] & \xrightarrow{\sim} & \mathbb{C}[\mathfrak{g}^*] \\ \uparrow & & \uparrow \\ \mathbb{C}[\mathfrak{g}/\mathfrak{b}] & \xrightarrow{\sim} & \mathbb{C}[\mathfrak{u}^*] \end{array}$$
Now given $\psi : U(\mathfrak{u}) \to \mathbb{C}$ a character, we wish to describe $\mathbb{C}_\psi$ as a $S(\mathfrak{u}) = \mathbb{C}[\mathfrak{u}^*] = \mathbb{C}[\mathfrak{g}/\mathfrak{b}]$-module. Indeed, $\mathbb{C}[\mathfrak{g}/\mathfrak{b}]$ has generators $x_i = (u_i, -)$, for u_i a basis for $\mathfrak{u}$. Then:
$$\mathbb{C}_\psi \simeq \mathbb{C}[x_i]/(x_i - (f, u_i)) = \mathbb{C}[x_i]/(x_i - x_i(f))$$
Therefore, for any $\mathbb{C}[\mathfrak{g}]$-module P, we have:
$$P \otimes_{\mathbb{C}[\mathfrak{g}/\mathfrak{b}]} \mathbb{C}_\psi \simeq P/(x_i - x_i(f))$$
Hence $\mathrm{supp}(P \otimes_{\mathbb{C}[\mathfrak{g}/\mathfrak{b}]} \mathbb{C}_\psi) = \mathrm{supp}(P) \cap V(x_i - x_i(f)) = \mathrm{supp}(P) \cap (\mathfrak{b} + f)$.

6. Fix a character χ of $Z(U\mathfrak{g})$. There is a HC bimodule $U_\chi = U(\mathfrak{g}) \otimes_{Z(U\mathfrak{g})} \mathbb{C}_\chi$. Under the PBW filtration we have:
$$\mathrm{gr} U_\chi \simeq \mathcal{O}(\mathcal{N})$$
and in particular $\mathrm{supp}(U_\chi) = \mathcal{N}$. Kostant's theorem on Whittaker module tells us U_χ are actually simple objects in the category of Whittaker modules, in particular they are nonzero.

7. Artin-Rees Lemma for noncommutative rings.

Proof. (First proof of full support implies nondegenerate). We wish to show
$$\mathrm{gr}(M \otimes_{U(\mathfrak{u})} \mathbb{C}_\psi) \neq 0$$

Note we have a spectral sequence:

$$\mathrm{Tor}_i^{\mathbb{C}[\mathfrak{u}^*]}(\mathrm{gr}M, \mathbb{C}) \Rightarrow \mathrm{gr}(\mathrm{Tor}_i^{U(\mathfrak{u})}(M, \mathbb{C}_\psi))$$

Consider the natural map $\mathcal{N} \to \mathfrak{g}/\mathfrak{b} \simeq \mathfrak{u}^*$. Restricting to the regular nilpotent orbit we have: $\mathcal{N}_0 \to \mathfrak{u}_0^*$. This is flat by Miracle Flatness. Note $\mathrm{gr}M$ is G-equivariant, hence it is a vector bundle over $\mathcal{N}_0$. Hence $\mathrm{gr}M$ is flat over $\mathfrak{u}_0^* \subset \mathfrak{u}^*$. From this we see:

$$\mathrm{supp}(\mathrm{gr}M \otimes_{\mathbb{C}[\mathfrak{u}^*]} \mathbb{C}) \cap \mathcal{N}_0 = \mathrm{supp}(\mathrm{gr}(M \otimes_{U(\mathfrak{u})} \mathbb{C}_\psi)) \cap \mathcal{N}_0$$

From this we see LHS equals $\mathcal{N}_0 \cap \mathfrak{b} \neq 0$ □

Proof. (Second proof of full support implies nondegenerate) Let M be an irreducible HC bimodule. Suppose the right $U(\mathfrak{g})$-action gives character χ. We wish to construct an injective left $U\mathfrak{g}$-morphism $U_\chi \to M$. Note this amounts to give an element $m \in M$. Injectivity will be implied by the injectivity on the level of associated graded, hence we need:

$$\mathcal{O}(\mathcal{N}) = \mathrm{gr}U_\chi \to \mathrm{gr}M$$

be an injective $\mathbb{C}[\mathfrak{g}]$-module map. Since $\mathrm{supp}(M) = \mathcal{N}$, we know the $\mathbb{C}[\mathfrak{g}]$-module $\mathrm{gr}M$ actually descends to a $\mathcal{O}(\mathcal{N})$-module. Fully support allows us to choose $\overline{m} \in \mathrm{gr}M$ such that $\mathrm{Ann}(m) = 0 \in \mathcal{O}(\mathcal{N})$. Any lift m of $\overline{m}$ will provide us with the left $U\mathfrak{g}$ morphism we desired.

In summary, we find a submodule $N \subset M$ which is nondegenerate. By Artin-Rees we know the functor:

$$P \mapsto \varprojlim P/(\ker\psi)^n P$$

is exact. Hence we have:

$$0 \neq \varprojlim N/(\ker\psi)^n N \subset \varprojlim M/(\ker\psi)^n M$$

This means $M/(\ker\psi)M \neq 0 \implies M$ nondegenerate. □

Proof. (First proof of nondegenerate implies full support). Suppose $\mathrm{supp}(M) \subsetneq \mathcal{N}$, we need to see $M \otimes_{U(\mathfrak{u})} \mathbb{C}_\psi = 0$.

As we remarked above we can put a compatible system of grading an filtrations on M. Now let's adjust the filtration:

$$F_i'(M_j) = F_{i+j}(M_j)$$

Hence

$$F_i'(M) = \oplus_{j\in\mathbb{Z}} F_{i+j} M_j$$

In particular, if $x \in F_i'(M_j) = F_{i+j}(M_j) = F_{i+j}M \cap M_j$, we have:

$$e_\alpha x \in M_{j-1} \cap F_{i+j+1}M = F_i'(M_{j-1})$$

namely $U(\mathfrak{u})$-action preserves the filtration F'. Therefore, under this filtration we have

$$\mathrm{gr}(M \otimes_{U(\mathfrak{u})} \mathbb{C}_\psi) \simeq \mathrm{gr}(M) \otimes_{\mathbb{C}[\mathfrak{u}^*]} \mathbb{C}_\psi$$

It has support: $\mathrm{supp}(M) \cap (\mathfrak{b} + f) = 0$ by the theory of Kostant slice. □

Proof. (Second proof of nondegenerate implies full support) We can check under the left $\mathfrak{g}$-action, $M \otimes_{U(\mathfrak{u})} \mathbb{C}_\psi$ is a Whittaker module. Hence by Kostant's theorem, we know there will be a $Z(U\mathfrak{g})$-module E such that:

$$E \otimes_{Z(U\mathfrak{g})} U(\mathfrak{g}) \otimes_{U(\mathfrak{u})} \mathbb{C}_\psi \simeq M \otimes_{U(\mathfrak{u})} \mathbb{C}_\psi$$

Now E has a quotient $\mathbb{C}_\chi$ for some character χ, this means $M \otimes_{U(\mathfrak{u})} \mathbb{C}_\psi$ has a quotient $U_\chi \otimes_{U(\mathfrak{u})} \mathbb{C}_\psi$. Therefore one sees:

$$\mathrm{supp}(M) \supset \mathrm{supp}(M \otimes_{U(\mathfrak{u})} \mathbb{C}_\psi) \supset \mathrm{supp}(U_\chi \otimes_{U(\mathfrak{u})} \mathbb{C}_\psi) = \mathfrak{u}$$

By invoking the first part. Now $\mathrm{supp}(M)$ is G-invariant, we see $\mathrm{supp}(M) = \mathcal{N}$. □